

Design and Analysis of Architectures and Protocols for WDM Optical Networks

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Abstract

Existing networks are becoming increasingly congested as a growing number of users seek access to the vast amount of information available on the Internet. At the same time, emerging multimedia applications, such as video conferencing, interactive television, and video-on-demand are expected to further test the limits of current network infrastructures. There is an immediate need for the development of new, evolutionary high-capacity networks which are capable of supporting these growing bandwidth requirements. Wavelength-division multiplexing (WDM) is a promising solution for satisfying this bandwidth bottleneck. As WDM deployment becomes widespread, new network architectures and protocols which take advantage of WDM's capabilities will need to be developed. This dissertation investigates network architectures and protocols for WDM networks, with the goals of measuring the performance of existing approaches and of developing new architectures and protocols which provide improved performance as well as support for services such as multicasting.

First, Rainbow, a circuit-switched WDM network prototype, is modeled and analyzed using a technique called equilibrium point analysis (EPA). The results show the existence of an optimal timeout duration for the protocol, and the analysis can be used to find this optimal point for a given set of system parameters. Second, a new class of protocols, called multiconfiguration multihop protocols, is developed and analyzed. This class of protocols is intended to provide a balance between the amount of time a transmitter or receiver spends tuning from one wavelength to another, and the number of intermediate nodes or hops traversed by a packet. The protocols provide a viable alternative to existing protocols which typically either require frequent tuning of transmitters and receivers, or require high average hop distances. Third, a number of algorithms for scheduling multicast transmissions in a way which optimizes the utilization of network resources is developed. By minimizing the amount of time a node spends waiting for a multicast message, the network may achieve higher throughputs. Finally, a new node architecture for wavelength-routed networks is presented. The architecture allows for low-cost scalability in a network while maintaining reasonable performance when compared to a wavelength-routing-switch node architecture.

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Chapter 1

Introduction

In recent years, the increasing popularity of the Internet and the World Wide Web has resulted in a rapidly rising demand for network bandwidth. It is expected that the current network infrastructure will be unable to support the massive wave of traffic resulting from emerging high-bandwidth multimedia applications. Already, users are seeing diminished performance from the Internet, including high delays and network outages. A new revolutionary technology is needed to support the bandwidth needs of network users and to provide support for emerging network services.

Optical fiber has been acknowledged as the best solution for meeting these demands, with a single fiber having a potential bandwidth on the order of 50 THz. In the past few decades, optical fiber has gradually replaced copper wires in many communications applications. Currently, links in long-haul networks consist almost entirely of optical fiber, with hundreds of thousands of miles of new fiber being deployed every year. In the local area network (LAN) environment, fiber-based protocols such as fiber distributed data interface (FDDI) have been deployed, and optical fiber is also starting to appear in traditionally copper-based LANs such as Ethernet.

Optical fiber offers increased bandwidth by allowing higher transmission rates over longer distances as compared to copper wires; however, the bandwidth is still limited by the electronic transmission and switching speeds at the fiber endpoints. Currently, transmission rates are limited to around 10 Gbps. The fiber bandwidth can be further exploited by dividing the bandwidth into a number of channels on different wavelengths, and by operating each channel at peak electronic rate. This technique is known as wavelength-division multiplexing (WDM) [35]. An optical WDM network not only provides a huge amount of bandwidth, but it may also offer the benefit of data transparency in which the network may carry signals of arbitrary format. Data transparency may be achieved through the all-optical transmission and switching of signals without electronic conversions at intermediate nodes.

The major challenge in deploying WDM in both local and wide area networks (WANs)

is to develop architectures and protocols which take full advantage of the benefits offered by WDM. At the same time, we must be careful to consider the limitations imposed by current optical device technology. The architectures and protocols must support critical services such as packet switching and multicasting, and must do so in a manner which efficiently utilizes the network resources. In a LAN, protocols must be designed in order to minimize the impact of the significant time required to tune transmitters and receivers from one wavelength to another. In designing wavelength-routed WDM networks, some of the major challenges include the complexity and scalability issues which arise from the need to demultiplex and individually route each wavelength at a node. In this work, we will focus primarily on architectures and protocols for LANs; however, one chapter will be devoted to a new switch architecture for wide area WDM networks. Before we address these issues, we will first present a brief overview of local and wide area network architectures.

1.1 WDM in Local Area Networks

As the need for bandwidth in the local area increases, new protocols with higher transmission rates need to be developed. At the same time, it is desirable for new protocols to be compatible with existing legacy protocols in order to minimize upgrade costs. Ethernet-based LANs are evolving from 10 Mbps Ethernet to 100 Mbps Fast Ethernet to 1 Gbps Gigabit Ethernet. Both of the faster Ethernet standards allow for the use of optical fiber as a transmission medium in addition to the traditional twisted-pair copper wires; however, it will be extremely difficult to scale Ethernet to the next generation beyond Gigabit Ethernet by simply increasing transmission rates. One limitation is that Ethernet uses the CSMA/CD MAC protocol which relies on the detection of packet collisions in order to resolve contention within an Ethernet LAN segment. To detect collisions at higher transmission rates over the same link distances, packets must be extended to some minimum size (as in Gigabit Ethernet). This carrier extension may result in very low channel utilization for small packets. Another limitation is that of achieving higher peak electronic transmission rates. A more practical approach to increasing bandwidth would be to utilize multiple wavelengths, with each wavelength operating at a rate compatible with existing protocols, such as legacy Ethernet.

WDM is an attractive solution for next-generation LANs since it offers the possibility of implementing traditional legacy LAN medium-access control (MAC) protocols on a subset of wavelengths, while running new WDM protocols on the remaining wavelengths. In the local area environment, to the best of our knowledge, WDM has yet to be implemented in commercial systems; however, a number of WDM LAN testbeds have been developed [24, 22, 12, 20], and commercial systems are expected to emerge soon.

A WDM LAN can be characterized by the network architecture, which includes the network topology and the node architecture, and by the protocol which is used to access

the network medium. These issues will also be examined in the following subsections.

1.1.1 Network Topology

A LAN can have a number of different possible physical topologies. Examples of physical topologies include mesh, star, bus, tree, and ring topologies (Fig. 1.1). The most common LANs in use today are Ethernet, which utilizes a broadcast bus topology, and the Token Ring, which utilizes a ring topology.

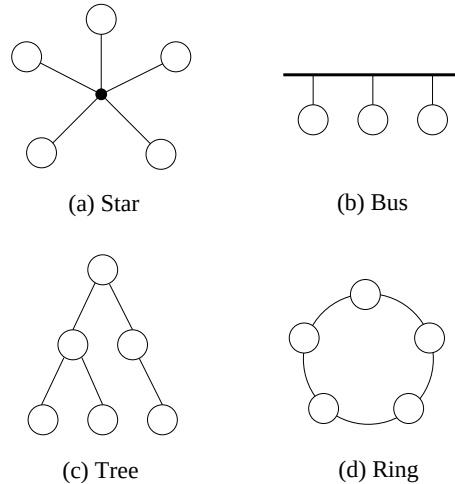


Figure 1.1: Physical topologies for LANs.

A promising architecture for WDM LANs is the broadcast-and-select architecture which is based on the star topology. The star topology can be physically implemented using an optical passive-star coupler (Fig. 1.2). Nodes connected to the star coupler may transmit an optical signal to the star coupler on one of multiple wavelengths. The star coupler combines the signals from all transmitters, and then splits the signal power, broadcasting the signal to all of the nodes. The nodes may then select a specific signal to receive by filtering out the appropriate wavelength. Because the star coupler is a passive component, it is fault-tolerant and relatively inexpensive. Also, the broadcast nature of the star coupler offers the possibility of implementing traditional broadcast protocols, such as Ethernet (and anticipated enhancements to Ethernet), over the architecture, thus possibly providing backward compatibility with existing legacy LAN equipment.

1.1.2 Node Architecture

In WDM networks, nodes are equipped with a number of transmitters and receivers. The transmitters consist of lasers which are modulated to transmit data. They may either transmit at a single fixed wavelength, or they may be tunable over a range of wavelengths. Receivers typically consist of photodetectors, and they may also be either fixed to receive on a single channel, or tunable to receive over a range of channels. Because of the high

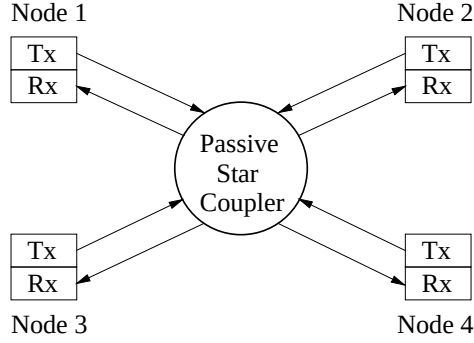


Figure 1.2: Architecture based on the passive-star coupler.

cost of tunable components, practical architectures are usually limited to a single tunable transmitter and single tunable receiver. In some architectures, tunable transmitters and receivers may be supplemented with fixed components, while in other architectures, only fixed components are used. Because the number of tunable transmitters and receivers at each node is limited, protocols must take full advantage of the tunability provided, and must also ensure that transmitter, receiver, and channel resources in the network are being utilized efficiently.

The network architecture can be characterized by the number of transmitters and receivers at each node, and whether these components are tunable or fixed. The notation $FT^iTT^j - FR^mTR^n$, introduced in [35], can be used to specify the node architecture, where i specifies the number of fixed transmitters, j specifies the number of tunable transmitters, m specifies the number of fixed receivers, and n specifies the number of tunable receivers. In some cases, the network protocol requires the use of a control channel which is a separate wavelength dedicated for the transmission of control and signaling information. The node architecture for such a network can be specified using the notation $CC - FT^iTT^j - FR^mTR^n$. Details on the device technologies utilized in various network components can be found in [5].

1.1.3 Access Protocol

Generally, WDM LAN protocols can be classified as being either circuit-switched or packet-switched, and either single-hop or multihop. In a *circuit-switched network*, a dedicated-bandwidth connection is established from the source node to the destination node for the duration of a call, while in a *packet-switched network*, information is broken into units of data, called packets, and these packets are transmitted from the source to the destination node. In a *single-hop network*, information is transmitted directly from the source node to the destination node in a single hop entirely in the optical domain. In a *multihop network*, information may traverse intermediate nodes (and experience electronic processing) before reaching the destination node.

Protocols can also be either contention-based or conflict-free. In a *contention-based protocol*, multiple users may contend for access to the network medium simultaneously, which may result in collisions, or transmissions which interfere with one another. In a *conflict-free protocol*, all transmissions are guaranteed to reach the intended destination. A conflict-free protocol may be implemented by properly allocating the bandwidth to the users in some dedicated manner. The allocation of bandwidth may be static, in which case each node is allocated a fixed portion of the bandwidth, or the allocation may be dynamic, in which case allocation is based on users' demands or reservations.

The dynamic reservation of bandwidth may require *pretransmission coordination* in which a source transmits control information to other nodes prior to transmitting data. Typically, pretransmission coordination schemes require a separate control channel for transmitting the control information; however, the need for a control channel may be eliminated through the use of in-band signaling, in which case the control information is sent over the data channels.

1.2 WDM in Wide Area Networks

WDM has been deployed commercially in long-distance networks (actually only in some long-haul links, as of today), with systems capable of operating up to 40 channels at rates of 2.5 Gbps or 10 channels at rates of 10 Gbps. These systems offer point-to-point WDM, where each channel must be converted to electronics at each node. The electronic processing and switching costs in such a system can be high if each node has a large number of input and output fiber ports. Emerging systems, which may contain optical switching and interconnection elements, will allow wavelengths to traverse a node entirely in the optical domain, thus reducing the amount of electronic processing required at each node. Such systems include ring networks which utilize optical add-drop multiplexers (ADMs) [18, 19], and wavelength-routed mesh networks, in which all-optical lightpaths are set up on specific wavelengths between pairs of nodes [1, 2, 14, 32, 36, 40].

1.3 Research Contributions

In WDM LANs, it is important to understand how system parameters such as the number of transmitters and receivers, number of wavelengths, transmitter and receiver tuning times, and propagation delays affect the network performance. It is also important to design protocols which appropriately account for system limitations and utilize the network resources in an efficient manner. In this dissertation, we investigate a number of WDM LAN protocols which are based on the optical passive-star coupler.

One network that is investigated is the Rainbow network testbed which was developed at IBM. Being a working prototype, this network illustrates some of the practical limi-

tations and challenges of optical device technology. An analytical model is developed for the Rainbow network in this dissertation, and for given network parameters, an optimal operating point for the protocol is found.

No previous study has directly compared single-hop networks to multihop networks. The primary obstacle has been the perception that single-hop networks and multihop networks have necessarily different hardware requirements. Single-hop networks typically require a small number of tunable transmitters and receivers, while multihop networks utilized a greater number of fixed transmitters and receivers. A framework for characterizing WDM LANs is also developed in this dissertation, and the framework allows for the direct comparison of certain single-hop and multihop networks. A new class of protocols, called multiconfiguration multihop protocols (MMPs), is then derived from this framework, and an analytical model for this class of protocols is developed.

In LANs, one type of service which efficiently utilizes network resources is multicasting. *Multicasting* is the transmission of data from a single source node to multiple destination nodes in a manner that attempts to minimize the consumption of network resources. Networks without multicasting support must rely on multiple one-to-one communications for sending data from a single source to multiple destinations. These multiple transmissions may waste transmitter resources at the source node as well as channel resources in the network. With multicasting support, the source node may send into the network just a single transmission which is then broadcast to multiple destinations or forwarded from one destination to another, thus conserving transmitter and channel resources. In WDM networks based on the passive-star coupler, the star coupler provides a natural mechanism for supporting multicast traffic. This dissertation will also present a simple multicasting algorithm and it will illustrate the tradeoff between the number of transmissions required and the amount of time a node in the multicast group must wait for a message.

It is important for a WDM network to be scalable, i.e., it must be able to support additional nodes or wavelengths at minimal costs. A number of WDM WAN testbeds have been deployed which utilize wavelength-routed switches. These switches typically support only a fixed number of wavelengths. As the final contribution of this dissertation, a novel switch architecture is proposed which allows the addition of new wavelengths to the system without requiring significant costs to upgrade existing hardware. The performance of a network utilizing the new switch architecture is analyzed and compared to the traditional wavelength-routed switch.

1.4 Organization of the Dissertation

This dissertation will investigate WDM LAN architectures and protocols which are based on the passive-star coupler. In Chapter 2, we develop and analyze a model of the single-hop protocol employed in the Rainbow WDM network prototype. The throughput and

delay characteristics of the network are studied, and the optimal timeout value for the protocol is found for a given set of system parameters. In Chapter 3, we propose a class of packet-switched protocols which are based on the same architecture and optical hardware as the Rainbow system. The protocols exploit the characteristics of both single-hop and multihop systems in order to minimize the effects of tuning delays and multiple hops. In Chapter 4, we turn to the problem of multicasting in single-hop systems. We show that partitioning multicast groups may provide significant performance gains over unicasting, in which separate identical transmissions are sent to each destination, and single-transmission multicasting, in which only a single transmission is sent for all of the destinations. Methods for partitioning large multicast groups into smaller groups are presented and compared. In Chapter 5, we investigate a new switch architecture for WDM WANs in order to support network scalability. In Chapter 6, we conclude the dissertation and discuss areas of future work.

Chapter 2

Performance Analysis of the Rainbow WDM Optical Network Prototype

2.1 Introduction

In a single-hop WDM LAN based on the passive-star coupler, stations communicate directly with one another without requiring that their information pass through any intermediate stations. The transmitter of the sending station and the receiver of the receiving station must both be tuned to the same wavelength for the duration of the information transfer in order for communications to take place. This may require stations to have either tunable transmitters and fixed receivers (TT-FR), fixed transmitters and tunable receivers (FT-TR), or both tunable transmitters and tunable receivers (TT-TR). (In some systems, such as LAMBDANET [20], stations may be equipped with a fixed transmitter and multiple fixed receivers, each tuned to a different wavelength (FT-FR^N).) Also, an efficient protocol must be employed to ensure that the tunable elements are tuned to the correct channel at the correct time.

A number of architectures and protocols for single-hop WDM networks have been proposed, and some prototypes have been built as well. A summary of architectures and protocols can be found in [33], and a more general overview of multiwavelength networks can be found in [39]. The Rainbow prototype is a single-hop WDM network testbed which was developed at IBM [16] [24]. The system supports up to 32 IBM PS/2 stations connected in a star topology (Fig. 2.1) over a range of 25 km (hence, it is referred to as a metropolitan area network (MAN)), and data can be transmitted on each channel at a rate of up to 300 Mbps. A follow-up testbed, Rainbow-II, was able to support data rates of 1 Gbps over a range of 10-20 km.

In the Rainbow architecture, each station is equipped with a single fixed transmitter,

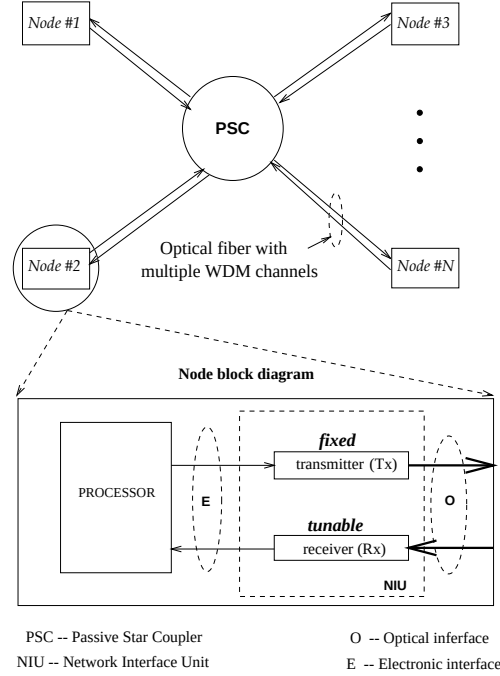


Figure 2.1: Passive-star topology for Rainbow.

which is tuned to its own unique wavelength, and a single tunable Fabry-Perot filter, which can be tuned to any wavelength (FT-TR) by adjusting the physical distance between two mirrors within the filter. Tuning to any particular channel may take up to 25 ms. This large filter tuning time results in a high connection setup time, making the system impractical for packet switching. Thus, the Rainbow system is intended primarily as a circuit-switched network with a long connection holding time.

This work provides an analytical model for the Rainbow medium-access protocol and presents a study of the system's behavior using the equilibrium point analysis (EPA) technique. To the best of our knowledge, an analysis of this protocol's performance has not been reported to date. The EPA technique [17] is a means of analyzing complex systems by assuming that the system is always at an equilibrium point. This technique has been successfully used to analyze a number of systems [31, 48, 26, 25], and has been found to provide accurate results.

The signaling protocol for Rainbow is described in Section 2.2. In Section 2.3, we present a model of the system. We then analyze the model using EPA in Section 2.4. Finally, in Section 2.5, we examine some representative numerical examples which show the effects that the timeout duration, offered load, and message size have on the throughput and delay of the system. Section 2.6 concludes the chapter.

2.2 Rainbow Protocol

The signaling protocol for Rainbow operates as follows. If a station is neither sending nor receiving a message, its tunable receiver scans across all of the channels, looking for connection requests from other stations. Each station is assigned its own unique channel on which its transmitter is fixed tuned; thus, an N -station network requires N wavelength channels. Upon the arrival of a message at Station A and destined for Station B , Station A first tunes its receiver to Channel λ_B (where λ_i is the unique channel assigned to station i) so that Station A will be able to pick up an acknowledgement signal from Station B . Station A then begins to send a continuous request signal on Channel λ_A . This request signal consists of a periodically repeated message which contains the identities of both the requesting station and the intended destination. If Station B 's receiver, which is continuously scanning across all channels, comes across the request on Channel λ_A , the receiver will stop on that channel, and Station B 's transmitter will send out an acknowledgement on Channel λ_B . Station A 's receiver, which is tuned to Channel λ_B , will receive the acknowledgement and will now know that Station B 's receiver is tuned to Channel λ_A . Station A 's transmitter will then begin transmitting the actual data message on Channel λ_A . This establishes a full-duplex connection.¹ Upon completion of the transmission, both stations resume scanning for requests. Note that all stations perform their operations asynchronously and independently.

With this protocol, there is the possibility of deadlock. If Station A is sending a connection setup request to Station B , Station B is sending a connection setup request to Station C , and Station C is sending a connection setup request to Station A , then all three stations will be waiting for an acknowledgement. Since each station is waiting for another station, acknowledgements will never be sent. To avoid this problem, the Rainbow protocol also includes a *timeout mechanism*. If an acknowledgement is not received within a certain timeout period measured from the message's arrival instant, the connection attempt is aborted, the message is blocked, and the station returns to scanning operation.

2.3 Model

Round-robin (or polling) systems have been modeled and analyzed extensively in the literature before (see, for example, [47]). In our current setting of the Rainbow protocol, although a station's receiver is performing a round-robin operation, that operation may

¹In the actual prototype system [24], the filters used do not allow round-robin type scanning. Instead the filter must scan back and forth across all wavelengths (the so-called elevator-type scanning). Also, the filters are not able to tune directly to a particular wavelength. They must perform "elevator scanning" to find the appropriate channel before stopping. In this work, we examine the original Rainbow system presented in [16]. We do not model the system in [24] because it is difficult to model elevator-type scanning, and also because we expect that newer filters, which do not have these limitations, may be employed in the future. In the remainder of the chapter, the term "scan" will be used to mean round-robin scanning, instead of elevator-type scanning.

be interrupted by the station's transmitter. "Vacation models" do not appear to be applicable here because the system's performance characteristics are now determined by the transmission ("vacation") process. The modeling challenge is to relate the transmitter and receiver operations at a station in a simple manner.

In our model, we make the following assumptions:

- There are N stations.
- There is no queueing. Each station has a single buffer to store a message, and any arrival to a non-empty buffer is blocked. A message departs from the buffer after it is completely transmitted.
- The sending station, upon a message arrival, tunes its receiver to the channel of the target station prior to sending the connection setup request.
- Stations monitor the channels in a round-robin fashion, in the sequence: 1, 2, ..., N , 1, 2, ...
- Time is slotted with slot length equal to $1 \mu\text{s}$. This value was chosen to provide a finer degree of granularity in the analysis, where other timing parameters, such as propagation delay, tuning time, and message length are expected to be in the range of tens of microseconds to hundreds of milliseconds.
- It takes a fixed amount of time, τ slots, to tune a receiver to any particular channel.
- Messages arrive at each station according to a Bernoulli process with intensity σ messages per slot.
- Message lengths are geometrically distributed with the average message length being $\frac{1}{\rho}$ slots. Message lengths are used to model connection holding times of circuits in the Rainbow prototype.
- The round-trip propagation delay between two station is R slots.
- Transmission times for request and acknowledgement messages are assumed to be negligible relative to the other timing components, such as tuning time, propagation delay, and message transmission time.
- The amount of time for which a station waits for an acknowledgement before timing out is specified as ϕ slots.

The state diagram for the model is given in Fig. 2.2. A station can be in any state during a given time slot. If the station is in the transmission state, TR , it remains in that state for a geometrically-distributed amount of time. A station departs from state TR with probability ρ at the end of a time slot and remains in state TR with probability $1 - \rho$. If

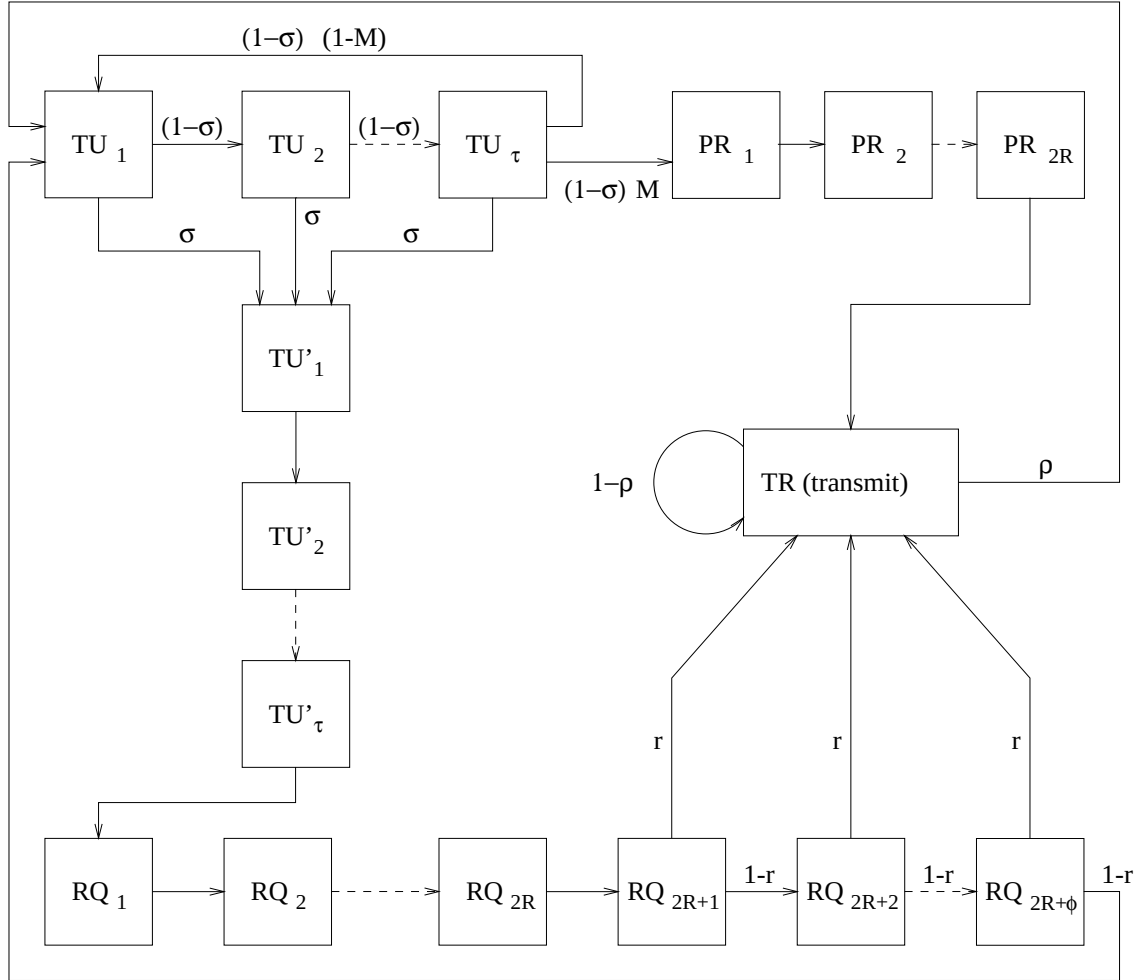


Figure 2.2: State diagram for the Rainbow model.

the station is in any other state, it remains in that state for a fixed amount of time (one slot). The states are defined as follows:

- $TU_1, TU_2, \dots, TU_\tau$: These are states during which a station's receiver is scanning across the channels for requests. It takes τ time slots to tune to a particular station. In each of these states, an arrival can occur with probability σ . From the state TU_τ , if there is no arrival, the station either finds a request on the channel to which it has just completed tuning with some probability M and proceeds to send an acknowledgement, or it doesn't find a request with probability $1 - M$ and proceeds to tune its receiver to the next channel.
- $TU'_1, TU'_2, \dots, TU'_\tau$: After an arrival occurs, the station immediately begins tuning its receiver to the channel of the destination. This takes τ time slots. After tuning, the station begins to transmit a request.
- $RQ_1, RQ_2, \dots, RQ_{2R+\phi}$: Upon sending a request, it takes R time slots of propagation delay for the request to reach the destination (see Fig. 2.3). At the earliest, an acknowledgement will be received after a propagation delay of $2R$ time slots. The station continues to send the request signal for a duration of ϕ time slots or until it receives an acknowledgement, whichever occurs first, where ϕ is the timeout duration. After sending a request for a duration of ϕ slots, the station must wait an additional $2R$ time slots of propagation delay for an acknowledgement. This ensures that all acknowledgements will result in a connection. If no acknowledgement is received, the current message is "timed out" and considered "lost," and the station returns to scanning operation. The probability of getting an acknowledgement is denoted by r , which is the same for each of the states RQ_{2R+1} to $RQ_{2R+\phi}$ since the system is memoryless, and an acknowledgement can be sent at any time by the acknowledging station. (The parameter r will be related to the probability M later in the chapter.) When an acknowledgement is received, the station immediately begins transmission of its message and goes into the transmission state TR .
- $PR_1, PR_2, \dots, PR_{2R}$: The station enters these states if it finds a request while scanning. Upon identifying the request, the station sends an acknowledgement to the requesting station. The acknowledgement takes R time slots of propagation delay to reach the station requesting the connection, after which the requesting station begins its transmission. It takes another R slots of propagation delay for the message to arrive at the destination station, after which the station goes into the transmission state, TR , to receive the message. A connection will always be established if an acknowledgement has been sent.
- TR (transmission): In this state, stations are either transmitting or receiving a message. Stations may stay in this state for a duration of more than one time unit

and depart with probability ρ at the end of a slot. Upon completion of message transmission or reception, the station returns to the scanning operation.

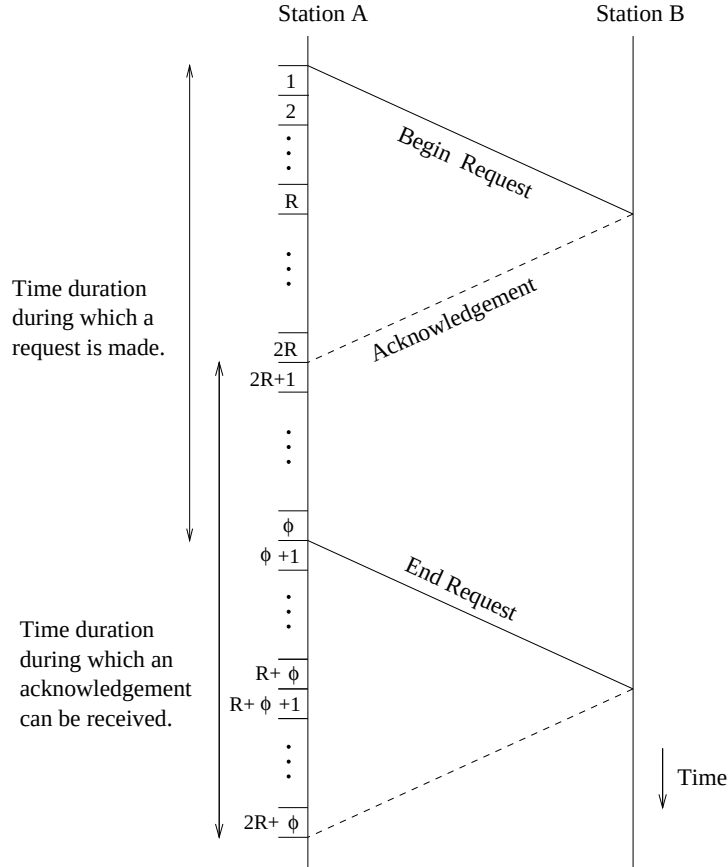


Figure 2.3: Timing for connection setup.

We define N_{TU_i} to be the expected number of stations in state TU_i , $N_{TU'_i}$ the expected number of stations in state TU'_i , N_{RQ_i} the expected number of stations in state RQ_i , N_{PR_i} the expected number of stations in state PR_i , and N_{TR} the expected number of stations in the transmission state.

The system can be modeled as a Markov chain with state space vector:

$$\mathbf{N} = \left\{ N_{TU_1}, \dots, N_{TU_\tau}, N_{TU'_1}, \dots, N_{TU'_\tau}, N_{RQ_1}, \dots, N_{RQ_{2R+\phi}}, N_{PR_1}, \dots, N_{PR_{2R}}, N_{TR} \right\}$$

It is difficult to analyze this system using Markov analysis techniques because of the very large state space. Therefore, we analyze the system at an equilibrium point using equilibrium point analysis (EPA).

2.4 Analysis

In EPA, the system is assumed to always be operating at an equilibrium point. This is an approximation since the system actually moves around the equilibrium point. At an equilibrium point of the system, the expected increase in the number of stations in any state is zero. Thus, the expected number of stations entering each state must equal the expected number of stations departing from each state in each time slot.

By writing the flow equation for each state, we obtain a set of K equations with K unknowns, where K is the number of states. Also, the flow equations can be written such that the expected number of stations in each state is ultimately expressed in terms of the expected number of stations in state TU_1 . Thus, if we are able to solve for N_{TU_1} , we will have the solution for the entire system. The flow equations are as follows (consult Fig. 2.2):

$$N_{TU_i} = (1 - \sigma)^{i-1} N_{TU_1} \text{ for } i = 2, 3, \dots, \tau \quad (2.1)$$

$$N_{PR_1} = N_{PR_2} = \dots = N_{PR_{2R}} = (1 - \sigma)^\tau M \cdot N_{TU_1} \quad (2.2)$$

$$N_{TU'_1} = N_{TU'_2} = \dots = N_{TU'_\tau} = N_{RQ_1} = N_{RQ_2} = \dots = N_{RQ_{2R}} = (1 - (1 - \sigma)^\tau) N_{TU_1} \quad (2.3)$$

$$N_{RQ_{2R+j}} = (1 - r)^{j-1} (1 - (1 - \sigma)^\tau) N_{TU_1} \text{ for } j = 1, 2, \dots, \phi \quad (2.4)$$

$$\begin{aligned} \rho \cdot N_{TR} &= N_{PR_{2R}} + \sum_{j=1}^{\phi} r \cdot N_{RQ_{2R+j}} \\ &= \left[(1 - \sigma)^\tau M + \left(1 - (1 - r)^\phi \right) \cdot (1 - (1 - \sigma)^\tau) \right] \cdot N_{TU_1} \end{aligned} \quad (2.5)$$

We now need to solve for the unknown variables N_{TU_1} , r , and M . Recall that M is the probability that a request will be found by a scanning station. This is equal to the probability that another station is in states RQ_{R+1} to $RQ_{R+\phi}$, and that the request is intended for the scanning station. That is,

$$M = \frac{1}{N - 1} \cdot \frac{1}{N} \left(\sum_{i=R+1}^{R+\phi} N_{RQ_i} \right) \quad (2.6)$$

Substituting (2.3) and (2.4) into (2.6) and simplifying yields:

$$M = \frac{1}{N - 1} \cdot \frac{1}{N} (1 - (1 - \sigma)^\tau) \cdot \left[R + \frac{1}{r} \left(1 - (1 - r)^{\phi-R} \right) \right] N_{TU_1} \quad (2.7)$$

The rate of flow into the active state from the request states must equal the rate of flow into the active state from the state PR_{2R} . This is because a station can only begin transmission if there is another station that will be receiving the transmission. This leads

to the equation:

$$N_{PR_{2R}} = \sum_{i=1}^{\phi} r \cdot N_{RQ_{2R+i}} \quad (2.8)$$

which, upon substitutions from (2.2) and (2.4), yields:

$$(1 - \sigma)^\tau M = (1 - (1 - \sigma)^\tau) \left(1 - (1 - r)^\phi\right) \quad (2.9)$$

In steady state, the sum of the stations in each state is equal to the total number of stations in the system, i.e.,

$$N = \sum_{i=1}^{\tau} N_{TU_i} + \sum_{i=1}^{\tau} N_{TU'_i} + \sum_{i=1}^{2R+\phi} N_{RQ_i} + \sum_{i=1}^{2R} N_{PR_i} + N_{TR} \quad (2.10)$$

or

$$\begin{aligned} N &= (1 - (1 - \sigma)^\tau) \cdot \left[\frac{1}{\sigma} + \tau + 2R + \left(\frac{1}{r} + \frac{1}{\rho} \right) \left(1 - (1 - r)^\phi\right) \right] N_{TU_1} \\ &+ \left[\left(2R + \frac{1}{\rho}\right) (1 - \sigma)^\tau M \right] N_{TU_1} \end{aligned} \quad (2.11)$$

Equations (2.7), (2.9), and (2.11) can be solved simultaneously for the variables r , M , and TU_1 , which can then be used to provide the steady-state solution to the entire system.

The primary measures of interest are throughput, delay, and timeout probability. The *normalized throughput* is defined as the expected fraction of stations in the active state (which is also the fraction of a channel's bandwidth that is utilized):

$$S = \frac{N_{TR}}{N} \quad (2.12)$$

Delay is defined to be the time from a message's arrival to the system until the time that the message completes its transmission. This consists of the time required to tune to the destination station's channel, the propagation delay for the request and acknowledgement signals, the time until an acknowledgement is received, and the message transmission time. Delay is measured in slots. Hence, we have,

$$D = \tau + 2R + \sum_{k=1}^{\phi} k \cdot (1 - r)^{k-1} \cdot r + \frac{1}{\rho} \quad (2.13)$$

The *timeout probability* is defined as the probability that a station will timeout after entering the request mode, i.e.,

$$p_{TO} = (1 - r)^\phi \quad (2.14)$$

Also of interest is the *blocking probability*, which is the probability of an arrival being blocked. This is equal to the probability that a station is not in scanning mode. The

equation for blocking probability is:

$$p_{BL} = 1 - \frac{\sum_{i=1}^T N_{TU_i}}{N} \quad (2.15)$$

2.5 Results

In our analysis, the following default parameters were used:

- $N = 32$ stations.
- slot length = $1 \mu s$.
- $R = 50$ slots (corresponding to a 10 km distance between each station and the star coupler).
- $\tau = 1000$ slots (corresponding to a 1 ms receiver tuning time).
- $\rho = 10^{-5}$ (corresponding to a mean message length of 100 ms).
- $\sigma = 10^{-4}$ (corresponding to a message arrival rate of 100 msg/second at each station).
- $\phi = 10^4$ slots (corresponding to a timeout of 10 ms).

In our numerical examples presented below, we shall study the effect of some of these parameters on the system performance by varying them around the default values.

In Fig. 2.4, we show normalized throughput versus message arrival rate for different values of timeout duration. As the arrival rate increases, the throughput will first increase as more messages become available for transmission, and then it will eventually begin to decrease as the number of stations in request mode begin to outnumber the stations that are available to acknowledge requests. Note that, for a given arrival rate, there is an optimum timeout duration at which the system achieves its maximum throughput. Also, with timeout duration equaling 10 ms or higher, and other parameters as chosen, the peak system throughput is approximately 0.45 which means that, if each channel is operating at 300 Mbps (as in the Rainbow prototype), the effective information rate on each channel equals 135 Mbps. To verify the accuracy of the analytical model, results from a simulation of the Rainbow protocol are also included in Fig. 2.4. We find that, for higher values of message arrival rate, there is excellent agreement between the analysis and simulation, but this is not so for low message arrival rates. An explanation on the cause of this discrepancy is provided later.

Normalized throughput versus mean message size ($\frac{1}{\rho}$) is plotted in Fig. 2.5. As expected, we see that the throughput approaches unity as the message size is increased to a thousand seconds or higher.

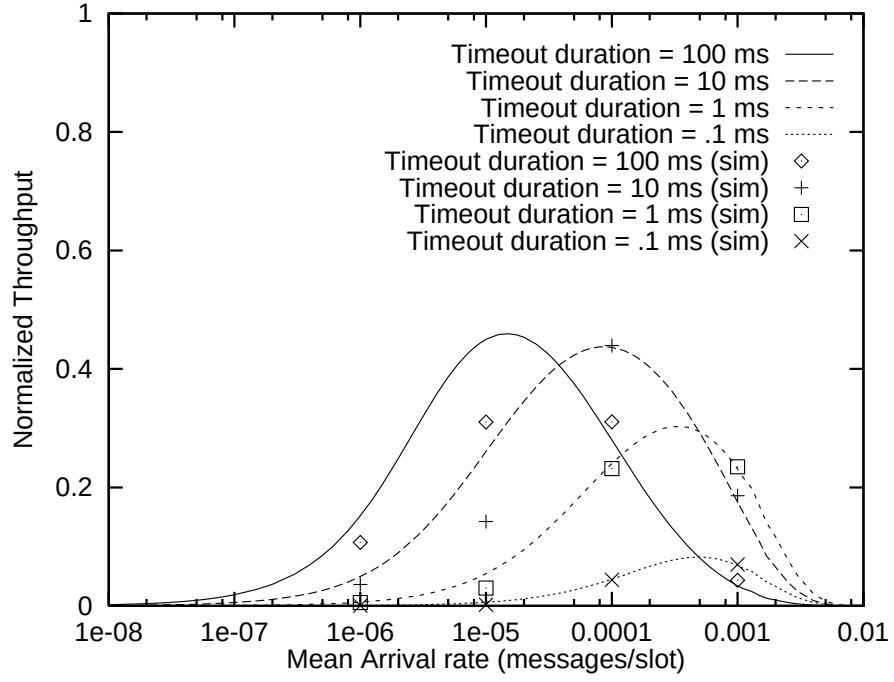


Figure 2.4: Throughput vs. arrival rate.
 $\text{slot} = 1 \mu\text{s}$, $N = 32$, $R = 50 \mu\text{s}$, $\tau = 1 \text{ ms}$, $\frac{1}{\rho} = 100 \text{ ms}$.

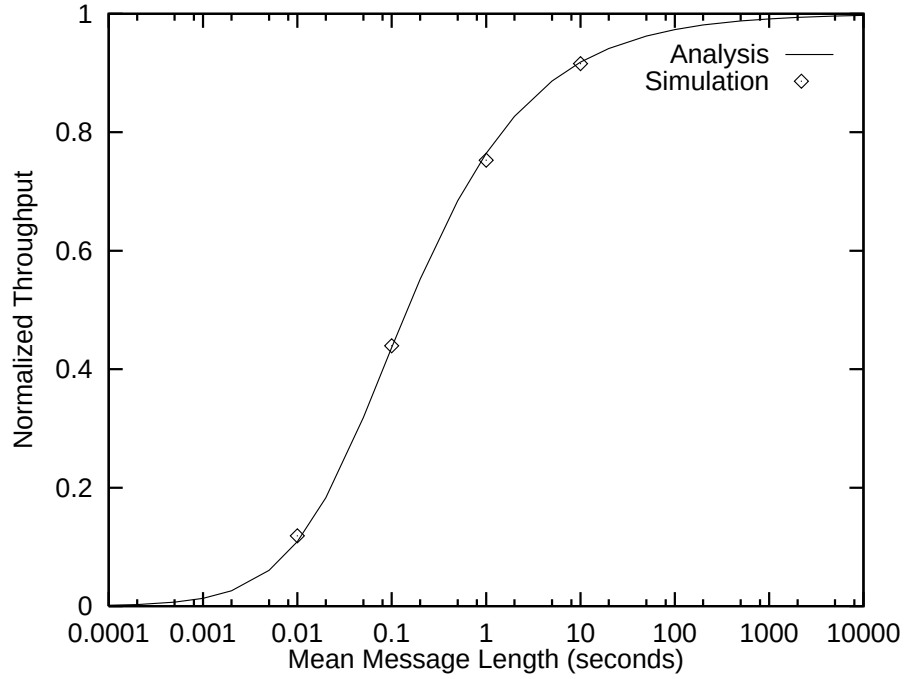


Figure 2.5: Throughput vs. message size.
 $\text{slot} = 1 \mu\text{s}$, $N = 32$, $R = 50 \mu\text{s}$, $\tau = 1 \text{ ms}$, $\sigma = 10^{-4} \text{ msg/slot}$, $\phi = 10 \text{ ms}$.

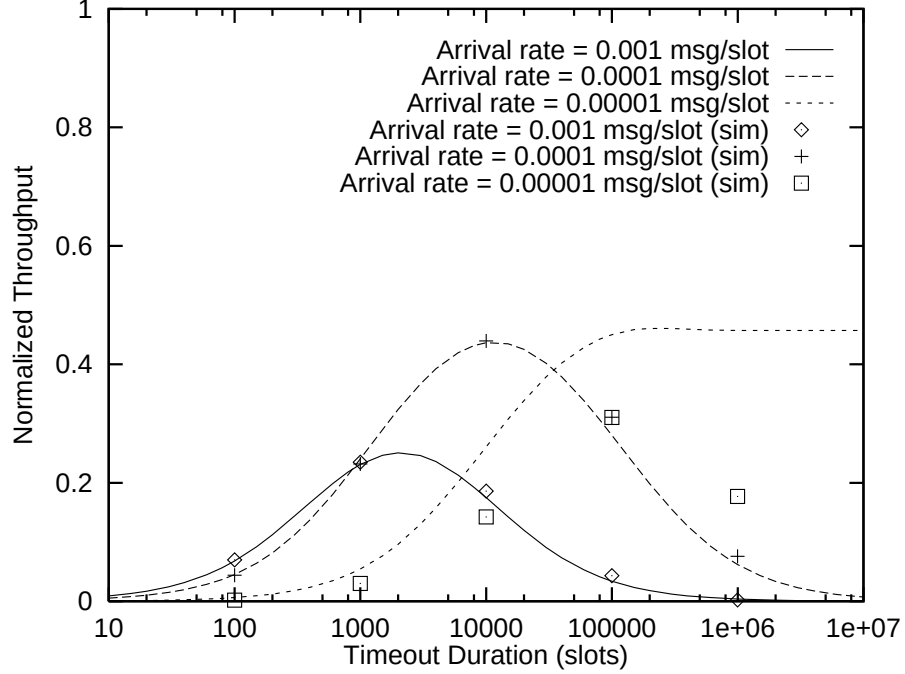


Figure 2.6: Throughput vs. timeout duration.
slot = 1 μ s, $N = 32$, $R = 50$ μ s, $\tau = 1$ ms, $\frac{1}{\rho} = 100$ ms.

In Fig. 2.6, we plot the normalized throughput versus timeout duration for various arrival rates. We observe that, for the higher arrival rates ($\sigma = 10^{-4}, \sigma = 10^{-3}$), as the timeout duration is increased, the throughput first increases and then decreases. For low timeout durations, requests are timing out too quickly, before they can be acknowledged. As the timeout duration increases, more requests will be acknowledged, resulting in higher throughput. As the timeout duration increases even further, stations are spending longer amounts of time in the request mode, resulting in fewer stations being available to acknowledge requests.

This phenomenon is confirmed in Fig. 2.7, where we examine the timeout probability as a function of the timeout duration. For low timeout durations, all of the requests are timing out, but as the timeout duration is increased, fewer requests time out. Eventually, the timeout probability levels off at some value.

We find that, for the lower arrival rates, the analytical and simulation results do not match very well. The analysis indicates that the throughput reaches some constant value as the timeout duration goes to infinity, while the simulation indicates that the throughput continues to decrease. This discrepancy is caused by the way deadlocks are modeled in the analysis.

In the analytical model, the deadlock probability is included in the timeout probability. A request times out when the target of the request is either in request mode or engaged in a connection (transmitting or receiving a message). For low arrival rates, most of the stations

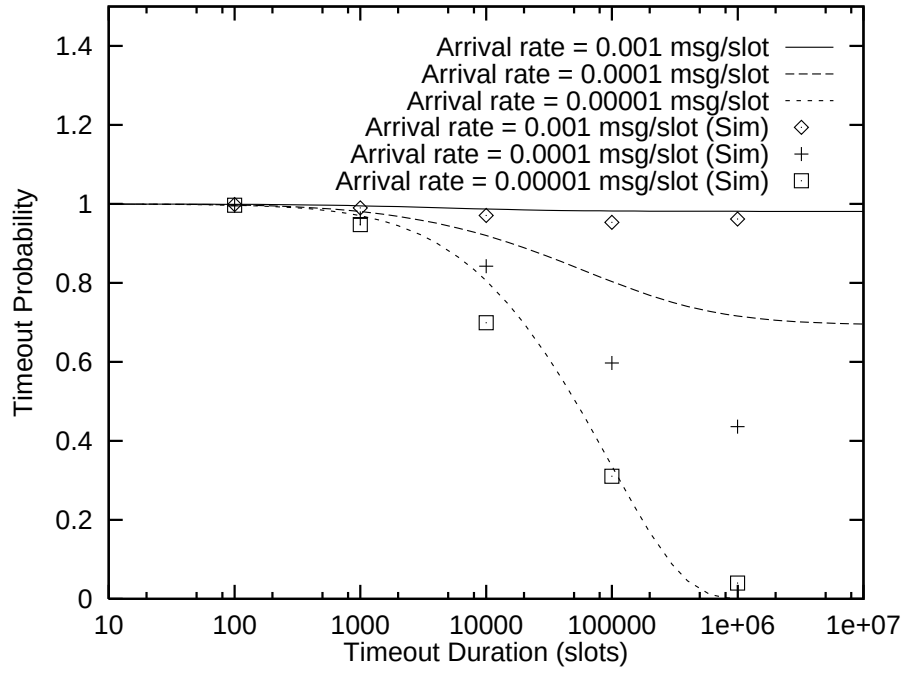


Figure 2.7: Timeout probability vs. timeout duration.
slot = 1 μ s, $N = 32$, $R = 50 \mu$ s, $\tau = 1$ ms, $\frac{1}{\rho} = 100$ ms.

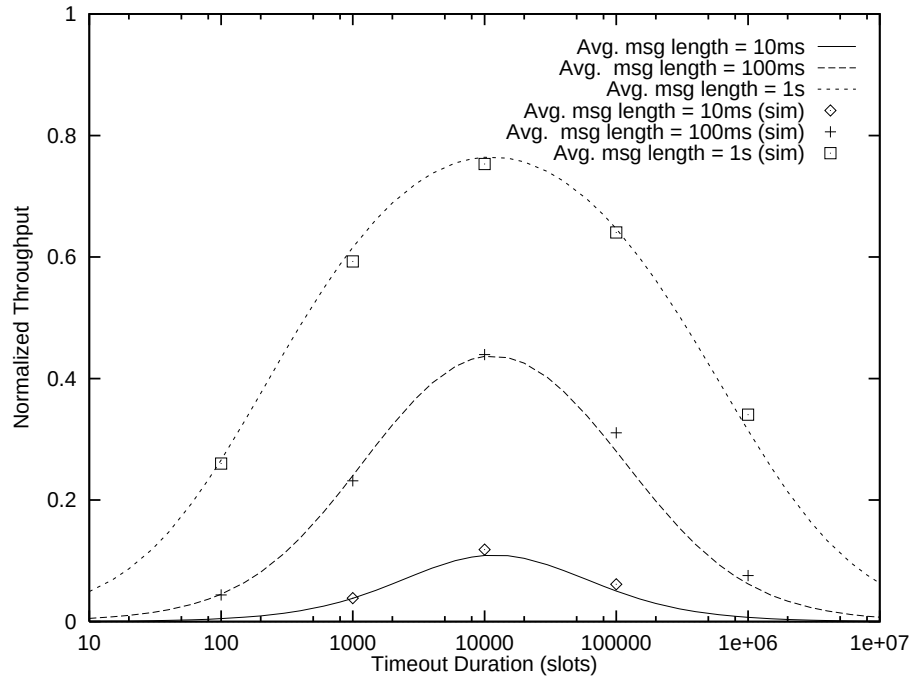


Figure 2.8: Throughput vs. timeout duration for different message lengths.
slot = 1 μ s, $N = 32$, $R = 50 \mu$ s, $\tau = 1$ ms, $\sigma = 10^{-4}$ msg/slot.

will be scanning for requests, and only a few stations will be requesting connections. The stations that are requesting connections will have a high probability of being acknowledged, especially as the timeout duration increases. However, in the simulation, a deadlock can occur if any station being requested is also requesting a separate connection of its own. This results in a higher probability of deadlock than the analytical model for low arrival rates.

A significant result is that, for a given arrival rate, there is a timeout duration which optimizes the throughput. It is further shown in Fig. 2.8 that this optimal timeout duration is not significantly affected by the message lengths. *This result leads to the conclusion that the protocol can be improved by having a dynamic timeout duration, i.e., by varying the timeout duration based on the arrival rates at each station, it may be possible to maximize the throughput for specific arrival rates.*

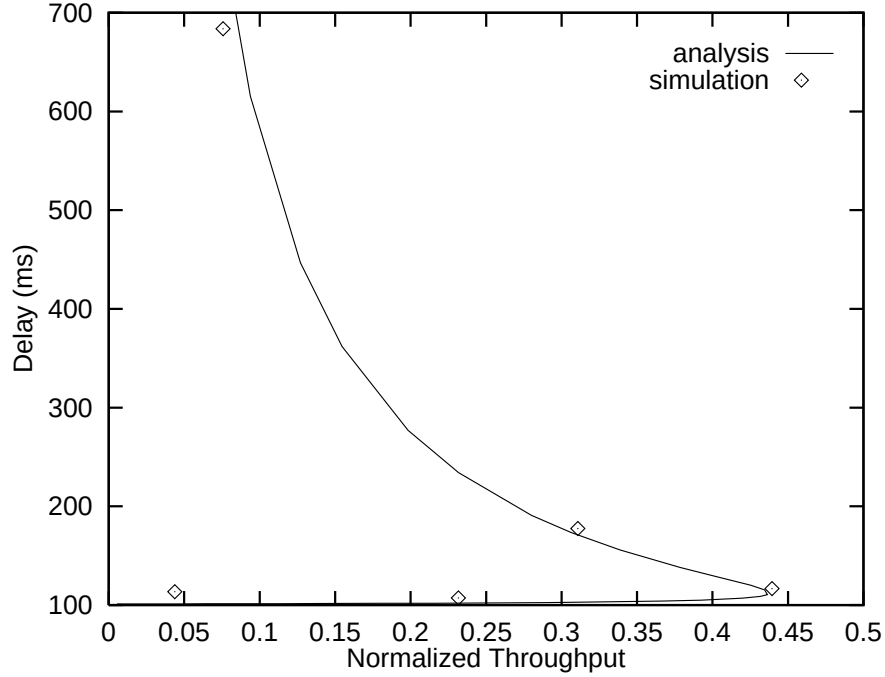


Figure 2.9: Delay vs. throughput with parameter ϕ .
slot = 1 μ s, $N = 32$, $R = 50$ μ s, $\tau = 1$ ms, $\sigma = 10^{-4}$ msg/slot, $\frac{1}{\rho} = 100$ ms.

In Fig. 2.9, we examine the average delay versus normalized throughput behavior of the system using the timeout duration ϕ as a parameter. As the timeout duration increases beyond a certain value, the throughput decreases while the delay continues to increase. For lower timeout durations, requests will either be serviced quickly or timeout quickly. The station will spend only a small amount of time in the request mode. Examining Eqn. (2.13), the bulk of the delay is equal to $\tau + 2R + \frac{1}{\rho}$. In this example, the tuning time and propagation delays are small compared to the message length, so the delay is approximately $\frac{1}{\rho}$ which is equal to 100 ms. As the timeout duration increases beyond the point of optimal

throughput, stations will spend longer periods in request mode before the requests are acknowledged. Thus, the summation term in (2.13) contributes to a larger portion of the delay.

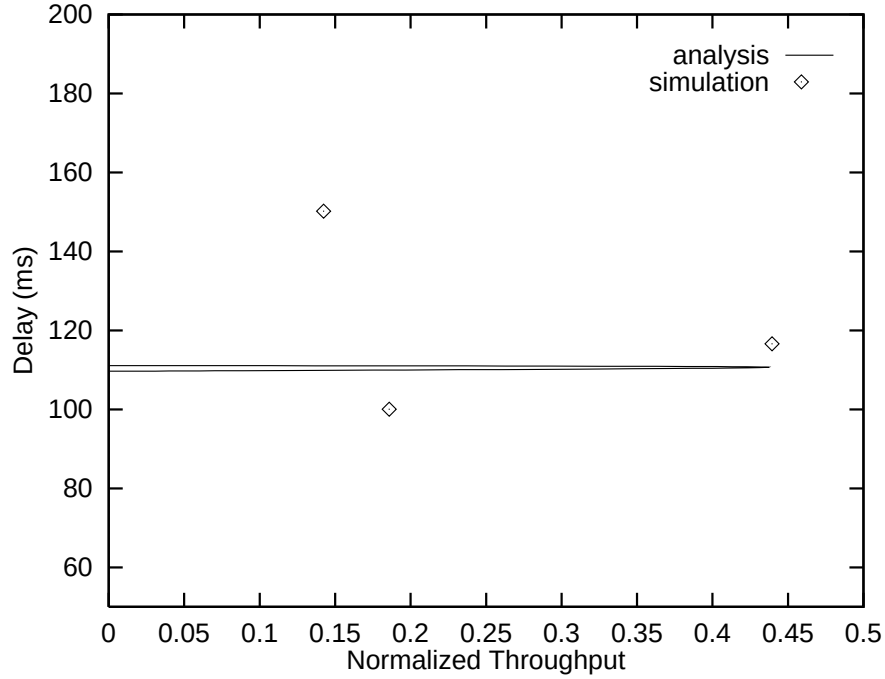


Figure 2.10: Delay vs. throughput with parameter σ .
 $\text{slot} = 1 \mu\text{s}$, $N = 32$, $R = 50 \mu\text{s}$, $\tau = 1 \text{ ms}$, $\frac{1}{\rho} = 100 \text{ ms}$, $\phi = 10 \text{ ms}$.

In Fig. 2.10, we study the average delay versus normalized throughput characteristics with increasing arrival rate, σ . Analytical results indicate that the arrival rate does not have a significant effect on the average delay of the system. This is because the only delay element that is changing is the amount of time spent in request mode. For a given timeout value ϕ , this duration will vary from 1 to ϕ . For higher ratios of ϕ to $\frac{1}{\rho}$, however, we expect that variations in arrival rate will have a greater effect on delays.

Figure 2.11 shows average delay versus normalized throughput for increasing message size. As expected, the delay increases for larger message sizes with the $\frac{1}{\rho}$ term dominating (2.13).

In Fig. 2.12, normalized throughput is plotted versus timeout duration for 16, 32, and 64 stations. This graph shows that the optimal timeout duration is not affected by the number of stations. This is because the optimal timeout duration depends on the ratio of stations that are scanning to the stations that are in request mode, rather than the total number of stations in the system. Also, we find that the per-channel (or per-station) throughput decreases for a larger number of stations. The receivers must tune across more channels to find a request. This indicates that the protocol is not very scalable. Increasing the number of stations reduces the per-channel throughput and requires more wavelengths.

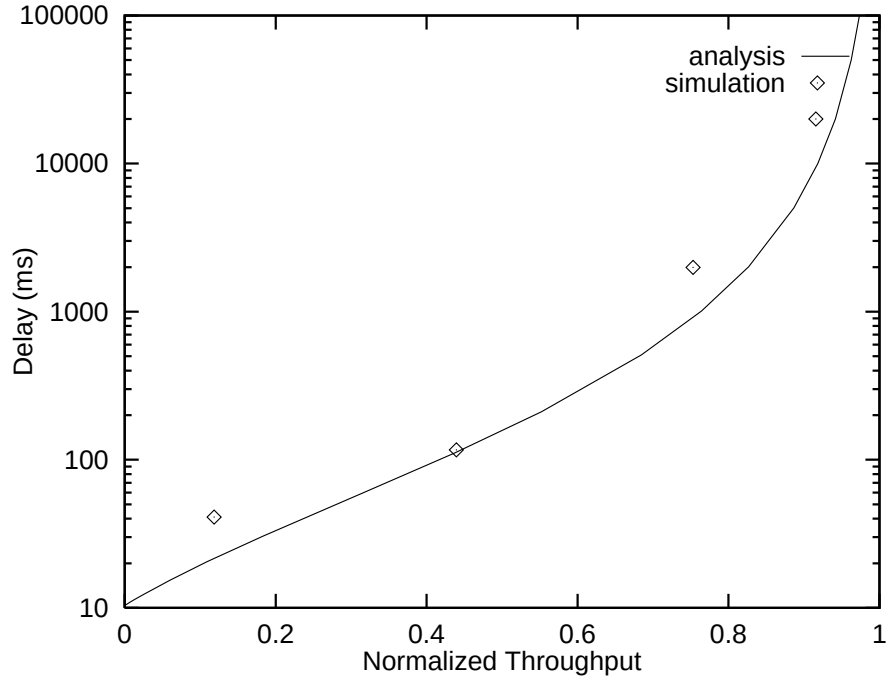


Figure 2.11: Delay vs. throughput for varying message size.
 slot = $1 \mu\text{s}$, $N = 32$, $R = 50 \mu\text{s}$, $\tau = 1 \text{ ms}$, $\sigma = 10^{-4} \text{ msg/slot}$, $\phi = 10 \text{ ms}$.

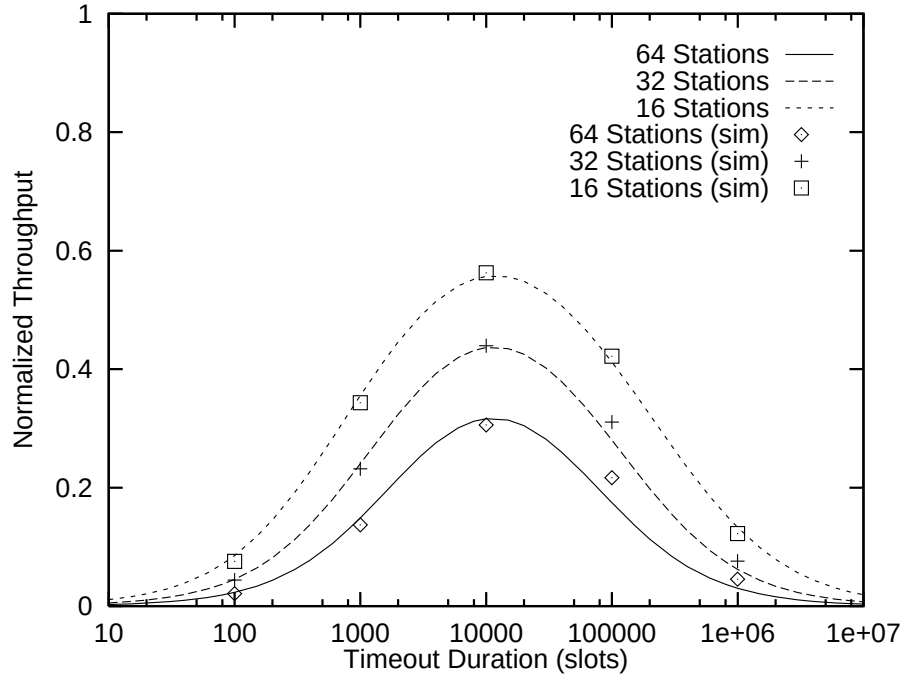


Figure 2.12: Throughput vs. timeout duration for different number of stations.
 slot = $1 \mu\text{s}$, $R = 50 \mu\text{s}$, $\tau = 1 \text{ ms}$, $\sigma = 10^{-4} \text{ msg/slot}$, $\frac{1}{\rho} = 100 \text{ ms}$.

2.6 Conclusion

The Rainbow system is difficult to analyze directly, because each station can be in any of a large number of states. The size of the state space therefore grows exponentially with the number of stations in the system. In this research, we provided a framework for analyzing the Rainbow protocol using the equilibrium point analysis (EPA) technique. By assuming that the system remains at its equilibrium point, we are able to reduce the complexity of the problem and obtain analytical results for an otherwise intractable system. In our analysis, we investigated the effect of system parameters, such as message arrival rate and timeout duration, on the performance of the system.

We determined that the EPA technique provides good results when the arrival rates are high, but it is not very accurate for lower arrival rates. This is because the model doesn't properly capture the deadlock phenomenon for low arrival rates.

We found that, for a given set of system parameters, there exists an optimal timeout duration which will maximize the throughput of the system. In general, this optimal timeout duration seems to be independent of the message length and the number of stations, but depends on the arrival rate of messages.

Chapter 3

A Class of Multihop Protocols for Packet-Switched WDM Networks

3.1 Introduction

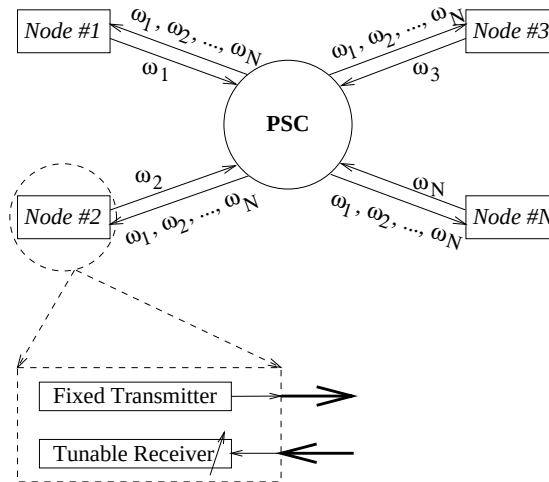


Figure 3.1: A passive-star-based local optical WDM network.

In the previous chapter we noted that, because of the high connection setup overhead, the Rainbow network is suited primarily for circuit-switched traffic; however, many of today's networks require packet-switching capabilities in order to optimize bandwidth utilization.

In this chapter, we present a class of protocols for handling packet-switched traffic over a network based on the FT-TR Rainbow architecture and optical hardware. We consider the Rainbow architecture for the following reasons. First, optical transmitters and receivers are expensive. By having only a single transmitter and receiver per node, network costs may be kept to a minimum. Second, the Rainbow network is a working prototype. By developing a protocol for the specified hardware, we ensure that the resulting network is

realizable with current optical device technology.

As discussed earlier, protocols for PSC-based WDM LANs (see Fig. 3.1) are classified as either single-hop or multihop. The basic goal of the protocol is to provide a certain degree of connectivity among the nodes in order to enable communications. A single-hop protocol typically achieves connectivity through the use of tunable components. In a sense, a single-hop network achieves full connectivity (a fully-connected *logical topology*) among nodes, since each node can communicate *directly* with every other node by appropriately tuning its transmitters or receivers; however, such connectivity is only meaningfully established for the packet transmission duration. A multihop network achieves connectivity through the use of additional transmitters and receivers at each node and typically requires no tuning, instead relying on a *static* logical topology. Since a multihop network doesn't rely on tuning, it doesn't incur the potentially high tuning overhead of a single-hop network. On the other hand, a multihop network's reliance on multiple transmitters and receivers, combined with the expense of such components, often results in low connectivity in the network, and, consequently, higher average hop distances (the number of nodes a message must traverse before reaching its destination).

In this chapter, we consider a class of protocols called *multiconfiguration multihop protocols (MMPs)*. These protocols are multihop protocols which not only rely on the number of transmitters and receivers to achieve a desired connectivity, but also *make use of tuning to increase connectivity*. These protocols use tunable components to cycle through a fixed number of configurations¹, and the logical topology of the network is then defined as the superposition of a number of configurations over the entire cycle. Using tuning in this manner, we can increase the logical connectivity of the network, thus reducing the average hop distance at the expense of additional tuning requirements.

In the following section, we describe a framework for classifying and comparing various protocols for the passive-star network architecture, and we present some previously proposed single-hop and multihop protocols. Section 3.3 describes the proposed class of MMPs and some of the design issues which must be considered when implementing such protocols. We apply the class of protocols to a specific network architecture in Section 3.4. Section 3.5 describes the analytical model. Section 3.6 presents some numerical examples, and Section 3.7 concludes this chapter and discusses areas for future study.

3.2 Framework and Previous Protocols

The most basic goal of any network is to provide connectivity. By providing connectivity, a network allows any of its nodes to communicate with any other node. One measure of

¹We define a configuration as the topology of the network at a given point in time as determined by the state of the tunable components in the network.

the level of connectivity in a network is the nodal degree². The nodal degree specifies the maximum number of direct links that one node may have to other nodes, and is inversely related to the average hop distance in a network: i.e., a higher nodal degree (high connectivity) results in fewer hops between a source and a destination, while a lower nodal degree (low connectivity) results in more hops between a source and a destination. In general, a logical nodal degree of at least $N - 1$ is required for a single-hop network, while a nodal degree of at least 1 is required for all networks. A network with a nodal degree which is less than $N - 1$ is considered to be a multihop network. The exact relation between nodal degree and the required number of hops depends on the actual logical topology of the network.

In a WDM PSC-based LAN, there are three approaches to increasing the nodal degree, and thus, connectivity, in the network. These approaches are outlined below:

1. *Additional transmitters and receivers*: The first approach to increasing connectivity is to increase the number of transmitters and receivers at each node. Each unit increase in the number of transceiver pairs per node increases the nodal degree by one. The practicality of this approach is limited by the high cost of optical transmitters and receivers.
2. *Channel sharing*: The second method for providing connectivity in a WDM LAN is to reduce the number of channels in the network, and allow channel sharing [50, 49, 28]. By limiting the number of channels in this manner, multiple transmitters and receivers (transceivers) from different nodes can be made to occupy the same channel, thus creating more direct connections between nodes. As an example, consider a network with N nodes and one transceiver pair per node. Assume that transmitters and receivers are uniformly distributed across all channels. If N channels are used, then one transmitter and one receiver may be tuned to each channel, thus the overall nodal degree is one. If two channels are used, then $N/2$ transmitters and $N/2$ receivers are tuned to each channel, resulting in a nodal degree of $N/2$. If only one channel is used, then we have a single-channel broadcast environment in which every node is directly connected to every other node (a fully-connected logical topology). In general, if w channels are used, then an average nodal degree of N/w can be achieved. The disadvantage of using shared channels is that the bandwidth (e.g., time slice) on each channel must be divided among all transmitters or receivers tuned to that channel. Thus, the transmitters and receivers will not be fully utilized at all times.
3. *Multiconfiguration multihop approach*: The final approach for increasing connectivity, that will be examined in this chapter, is to use tunable components to increase the

²In this work, the term *nodal degree* will refer to the *logical* nodal degree, or the number of connections each node has to other nodes in the logical topology, rather than the *physical* nodal degree which specifies the number of transmitters and receivers at each node. We will assume that the logical degree is the same for each node in the network, and that the physical degree is the same for each node in the network.

connectivity in the time domain. At any given point in time, the nodal degree is determined by the number of transceivers at each node; however, if the transceivers are tunable, the nodal degree may be increased by considering connections made over a period of time. For example, consider a network with t transceivers per node. Suppose that time is divided into cycles which are repeated indefinitely, and each cycle is divided into two periods. During the first period, the tunable transmitters and tunable receivers are configured according to some configuration C_1 . During the second period, the tunable transmitters and tunable receivers are reconfigured according to a different configuration C_2 . Within each period, the nodal degree is equal to t , since each node can have a direct connection to only t other nodes. However, over an entire cycle, the nodal degree can equal $2t$, if each node connects to t nodes in the first period and t other nodes in the second period. In general, the tunability in a network may be used to achieve an arbitrary degree of connectivity by adjusting the number of periods or configurations per cycle. The cost of using tunable components to provide connectivity, however, is that tuning a component from one channel to another may incur some significant amount of delay. Also, tunable components generally cost more than non-tunable components depending on the technology used to build the components.

The concept of using a number of different network configurations over time has been previously studied for mesh networks [52] and for networks with centralized switches [30], but has not been applied to WDM-based LANs in which nodes are equipped with tunable components.

In summary, a network may be characterized by its degree of connectivity, specified by the nodal degree and the methods by which it achieves this connectivity. We employ the following parameters to characterize such a network:

- N : Number of nodes in the network.
- d : Nodal degree.
- t : Number of transceiver pairs per node. (The number of transmitters is equal to the number of receivers and is the same for all nodes.)
- w : Number of wavelengths used in the network.
- c : Number of configurations or tunings required to achieve a desired logical connectivity.

From the above description, we observe that the number of transceivers per node, the level of tunability, and the number of channels in the network can all be related by the amount of connectivity they provide. This connectivity, expressed in terms of the maximum

possible nodal degree, can be represented as:

$$d = t \cdot c \cdot \frac{N}{w} \quad (3.1)$$

A graph depicting this relationship is shown in Fig. 3.2. The graph also illustrates a relation between single-hop and multihop networks. In a single-hop network, a nodal degree of at least $N - 1$ is required, which implies that:

$$t \cdot c \cdot \frac{N}{w} \geq N - 1$$

or,

$$\frac{t \cdot c \cdot N}{w \cdot (N - 1)} \geq 1. \quad (3.2)$$

Similarly, we have

$$\frac{t \cdot c \cdot N}{w \cdot (N - 1)} < 1 \quad (3.3)$$

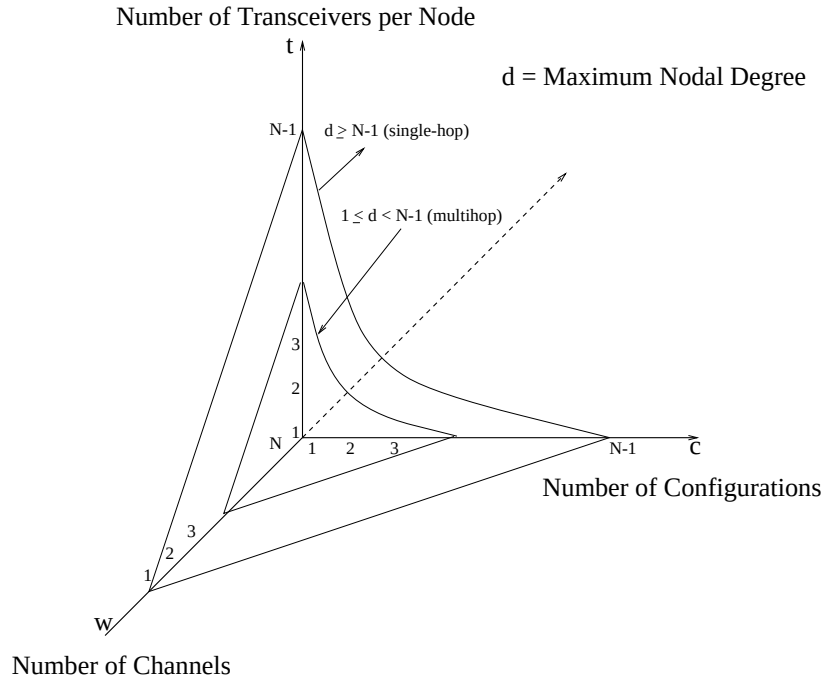
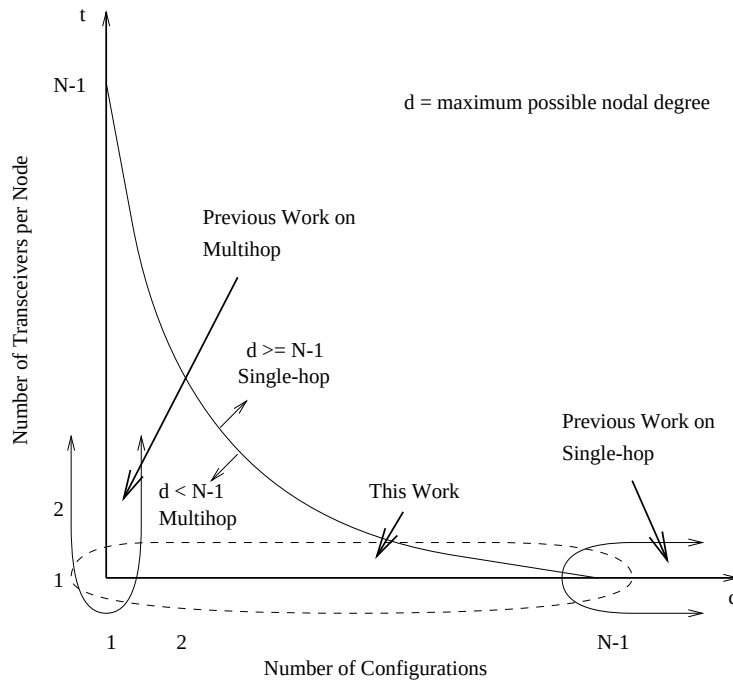
for a multihop network. In Fig. 3.2, the outer curve indicates the surface for which $d = N - 1$, and to the right of this curve is the region for which $d > N - 1$. In these areas, we have networks which can be classified as single-hop networks. When the maximum nodal degree is less than $N - 1$ (the area of the graph closer to the origin), then we have a multihop network. The origin of the graph indicates a nodal degree of one, in which case we have a static ring network.

Figure 3.3 illustrates the relation of connectivity to the number of transceivers and the tunability requirements when the number of channels is equal to N . Previous works on single-hop protocols have typically relied on a single tunable transceiver pair at each node (not including transceivers for a control channel) and have employed tuning to achieve connectivity. Thus, previous single-hop protocols can be classified in the region $c \geq N - 1$ of Fig. 3.3. On the other hand, previous works on multihop protocols have typically relied on multiple transceivers for connectivity, and can therefore be classified along the t -axis of Fig. 3.3. In this work, we wish to focus on the parameter c , which ranges from 1 (for a traditional multihop network) to $N - 1$ (for a traditional single-hop network).

We now briefly describe previous work on single-hop and multihop packet-switched protocols. Reviews of single-hop and multihop protocols may be found in [33] and [34].

3.2.1 Single-Hop Protocols

In order to accommodate packet switching in a single-hop network, a protocol must overcome the potentially high tuning delays of transceivers and obtain the required level of coordination between nodes. We note that tunable receivers may be used for two purposes: 1) to provide increased connectivity in a network, and 2) to provide an adaptive,


 Figure 3.2: Achievable connectivity vs. t , c , and w .

 Figure 3.3: Achievable connectivity vs. t and c , with $w = N$.

dynamic network in which the logical connectivity changes appropriately in order to accommodate varying traffic patterns. In a dynamic single-hop network, the network is often reconfigured on a packet-by-packet basis. In general, setting up connections on a packet-by-packet basis may be very inefficient. Also, if each node only has a single receiver, it may only listen to a single channel at any time. Thus, a node cannot obtain information regarding each of the other nodes' traffic demands without tuning to each channel.

Packet switching may be provided in a static manner in a single-hop network by using time-division multiplexing (TDM) [15, 8]. In the static WDM/TDM approach, time on each channel is divided into a number of cycles, with each cycle subdivided into slots. Nodes are equipped with a fixed transmitter and a tunable receiver (FT-TR); however, such protocols may be easily modified to accommodate a tunable-transmitter, fixed-receiver system (TT-FR). During a slot, each node's receiver is assigned to a particular channel, allowing any transmitter on that channel to transmit packets to a given receiver. Each TDM cycle contains a sufficient number of slots which are assigned such that each node is able to communicate with every other node. In this situation, the tuning property of a transceiver is used only to provide connectivity and not used to provide adaptivity in the network.

A dynamic single-hop approach requires nodes to exchange information regarding packet transmission requests. In some cases, this control information is exchanged by equipping each node with an additional transmitter and receiver, and having these components fixed-tuned to operate on a control channel which is separate from the data channels, e.g., [11, 21, 27]. Alternatively, the control information may be exchanged over the data channels using in-band signaling [46]. The dynamic single-hop approach requires more tuning and overhead, but is able to accommodate varying traffic patterns. Both static single-hop protocols and dynamic single-hop protocols may perform poorly if tuning times are high.

3.2.2 Multihop Protocols

In a multihop protocol, a static logical topology is established over the physical topology. In previous works, the logical topology is created by fixing each transmitter and each receiver to an appropriate channel [3]. In the case of a single transmitter and a single receiver per node, the only possible fully-connected topology would be a ring. A ring allows all nodes to communicate with one another, but requires each packet to traverse an average of $N/2$ hops, where N is the number of nodes in the network. For a large network, a ring topology may be very inefficient.

A number of dynamic multihop protocols have been proposed [43, 29]. These networks reconfigure their logical topology based on prevailing traffic conditions. However, each logical topology is still constrained by the number of transmitters and receivers at each node. In this case, tuning is used to create an adaptive network, but not to increase the connectivity of the network.

3.2.3 Our Proposal: Multiconfiguration Multihop Protocols

In this work, we introduce a new class of multiconfiguration multihop protocols which employs tunable components to obtain a desired logical topology. Instead of defining the logical topology over a single configuration of the transmitters and receivers, these protocols define the logical topology over a number of configurations in the time domain. This approach results in a lower average number of hops than in a single-configuration (i.e., standard) multihop network; also it requires less tuning than in a single-hop network.

3.3 Multiconfiguration Multihop Protocols (MMP)

In this section, we present a high-level description of the proposed multiconfiguration multihop protocols (MMP) and discuss the corresponding architectural issues that need to be considered when designing such protocols. The protocols may be implemented either statically or dynamically.

3.3.1 Static MMPs

In a static MMP, time is divided into frames, and in each frame the network cycles through a series of c configurations in a TDM fashion. The topology of each of the configurations is superimposed to determine the logical topology over the entire frame. For example, in Fig. 3.4, each node is equipped with a single fixed transmitter and a single tunable receiver (FT-TR). Two configurations, C_1 and C_2 , are used to construct the logical topology of a bidirectional ring (even though each of the N nodes has only one transmitter and one receiver). The corresponding TDM frame for the protocol is given in Fig. 3.5. The ω 's in the schedule represent the wavelengths to which transceivers must tune during each configuration. For example, in Configuration C_1 , Node 1's transmitter and Node 2's receiver must both be tuned to channel ω_1 . Between configurations, each node must retune its transceivers appropriately to generate the next configuration. The time required to tune the transceivers is given by τ . The receivers cycle through the configurations specified by the TDM schedule indefinitely. In this manner, any arbitrary virtual topology may be embedded onto a network which has tunable components.

3.3.2 Dynamic MMPs

A dynamic version of the MMP may also be implemented. The primary difference in the dynamic protocols is that the logical topology is reconfigured based on prevailing traffic patterns, while in the static protocol, the logical topology is selected based on known (or assumed) traffic patterns.

Among dynamic protocols, the time scale on which the logical topology can change may vary widely. The logical topology can change infrequently based on long-term traffic

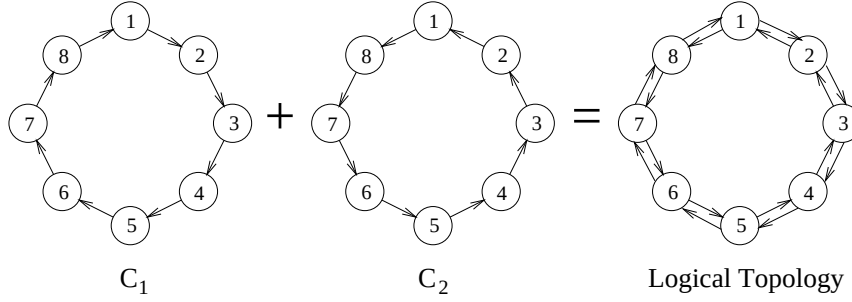


Figure 3.4: A bidirectional ring with two configurations.

	C_1	τ	C_2	τ
ω_1	1->2		1->8	
ω_2	2->3		2->1	
ω_3	3->4		3->2	
ω_4	4->5		4->3	
ω_5	5->6		5->4	
ω_6	6->7		6->5	
ω_7	7->8		7->6	
ω_8	8->1		8->7	

Figure 3.5: TDM frame for bidirectional ring topology.

patterns, or it can change based on immediate traffic demands. In either case, some amount of coordination is required. This coordination may be provided either through a separate control channel, or through the use of in-band signaling over the data channels (see, for example, [46]).

The case in which the logical topology changes infrequently is a straightforward extension of the static protocol. It may be implemented by having each node accumulate traffic information over a period of time, and then having the nodes exchange this information at predetermined intervals. Once the information is exchanged, each node can determine the new logical topology. Rather than generating topologies on-line, an alternative implementation is to generate a number of logical topologies off-line and simply have the nodes switch from one logical topology to another depending on the traffic load. For example, under light loads, the network may simply maintain a ring topology which doesn't require any tuning. If the queues at the nodes start to build up, the network may then switch to the bidirectional ring topology, as shown in Fig. 3.4, in which the nodes cycle between two unidirectional ring configurations. The basic problem in the case of both static and dynamic MMPs is then to find an appropriate logical topology for a given traffic matrix.

3.3.3 Logical Topology Design

In general, when selecting a logical topology and the corresponding configurations, the objective is to minimize delay or maximize throughput for a given traffic matrix. In [3],

the problem of finding an optimal logical topology for a given traffic matrix and a given distance matrix is addressed. The problem presented in this chapter differs from [3] in that we are not constrained by the number of transmitters or receivers at each node, and we are allowed to add as many logical links as we wish to the logical topology with the condition that each increase in the nodal degree of the logical topology also contributes to an increase in the average delay (due to additional tuning requirements). We can then modify the problem statement in [3] to accommodate networks in which multiple configurations are employed to create a logical topology. The problem of finding an optimal logical topology to minimize the average packet delay can then be stated as follows:

Given:

Traffic matrix

Receiver tuning time

Number of transceivers per node

Objective:

Minimize average packet delay

Design variables:

Logical topology

c : Number of configurations

K : Duration of each configuration

Composition of each configuration (transmitter and receiver tuning schedule)

Such topological optimization problems have been shown to be hard to solve [3]. In this work, we will simplify the problem by assuming that the traffic matrix is uniform, and that each node is equipped with one fixed transmitter and one tunable receiver (FT-TR). Since the traffic matrix is uniform, we will consider only regular logical topologies. One class of regular logical topologies which is scalable and is able to accommodate an arbitrary nodal degree is GEMNET [23]. GEMNET also compares favorably to other classes of topologies in terms of average hop distance.

3.3.4 Configuration Design

Once a logical topology is selected, another issue to consider is how to partition the logical topology into individual configurations, and how long the network should remain in any given configuration. For any given logical topology, there may be a number of ways to partition the topology into individual configurations. For example, the bidirectional ring topology in Fig. 3.4 may also be partitioned into the configurations shown in Fig. 3.6. Although different sets of configurations may result in the same logical topology, they may affect the performance of the network in different ways, specifically with regard to multihop traffic. In some instances, a multihop packet may be able to traverse multiple hops within a

single configuration. For example, in Fig. 3.4, a packet from Node 1 to Node 3 may traverse the link from Node 1 to Node 2 and the link from Node 2 to Node 3 during Configuration C_1 , if the duration of the configuration is long enough. If the configuration is not long enough, or if the packet arrived to Node 2 towards the end of the configuration, then the packet will have to wait for the network to cycle through Configuration C_2 and then back to Configuration C_1 before traversing the link from Node 2 to Node 3. On the other hand, if the logical topology is partitioned into the configurations as shown in Fig. 3.6, then the packet will only be able to traverse one hop in any configuration. Thus, the same packet will traverse the link from Node 1 to Node 2 during Configuration C_1 , wait for the network to switch to Configuration C_2 , and then traverse the link from Node 2 to Node 3. The difference between the two possible partitionings becomes greater when the average hop count is high and the duration of a configuration is longer.

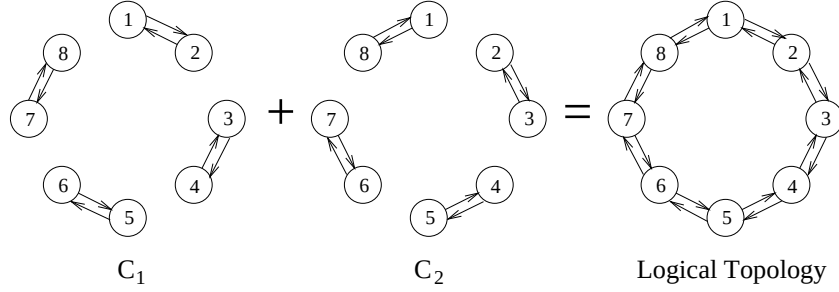


Figure 3.6: Alternative configurations for the bidirectional ring.

In the topologies that we will study, we will set each configuration to be of equal duration. However, there are situations in which it may be beneficial for some configurations to have longer durations than other configurations. Such situations include non-uniform traffic and cases in which the logical topology leads to unbalanced link loads. In extreme cases of link load imbalance, it may also be feasible to employ the same logical link in more than one configuration. We do not consider such situations in this work.

3.4 Network Architecture

The architecture consists of N nodes connected to a passive-star coupler (Fig. 3.1). Each node has a single fixed transmitter and a single tunable receiver (FT-TR), and each node's transmitter is fixed to an unique wavelength channel. Thus, each node is uniquely identified with the channel on which it transmits (as in the Rainbow network). Receivers are tunable across all channels and have a tuning time of τ . Although we have selected an FT-TR architecture, the proposed protocol will also work for a network in which each node is equipped with a tunable transmitter and a fixed receiver (TT-FR) with straightforward adjustments.

One example of the above architecture is the Rainbow-I network [24] developed at

IBM. The tunable receivers in the Rainbow network have significant tuning delays which are on the order of tens of milliseconds. Due to these high tuning times, the Rainbow protocol described in [24] is suited only for circuit-switched traffic. However, by replacing the Rainbow protocol with an MMP while using the same optical hardware, the network may also be able to handle packet-switched traffic.

3.4.1 Synchronization

In order to implement MMPs on this network architecture, we require some method of synchronization to ensure that all of the nodes are tuned to the appropriate channel at the beginning of each configuration. One approach to achieving synchronization is through the use of a separate control channel. Each node may be equipped with an additional transmitter and receiver which are fixed to a control channel, and one of the nodes may broadcast a global synchronization signal to all of the other nodes.

3.4.2 Routing

Each node in the network maintains a separate queue for each configuration, and during a given configuration, only packets from the appropriate queue are transmitted. When a packet arrives at a node, the node must choose the appropriate queue in which to place the packet. This decision is based on the routing mechanism. The routing mechanism may be dynamic, in which case the routing decision is based on the current network configuration as well as the network's logical topology, or it may be static, in which case the routing decision is based only on the logical topology.

A simple dynamic routing mechanism is one in which a packet is placed in the queue for the current configuration if the topology of the current configuration has a path to the packet's destination. The problem with this type of mechanism is that, if a packet doesn't reach its destination by the end of a configuration, then the routing decisions for subsequent hops may be based on different configuration information. As a result, the packet may end up being routed back to nodes which have already been visited. For example, consider the network in Fig. 3.4. If a packet destined for Node 7 arrives at Node 1, and if the network is currently in Configuration C_1 , then a dynamic routing mechanism may choose to send the packet to Node 2 with the expectation that the packet will traverse Nodes 3, 4, 5, and 6 before finally reaching Node 7. However, if the network switches to Configuration C_2 immediately after the packet has been transmitted to Node 2, then Node 2 may decide to send the packet back to Node 1, since a path exists from Node 1 to Node 7 via Node 8.

In a static routing mechanism, a single fixed path is chosen for every source-destination pair. If the links in each configuration are unique, then every link on the chosen path will exist in exactly one configuration, and thus correspond to a specific queue at each node. In this work, we will consider a static shortest-path routing mechanism.

3.5 Throughput and Delay Analysis

We now develop an analytical model for measuring the effectiveness of a given logical topology in terms of delay and throughput. Each node is modeled as shown in Fig. 3.7, and each outgoing link in the logical topology at a node corresponds to a queue at that node. The following parameters are given:

- N : Number of nodes.
- w : Number of channels (assumed to be equal to N).
- t : Number of transceivers per node (assumed to be equal to one).
- c : Number of configurations.
- Logical topology for above N, w, t , and c .
- τ : Tuning time in slots.
- K_{tot} : Total number of data slots per frame.
- K : Total number of data slots per configuration (equal to $\frac{K_{tot}}{c}$ if configurations are of equal duration).
- λ : Arrival rate of new packets to the system across all nodes (measured in packets per slot).
- μ : Service rate of packets (assumed to be normalized to one packet per slot).
- p : Propagation delay in slots.

Given the logical topology, we can calculate the following terms:

- h : Average hop distance.
- L_i : Link load for link $i = 1, 2, \dots, N \cdot c$ as a fraction of total load in the network.

In this work, we calculate h and the L_i 's using a direct approach. We assume one unit of traffic flow from each source to each destination, and shortest-path routing. The shortest path from each source to each destination is calculated, and the appropriate flows are assigned to each link in the shortest path. In the case of multiple shortest paths, the flow is equally divided among the shortest paths.

Since λ packets arrive to the system per slot and each packet traverses an average of h hops, the total arrival rate of both new and forwarded packets to the system is $\lambda \cdot h$ packets per slot. Each of the $N \cdot c$ links in the network is modeled as a single queue (Fig. 3.7). The packet arrivals to each queue i are assumed to be according to a Poisson process with rate $\lambda \cdot h \cdot L_i$ (this is an approximation since only the arrivals of new packets are Poisson, while

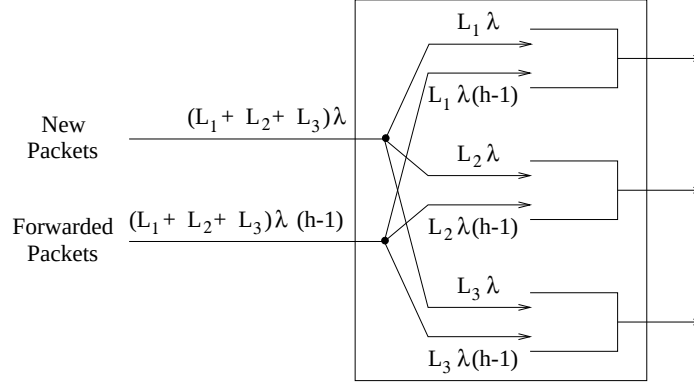
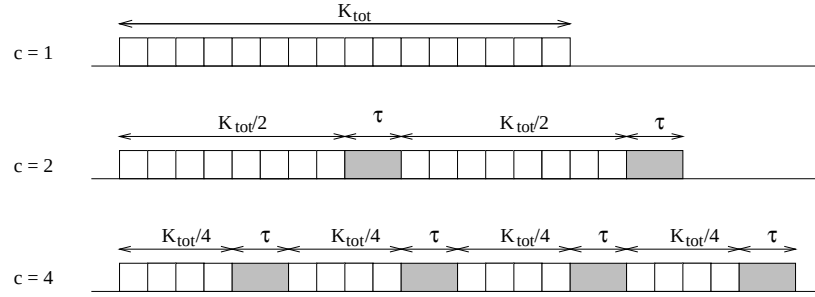
Figure 3.7: Model for a node with a logical nodal degree equal to three ($t = 1$, $c = 3$).

Figure 3.8: Frames for different number of configurations.

the arrivals of forwarded packets are non-Poisson). The service discipline is conducted in a generalized TDM fashion, with the link transmitting one packet per slot for the duration of a configuration ($\frac{K_{tot}}{c}$ slots), and transmitting no packets during the remainder of the frame, where the duration of an entire frame is $K_{tot} + \tau \cdot c$ slots (see Fig. 3.8). The queueing model can then be solved to find the queueing delay, D_i , for each individual link using the generalized TDM model from [41].

The queueing delays, D_i , consist of three components. The first is the time required for an arriving packet to synchronize to the appropriate configuration. This term can be calculated as the expected number of slots that a packet must wait in order to reach the beginning of a slot of the desired configuration. If the packet arrives outside of the desired configuration, then it must wait an average of $\frac{K_{tot} + \tau \cdot c - K}{2}$ slots to reach the beginning of the proper configuration. If the packet arrives within the desired configuration, then it must wait an average of $\frac{1}{2}$ slot to reach the next slot in the same configuration. The second component of delay is the time required to transmit packets already in the queue. This component consists of a number of complete frames followed by a partial frame, and is calculated by first finding the expected number of packets in the queue. After finding the expected number of packets in the queue, we can then find the expected value for the number of full frames and the expected number of remaining packets in the partial frame. The final component of the delay is the transmission time of the packet itself, which is equal

to one slot. The average link delay for the entire system can then be found by weighting the delay of each link according to its link load, summing the weighted delays, and dividing by the total number of links. The average packet delay is found by adding the propagation delay to the average link delay and multiplying by the average hop count, i.e.,

$$D_{avg} = h \cdot (p + \sum_{i=1}^{N \cdot c} D_i \cdot L_i). \quad (3.4)$$

The normalized throughput of the system is defined as the fraction of time spent transmitting new packets, i.e.,

$$S = \frac{\lambda}{N} \quad (3.5)$$

The maximum normalized throughput of the system is the throughput at which the system becomes unstable and can be calculated as follows:

$$\rho_i = \lambda L_i h \cdot \frac{K_{tot} + \tau c}{K} \quad (3.6)$$

where ρ_i is the fraction of time that link i is busy. In order to maintain stability, ρ_i must be less than one for all i ; we then have

$$\lambda < \frac{K}{L_{max} h (K_{tot} + \tau c)} \quad (3.7)$$

where

$$L_{max} = \max_i L_i.$$

The maximum throughput is then given by

$$S_{max} = \frac{K}{N L_{max} h (K_{tot} + \tau c)}. \quad (3.8)$$

3.6 Illustrative Numerical Examples

c	Logical Topology	h	L_{max}
1	Unidirectional ring	16.0	0.0313
2	(2,16,2) GEMNET	3.564	0.0195
4	(2,16,4) GEMNET	2.258	0.0085
8	(1,32,8) GEMNET	1.75	0.0052
16	(2,16,16) GEMNET	1.484	0.0020
32	Fully-connected (Single-hop)	1.0	0.0010

Table 3.1: Logical topologies, average hop distance, and maximum link load for a system with $N = 32$ and an uniform traffic matrix.

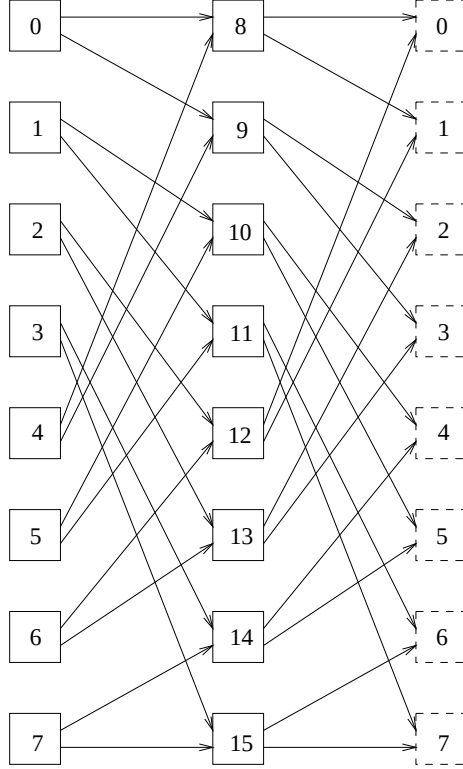


Figure 3.9: A (2,8,2) GEMNET topology.

In most of the examples presented below, we plot average delay versus throughput for various 32-node GEMNET logical topologies, with $K_{tot} = 32$, and $\tau = 10$. A topology consisting of a single configuration ($c = 1$) corresponds to a unidirectional ring, while a 32-configuration topology corresponds to a single-hop network. The remaining topologies correspond to one-column or two-column GEMNETs. GEMNET is a general class of shuffle-exchange networks which includes regular topologies such as ShuffleNet and de Bruijn graphs, and a GEMNET topology can be specified by three parameters: x , y , and z . An (x, y, z) GEMNET has x columns, y rows, and a nodal degree of z (see Fig. 3.9 for an illustrative example). We select one or two columns for the GEMNETs since these configurations yield the lowest average hop distances [23]. The details of each logical topology are given in Table 3.1. In these examples, we also assume that the propagation delay is negligible, since we are considering a LAN environment. The analytical results are verified through simulation.

We observe in Fig. 3.10 that for low loads, a ring topology performs the best, while for higher loads, it is better to introduce tuning. In low-load situations, the queuing delays at intermediate nodes are small compared to the tuning times. Therefore, reducing the hop count by increasing the amount of configurations doesn't improve network performance for low loads and may actually increase delays due to the additional tuning overhead. As the load increases, the delay for the ring topology increases quickly. The high average hop count

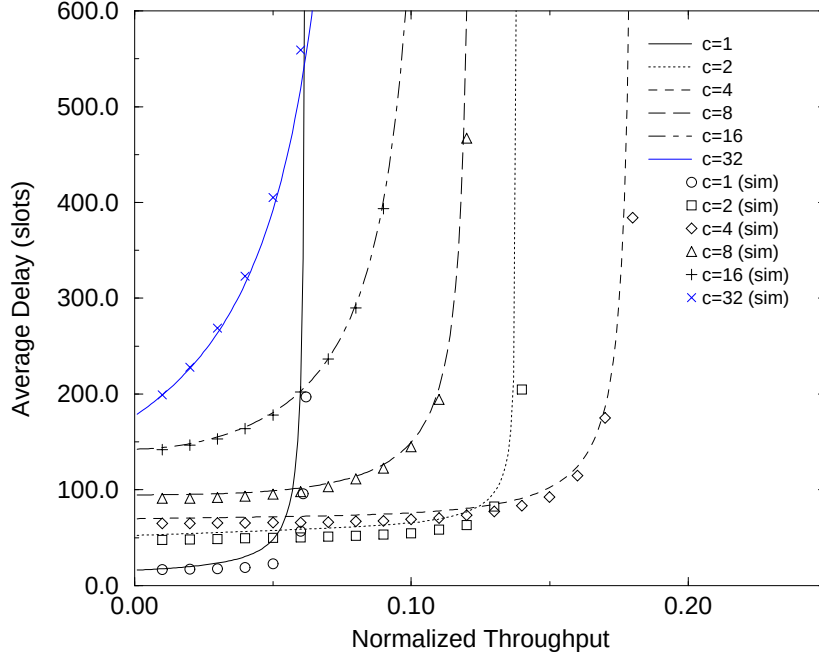
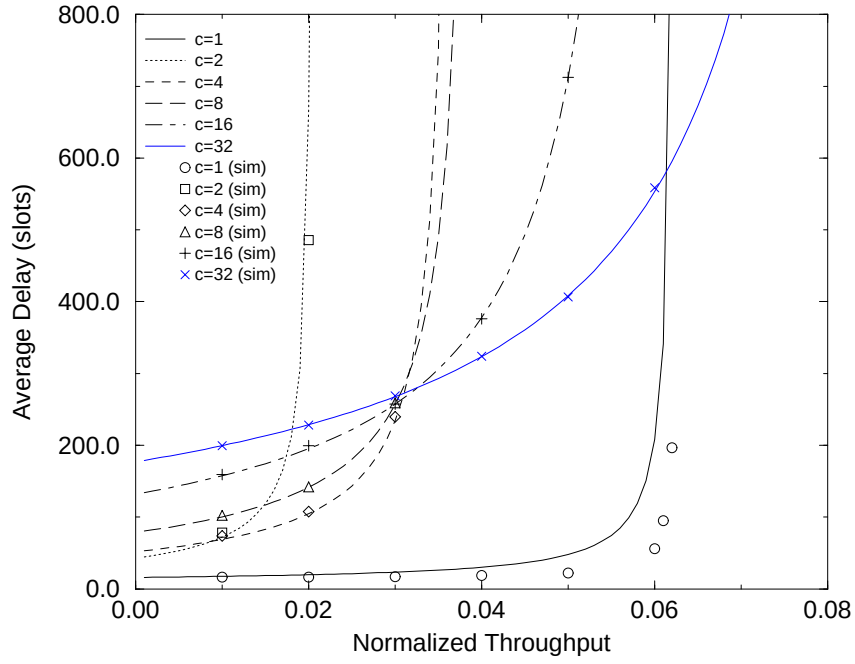
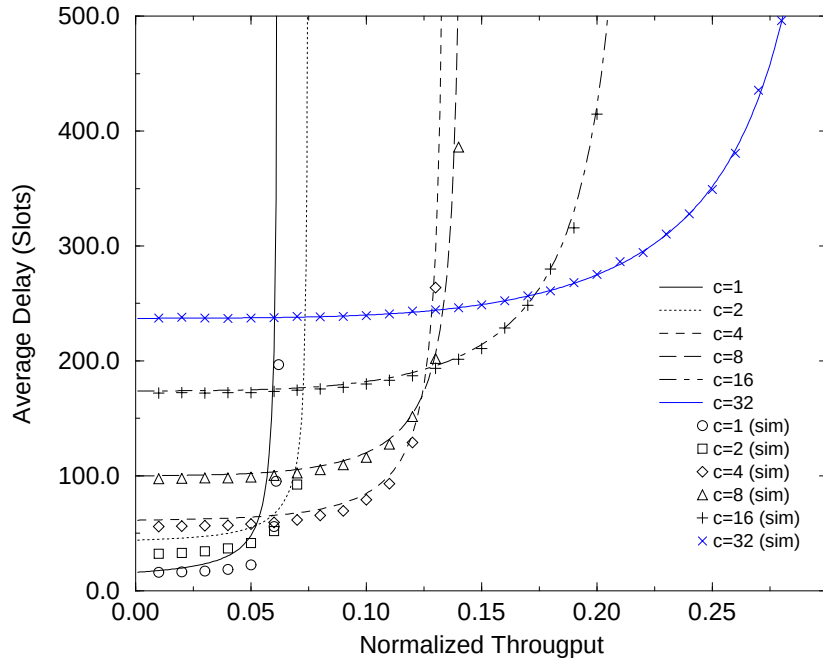


Figure 3.10: Delay vs. normalized throughput ($N = 32, K_{tot} = 32, K = \frac{32}{c}, \tau = 10$).

of a ring topology leads to a larger number of forwarded packets, thus resulting in a higher overall load on the system. Figure 3.10 verifies the maximum throughput derived in Eqn. (3.8). Also, we observe good agreement between the analysis and simulation, indicating that the Poisson arrival approximation for forwarded traffic is not very detrimental.

Note that, for any given load, there is an optimal topology which minimizes the average packet delay. Consequently, a simple, but effective dynamic protocol can be developed which switches between different topologies. In the above example, in order to achieve minimum packet delay, the network can switch between an unidirectional ring ($c = 1$), a (2,16,2) GEMNET ($c = 2$), or a (2,16,4) GEMENT ($c = 4$), depending on the traffic load (see Fig. 3.10).

Figures 3.11 and 3.12 also plot delay versus throughput for the same basic network parameters; however, instead of keeping K_{tot} constant over all of the topologies, we keep K constant. We see that setting $K = 1$ (Fig. 3.11) greatly reduces the performance of those topologies with $c = 2, 4, 8$, and 16. When $K = 1$, each packet may only traverse one hop per configuration. Thus, each additional hop after the first hop will incur at least one tuning delay. In a single-hop system ($c = 32$), all packet traverse only a single hop and do not incur such additional delays. The single-configuration system doesn't require any tuning, thus it also does not incur these tuning delays. In Fig. 3.12, we see that increasing K to 5 significantly increases the maximum throughput of those systems which make use

Figure 3.11: Delay vs. normalized throughput ($N = 32, K_{tot} = K \cdot c, K = 1, \tau = 10$).Figure 3.12: Delay vs. normalized throughput ($N = 32, K_{tot} = K \cdot c, K = 5, \tau = 10$).

of tuning.

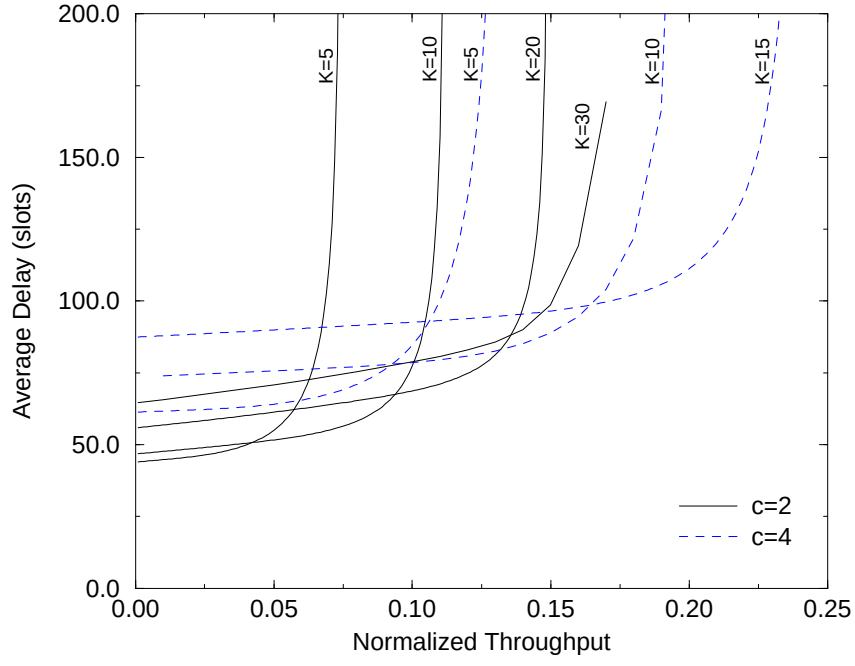
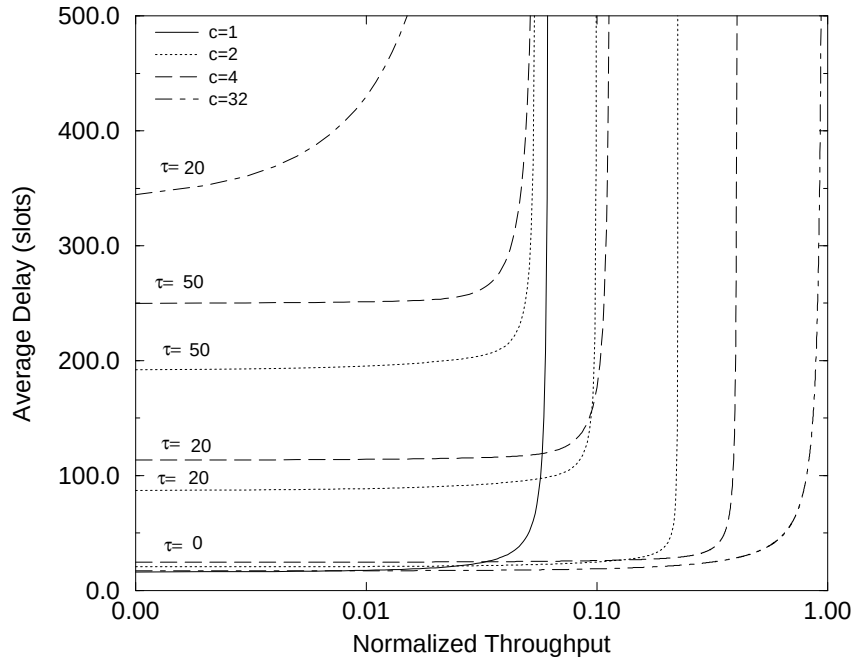
In Fig. 3.13, we plot delay versus throughput for different values of K , and $c = 2$ and 4. Only analytical results are shown. (For the remainder of the chapter, only analytical results will be shown unless otherwise specified.) We see that, by increasing the parameter K , we can achieve higher throughput at the expense of slightly higher delays under light loads. Increasing K reduces the overhead of tuning by allowing more packets to be transmitted per configuration. However, if the load is low, a packet may potentially have to wait a longer time before the network reaches the appropriate configuration, since each configuration has a longer duration. The result also provides another approach to implementing a dynamic protocol. Instead of switching between different logical topologies, we can simply (dynamically) extend or reduce the length of each configuration.

In Fig. 3.14, we plot delay versus throughput for different values of τ , and $c = 1, 2, 4$, and 32. Note that, since the static ring topology ($c = 1$) requires no tuning, the curve for $c = 1$ remains unchanged for different values of τ . As expected, in a system with low tuning times (e.g., $\tau = 0$), topologies with a higher number of configurations offer better performance, while in a system with high tuning times, systems with fewer configurations perform better. We observe that, when $\tau \geq 50$, the static ring topology ($c = 1$) will provide the best performance for any offered load. However, by increasing K , the maximum throughput can be increased for $c = 2$ and $c = 4$ (as in Fig. 3.13).

Figure 3.15 shows the upper bound on the system throughput for topologies consisting of various number of configurations. The single-hop topology ($c = 32$) is able to achieve the highest throughput for a given K , whereas the ring topology ($c = 1$) achieves the lowest throughput for $K = 5$ and $K = 10$. In general, a lower average hop distance results in a higher maximum throughput, assuming that the links in the logical topology have balanced loads. If the link loads are unbalanced, then the throughput is limited by the most heavily loaded link in the network. For the case of $c = 16$, the chosen topology results in evenly loaded links, resulting in a much higher maximum throughput than for surrounding points ($c = 15, c = 17$, etc.) which have unequally balanced link loads.

Although a lower average hop distance results in higher throughput, the tuning required to reduce the hop distance increases the delay for low system loads. In Fig. 3.16, we show the delay for very light loads. These delays are calculated by setting $D_i = \frac{K_{tot} + \tau \cdot c - K}{2} + 1$ in Equation 3.4. We observe that the ring topology yields the lowest delay, while the single-hop topology has the highest delay. In a lightly-loaded system, most of the delay is encountered when the packet must wait for the correct configuration before being transmitted. In general, if there are c configurations, a packet must wait an average of $c/2$ configurations per hop. In the case of a ring topology, there is only a single configuration; therefore, the delay is caused only by the transmission time of the packet at each hop.

In Fig. 3.17, we study the effect of implementing the same logical topology with different sets of configurations. We consider the two implementations of an 8-node bidirectional

Figure 3.13: Delay vs. normalized throughput ($N = 32, \tau = 10$).Figure 3.14: Delay vs. normalized throughput ($N = 32, K = \frac{32}{c}$).

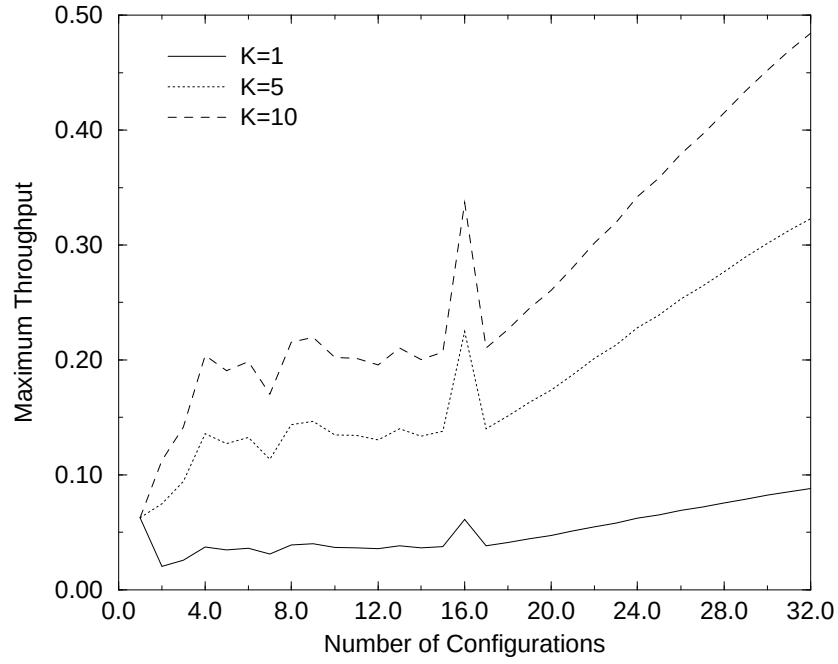


Figure 3.15: Maximum throughput vs. number of configurations ($N = 32, \tau = 10$).

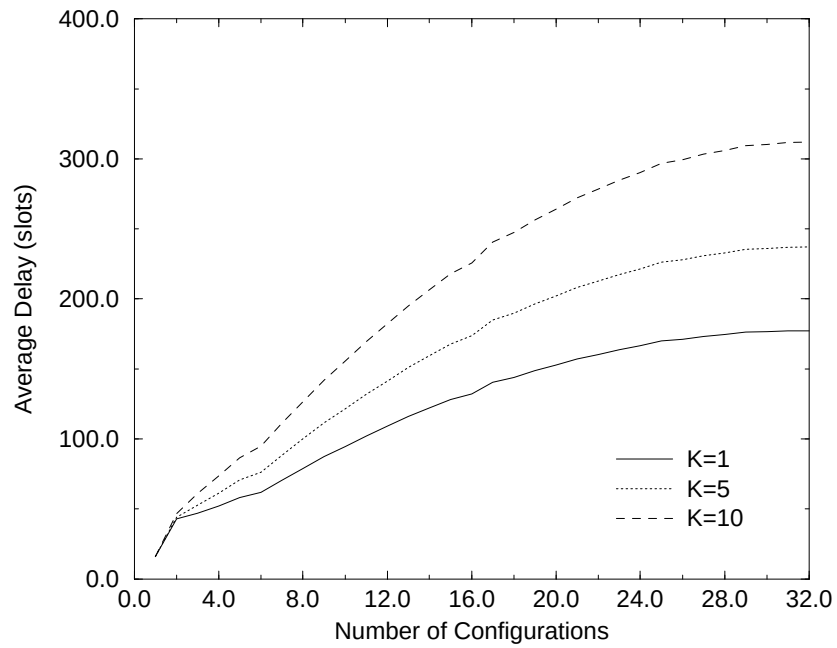


Figure 3.16: Delay vs. number of configurations for light loads ($N = 32, \tau = 10$).

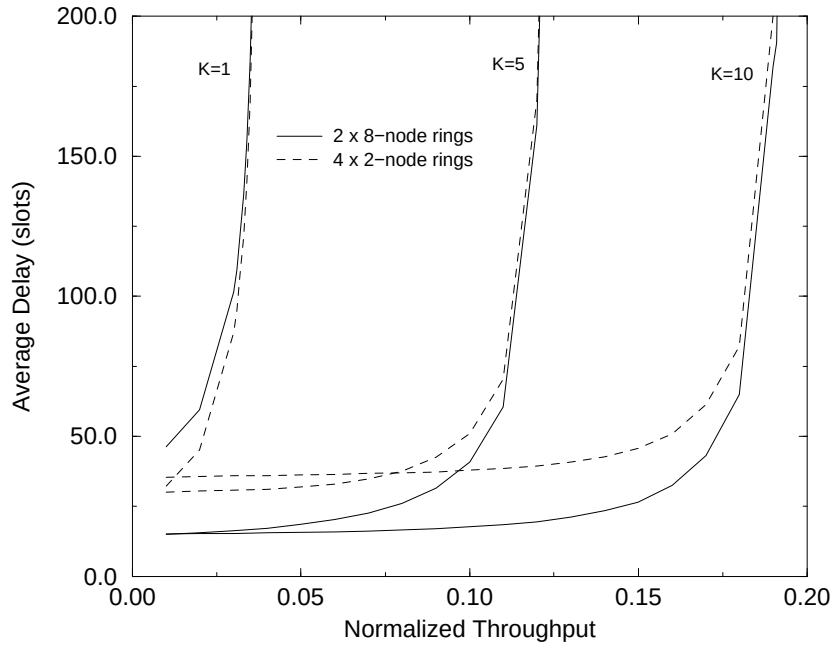


Figure 3.17: Delay vs. normalized throughput for different sets of configurations ($N = 8, \tau = 10$).

ring given in Figs. 3.4 and 3.6. In Fig. 3.4, each configuration consists of an 8-node unidirectional ring, while in Fig. 3.6, each configuration consists of four 2-node rings. Since the analysis doesn't model individual configurations, we present only simulation results. We observe that, for $K = 5$ and $K = 10$, the network consisting of two 8-node ring configurations performs better at light loads. The reason for this performance is that the 8-node rings allow packets to traverse multiple hops in a single configuration, whereas the 2-node rings only allow a single hop per configuration. The number of hops that a packet may traverse per configuration is not only limited by the topology of the configuration, but also by the duration of the configuration. A packet is limited to at most K hops per configuration, where K is the duration of the configuration in slots. Therefore, when K is small, the difference between the two different sets of configurations is not as high. For $K = 1$, we observe that the network consisting of 2-node-ring configurations performs better than the network consisting of 8-node-ring configurations. In both cases, all packets are limited to a single hop per configuration; however, consecutive hops in the 2-node-ring configurations may occur in consecutive configurations (i.e., the first hop occurs during Configuration 1, the second hop during Configuration 2, etc.), while in the 8-node-ring configurations, each consecutive hop must wait for the same configuration (i.e., the first hop takes place during Configuration 1, the second hop must wait for the network to cycle

around back to Configuration 1, etc.). Thus, under low loads, a packet traversing h hops would require approximately h configurations under the 2-node-ring case, but would require approximately $2h$ configurations under the 8-node-ring case.

3.7 Conclusion and Future Work

One of the key points in this work is that tunable components not only provide the capability to increase the connectivity of the network, but also provide dynamic capabilities. Many networks take advantage of only one of these benefits while ignoring the other. In this work, we proposed a class of multiconfiguration multihop networks (MMPs) which takes advantage of tunable receivers to increase connectivity and to reduce the average hop distance, without requiring as much tuning as in a single-hop network. We also showed how these networks can be designed to accommodate dynamic traffic.

We have addressed some of the issues involved in finding a logical topology for an uniform traffic matrix and have shown that the offered load as well as the tuning time play a significant role in deciding on a logical topology. We have also shown that, by appropriately selecting the number of configurations and the duration of these configurations, we can obtain a trade-off between the amount of tuning required and the average hop distance. Reduced tuning requirements lead to lower delays at low loads, while lower hop distances result in a higher amount of network utilization at high loads.

Further work is needed to refine and solve the problem of selecting an optimal logical topology for a given non-uniform traffic pattern. An open problem is to find an appropriate logical topology, as well as the composition and duration of each configuration in the logical topology, such that the average packet delay is minimized.

Another extension to this work is to introduce channel sharing. With channel sharing, the number of channels, w , is less than the number of nodes. In [50], it has been shown that channel sharing increases the connectivity of the network by using fewer channels. The study assumed a single fixed transmitter and a single fixed receiver per node. By replacing the fixed receiver with a tunable receiver, we may obtain very high connectivities through both channel sharing and receiver tuning, and we will also gain the ability to dynamically adjust the network to meet changing traffic demands.

Chapter 4

The Advantages of Partitioning Multicast Transmissions in a Single-Hop Optical WDM Network

4.1 Introduction

In modern networks, the demand for bandwidth and high quality of service requires the efficient utilization of network resources such as transmitters, receivers, and channel bandwidth. One method for conserving these resources is to employ efficient implementations of multicasting wherever possible.

Traditionally, networks have supported unicast transmissions in which a single source may transmit data to a single destination. However, if a message is intended for multiple destinations, sending a separate transmission to each destination may result in an inefficient use of resources. Using multicasting, a source sending a message to multiple destinations may schedule a single transmission which can then be broadcast to multiple destinations or forwarded from one destination to another, thus conserving transmitter resources and channel bandwidth.

Methods for implementing multicasting in networks have been gaining much attention in recent years. Many previous efforts in multicasting have focused on the multihop multicasting problem. The multihop multicasting problem is to construct a minimal cost tree which is rooted at the source node and spans all of the destination nodes. The cost of the tree is determined by the amount of network resources consumed, such as the number of links in the tree. This problem has traditionally been formulated as the Steiner Minimal Tree problem and has been studied in a number of previous works [51, 38].

A problem that has received less attention is the single-hop multicasting problem. Since single-hop optical networks are based on a broadcast medium (Fig. 4.1), multiple destinations may receive the same transmission simultaneously, thereby reducing the number of

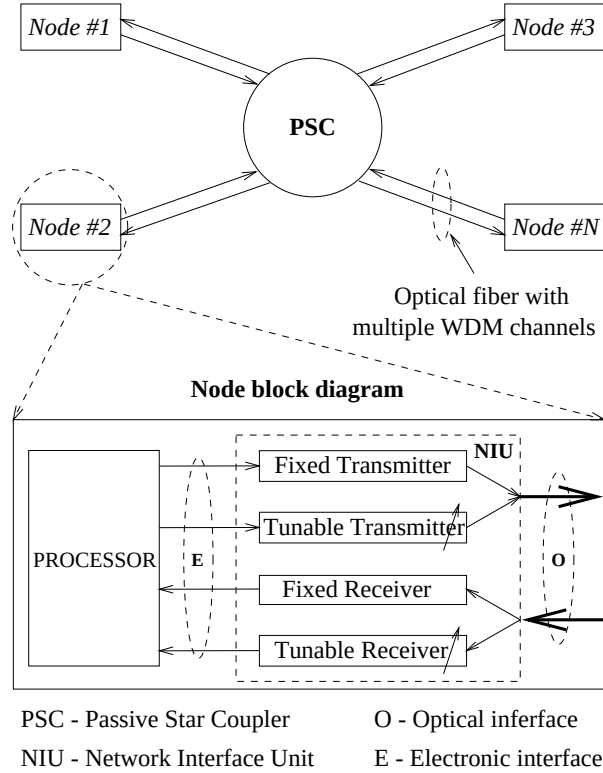


Figure 4.1: A passive-star based network.

times that a message is transmitted. However, in a single-hop network with multiple channels, each destination may have its receiver tuned to a different channel. Thus, we have the additional problem of coordinating the source node and the destination nodes such that they are tuned to the same channel during a transmission. The single-hop multicasting problem in a multiple-channel network is then to schedule the transmission of a message in a way which not only ensures that each of the destinations receives the message, but also minimizes the total amount of network resources being used.

In this chapter, we study the single-hop multicast problem as applied to a WDM passive-star coupler network in which nodes access different channels through the use of tunable transmitters and receivers. In a multicast transmission, the source node must coordinate with multiple destination nodes such that the destination nodes' receivers are tuned to the appropriate channel at the appropriate time. The single-hop multicasting problem in WDM LANs has been addressed in a few previous works [42, 6, 44].

In [42], a TDM-based approach is presented. Time on each channel is divided into frames, with each frame consisting of a number of slots. Each node is allocated slots in which it may transmit multicast packets to specified destinations. A heuristic is used to generate the TDM schedule based on known average traffic patterns.

A reservation-based, on-line scheduling protocol is studied in [6], in which nodes send out transmission requests prior to transmitting packets. Based on this information and

information maintained by the nodes, the transmission is scheduled at a point in time when the source node's transmitter and the destination nodes' receivers are all able to tune to the same channel.

In [6], only a single transmission is scheduled for each multicast message, resulting in a minimal use of bandwidth and transmitter resources. However, since receivers at each destination node may become available at different times, some destination nodes may have to wait longer than others to receive the message. This may lead to an inefficient use of receiver resources.

In this chapter, we extend the work in [6] and show that it may be more efficient for a source node to transmit a multicast message multiple times to subsets of the multicast group rather than transmitting the message a single time to the entire multicast group. For example, a subgroup of nodes which becomes available earlier than the other nodes in the multicast group would receive one transmission from the source node, while a subgroup of nodes which becomes available later would receive a separate transmission at a later time. These separate transmissions reduce the amount of time that the nodes in the first subgroup have to wait before receiving the message from the source. The problem is then to find a partitioning of the multicast destination nodes which most efficiently utilizes receiver resources in the network. In this chapter, the measure for receiver resources will be the average receiver waiting time.

The *receiver waiting time* is defined from the time that the receiver becomes free to receive the transmission to the time that the receiver actually begins receiving the transmission. By minimizing the average receiver waiting time, we are minimizing the amount of time a receiver is idle, and thus increasing receiver utilization.

In Section 4.2, we describe the system architecture. In Section 4.3, we define the Multicast Partitioning Problem (MPP). In Section 4.4, we provide heuristics for determining an efficient partitioning of the multicast destination set. The heuristics are tested on randomly-generated test cases in Section 4.5, and in Section 4.6, the heuristics are incorporated into a protocol. Section 4.7 concludes the chapter, summarizing its contribution and stating some topics for future research.

4.2 System Architecture

We consider a network in which N nodes are connected via fiber to a passive-star coupler. The fiber supports a number of WDM channels, each channel corresponding to a unique wavelength. W channels are used for communicating data, while one channel is reserved for use as a control channel. The data channels are accessed by a single tunable transmitter and a single tunable receiver at each node. The tunable transmitters and receivers may tune to any of the data channels in the system, but may selectively access only a single channel at a time. Each node is also equipped with a single fixed transmitter and a single

fixed receiver which are always tuned to the control channel. Transmissions on each channel are broadcast by the star coupler to all nodes, and a node may receive a transmission by tuning its receiver to the appropriate channel.

In order to schedule data transmissions, each node maintains the following state information:

- $T_i, i = 1, 2, \dots, N$: Time at which the transmitter at node i becomes available to transmit messages.
- $R_i, i = 1, 2, \dots, N$: Time at which the receiver at node i becomes available to receive transmissions.
- $C_i, i = 1, 2, \dots, W$: Time at which channel i becomes available.

The availability time is defined as the time at which the last scheduled transmission for a particular transmitter, receiver, or channel has ended. This time may also include propagation delays or the time required to tune a transmitter or receiver to a particular channel.

The control channel is accessed in a contention-free manner via round-robin TDM. When a message arrives at a node, the node sends a request packet on the control channel in its specified control slot. The request packet indicates the source node, the intended destination nodes, and the length of the message. Each node continuously monitors the control channel for transmission request packets. When a request is received, each node runs an identical scheduling algorithm to schedule transmissions of the message. The scheduling algorithm specifies when the transmissions will take place, the channels on which they will take place, and which receivers should be tuned to which channels at the specified transmission times. After a message is scheduled, all nodes appropriately update the state information, T_i , R_i , and C_i , for the transmitters, receivers, and channels involved in the transmissions. Since the same algorithm is executed at each node, each node will generate the same schedule and have the same consistent global information about the state of the system.

Because each node only maintains the times at which the last scheduled transmission for each transmitter, receiver, and channel has ended, transmissions for new messages can only be scheduled after these times. This may lead to idle times or “gaps” in the schedule which cannot be filled. To fill these gaps would require that nodes maintain the starting and ending times of all previously scheduled transmissions, which would considerably increase the complexity of the scheduling algorithm. In order to have a simple, on-line algorithm, we have chosen not to schedule transmissions in gaps.

When a multicast request is received, the multicast may be scheduled in a number of ways. One approach is to transmit the message separately to each destination. However, this may result in an inefficient use of bandwidth. Also, because the transmitting node may only transmit one message at a time, some receivers may have to wait for the completion

of previous transmissions before receiving the message. Figure 4.2(a) shows how a multi-destination message would be scheduled using the traditional unicast approach. The X_i 's are the receiver availability times for the nodes in the multicast destination group, and the S_j 's are the scheduled transmission times for the message. We assume in this case that the transmitter at the source node becomes available before the first receiver becomes available, and that there is also a channel available before this time. The waiting time for each receiver is indicated by w_i . We clearly see that this schedule is inefficient, since at the very least, Nodes 4 and 5 could have both received the transmission S_4 , reducing the number of transmissions and reducing the waiting time for Node 5.

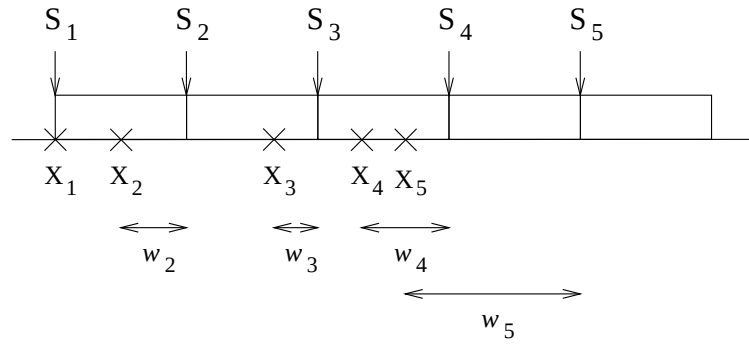
Another extreme approach is the traditional multicast approach [6] which schedules a single transmission at the time when the last receiver becomes available, as shown in Figure 4.2(b). All nodes receive the transmission at time S_1 . This efficiently utilizes the bandwidth in the network since only one transmission takes place, but if all receivers do not become available at the same time, then some of the receivers must wait until the last receiver becomes free. During this time, receivers that are waiting may be unable to receive transmissions from other sources. Thus, this approach may result in low receiver utilization.

The approach proposed and examined in this work is a generalization of the previous two approaches. For each multicast request, the group of multicast destinations is partitioned into a number of subgroups, and a separate transmissions is scheduled for each subgroup. Each receiver in the same subgroup receives the same transmission. In Figure 4.2(c), Nodes 1 and 2 both receive the transmission at time S_1 , while the remaining nodes receive the transmission at time S_2 . This approach allows some destination nodes which become available earlier to avoid excess waiting times, and also makes better use of bandwidth than scheduling a separate transmission for each destination.

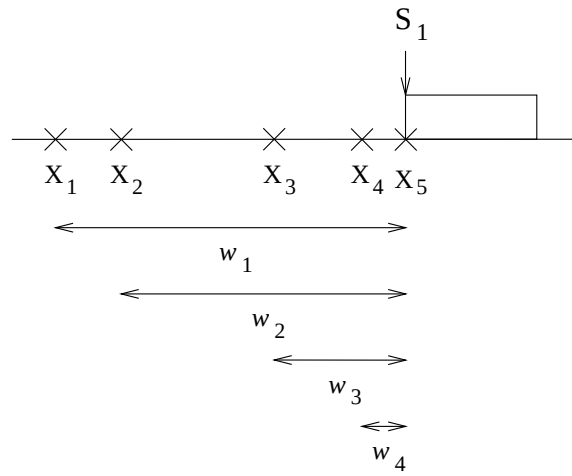
In order to maximize the performance of the network, we need to determine an efficient method for partitioning the multicast group. In the following sections, we present the criteria for choosing a partitioning, formulate the partitioning problem, and provide a number of heuristic approaches to solve the problem.

4.3 Multicast Partitioning Problem (MPP)

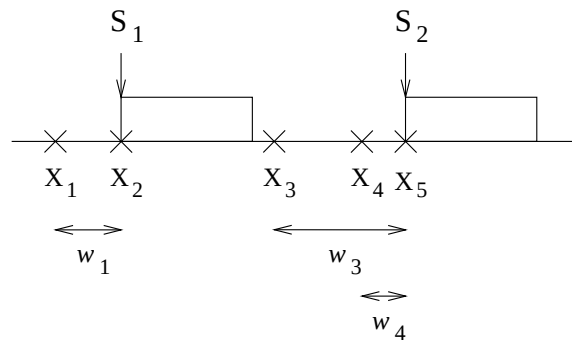
When a node sends a request on the control channel, the request specifies the intended destinations of the message as well as the length of the message. Each node maintains information on the available times of the transmitters, receivers, and channels in the system. With this information, the nodes should schedule one or more transmissions such that all of the intended destinations receive the message, and the average waiting time for the destinations is minimized. We refer to this problem as the Multicast Partitioning Problem (MPP).



(a) Scheduling one unicast transmission per destination



(b) Single-transmission (traditional) multicast



(c) Multiple-transmission multicast (this work)

Figure 4.2: Scheduling a multicast message.

Our formulation assumes that the source node's transmitter and at least one channel is available before the first receiver in the multicast destination group becomes available. This assumption can be justified by noting that multicast messages consume an equal amount of transmitter and channel resources, but consume a higher amount of receiver resources, since each transmission is received by multiple receivers. Thus, receivers are likely to be the bottleneck in the network. We also assume that receiver tuning times are negligible. This assumption is valid as long as tuning times are small relative to message lengths.

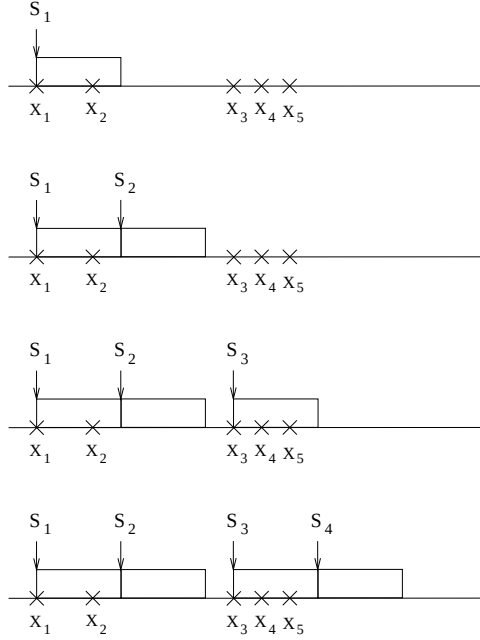


Figure 4.3: Earliest Available Receiver (EAR).

The problem has the following inputs:

- L : Length of the message.
- M : Number of destinations.
- $X_i, i = 1, 2, \dots, M$: Time when receiver at the destination node i becomes available. Without loss of generality, the X_i 's are ordered such that $X_1 \leq X_2 \leq \dots \leq X_M$.

The problem has the output S_j for $j = 1, 2, \dots, M$, which is the time at which transmission j is scheduled¹. The problem is then to minimize the average receiver waiting time given by

$$\bar{w} = \frac{1}{M} \cdot \sum_{i=1}^M \min_{j \ni S_j \geq X_i} (S_j - X_i) \quad (4.1)$$

¹Although the problem outputs M scheduled times, a transmission is not necessarily scheduled at all of these times. For example, in Fig. 4.3, $M = 5$, but only four transmissions are scheduled (as we shall observe shortly).

with respect to the transmission starting times, $S_j, j \in \{1, 2, \dots, M\}$, such that

$$S_M \geq X_M \quad (4.2)$$

and

$$S_j + L \leq S_{j+1} \quad \text{for } j = 1, 2, \dots, M-1. \quad (4.3)$$

The constraint in (4.2) guarantees that all nodes will be able to receive at least one of the scheduled transmissions. The constraint in (4.3) is due to the fact that there is only one transmitter at the source. This prevents more than one transmission from occurring at the same time.

Note that $\overline{w}$ is discontinuous at the points where $S_j = X_i$ for all $j \in \{1, 2, \dots, M\}$ and $i \in \{1, 2, \dots, M\}$. These points correspond to local minima.

Also, if $S_j < X_1$ or $S_j \geq X_M + L$, then no transmission will be scheduled at time S_j . This is because, if $S_j < X_1$, no receiver will be ready in time to receive the transmission. If $S_j \geq X_M + L$, then there must be an earlier transmission which can be received by each node. For example, in Figure 4.2(c), transmissions are only scheduled at times S_1 and S_2 , since S_3, S_4 , and S_5 are all greater than $X_5 + L$.

We now attempt to reduce the solution space of the problem by providing the following theorems.

Theorem 1. *The set $\mathbf{S} = \{S_1, S_2, \dots, S_M\}$ is an optimal solution to MPP only if $S_i = X_j$ for at least one (i, j) pair.*

Proof by contradiction: Suppose an optimal solution existed for which $S_i \neq X_j$ for all (i, j) pairs. There must exist at least one S_i for which $S_i > X_1$ because of the constraint $S_M \geq X_M$. For each $S_n < X_1$, we can set $S_n = -\infty$ without affecting the cost function. Next, if $S_1 > X_1$, then choose $S_i = S_1$; otherwise, choose the S_i such that $S_i > X_1$ and $S_{i-1} < X_1$. Either $X_j < S_i < X_{j+1}$ for some $j, j < M$, or $S_i > X_M$. If the former is the case, then setting $S_i = X_j$ will reduce the cost function. If the latter is the case, then setting $S_i = X_M$ will reduce the cost function. This contradicts the original assumption that the solution was optimal.

Theorem 2. *If the set $\mathbf{S} = \{S_1, S_2, \dots, S_M\}$ is an optimal solution to MPP and there exists an $S_i, 1 < i \leq M$, for which $S_i \notin \{S_{i-1} + L, X_1, X_2, \dots, X_M\}$, then there exists a corresponding equivalent solution $\mathbf{S}'$ for which $S'_i \in \{S'_{i-1} + L, X_1, X_2, \dots, X_M\}$ which yields the same cost.*

Proof by contradiction: Suppose $\mathbf{S}$ is an optimal solution in which $S_i \notin \{S_{i-1} + L, X_1, X_2, \dots, X_M\}$. Then, the transmission at S_i is not received by any node. This can be proved by contradiction. If the transmission at S_i were received by a node, then the cost may be reduced by setting $S_i = k$ where k is an element in the set $\{S_{i-1} + L, X_1, X_2, \dots, X_M\}$ which is the closest to S_i and less than S_i . This contradicts the optimality assumption. Since S_i is not received by any node, an equivalent solution may be

obtained by setting $S'_1 = S_1, S'_2 = S_2, \dots, S'_{i-1} = S_{i-1}, S'_i = S_{i+1}, S'_{i+1} = S_{i+2}, \dots, S'_{M-1} = S_M$, and $S'_M = S_M + L$ if $S_i < X_M$; or by setting $S'_1 = S_1, S'_2 = S_2, \dots, S'_{i-1} = S_{i-1}, S'_i = S'_{i-1} + L, S'_{i+1} = S'_i + L, \dots, S'_M = S'_{M-1} + L$ if $S_i > X_M$.

From Theorems 1 and 2, an optimal – but not necessarily unique – solution $\mathbf{S}$ exists such that

$$S_1 \in \{X_1, X_2, \dots, X_M\} \quad (4.4)$$

and

$$S_i \in \{S_{i-1} + L, X_i, X_{i+1}, \dots, X_M\} \quad \text{for } i = 2, 3, \dots, M. \quad (4.5)$$

4.4 Scheduling Algorithms

Scheduling problems such as MPP generally tend to be hard; therefore, we investigate a number of heuristic approaches. Based on the conclusions in the previous section, a number of heuristics for finding an efficient partitioning have been developed. Because the scheduling algorithm is an on-line algorithm, the heuristic should be simple and fast. Therefore, we first examine two simple greedy approaches to MPP. Then, we examine more complex approaches which attempt to search a larger solution space in order to find a solution which is closer to the optimal solution.

4.4.1 Earliest Available Receiver (EAR)

Earliest Available Receiver (EAR) is a greedy algorithm. It schedules the first transmission at the time at which the earliest receiver in the multicast group becomes available. The next transmission is either scheduled immediately after the first transmission if any of the remaining receivers become available during the first transmission, or, if no receivers became available during the first transmission, whenever the next receiver becomes available. The heuristic proceeds to schedule transmissions at the earliest possible time for each of the remaining receivers. The running time for this heuristic is $O(M)$.

Figure 4.3 shows an example of EAR. The first transmission is scheduled at time S_1 . Since Receiver 2 becomes ready during the first transmission, the second transmission takes place immediately after the first. No receivers become available during the second transmission, so the next transmission is scheduled when the next receiver becomes available at time X_3 . Receivers at Nodes 4 and 5 become available during the third transmission, so the next transmission is scheduled to take place immediately following the third transmission. Nodes 4 and 5 are both able to receive the transmission starting at S_4 , so no further transmission is necessary. Pseudocode for this heuristic follows.

EARLIEST AVAILABLE RECEIVER

Given: X_i, L, M

Step 0

Set $S_1 = X_1$, $endtx = S_1 + L$, $i = 2$, $j = 2$

Step 1

If $X_i \geq endtx$

$$S_j = X_i$$

$$endtx = S_j + L$$

$$j = j + 1$$

If $X_i < endtx$ and $i = M$

$$S_j = endtx$$

$$endtx = S_j + L$$

$$j = j + 1$$

If $X_i < endtx$ and $X_{i+1} > endtx$

$$S_j = endtx$$

$$endtx = S_j + L$$

$$j = j + 1$$

$i = i + 1$

If $i \leq M$ go to **Step 1**

end

4.4.2 Latest Available Receiver (LAR)

Another greedy approach is Latest Available Receiver (LAR). LAR begins by scheduling a transmission at the time when the latest receiver becomes available, X_M (e.g., X_5 in Fig. 4.4). Receiver $M - 1$ is considered next. If a transmission to this receiver would conflict with the previously-scheduled transmission, then Receiver $M - 1$ is placed in the same group as Receiver M . If a transmission starting at X_{M-1} does not conflict with the transmission at X_M , then a separate transmission is scheduled at time X_{M-1} . The heuristic proceeds backwards through the receivers in a similar manner until transmissions for all receivers have been scheduled. This approach guarantees that all transmissions for a given multicast group will be completed no later than L time units after the last receiver becomes available. The running time for this heuristic is $O(M)$.

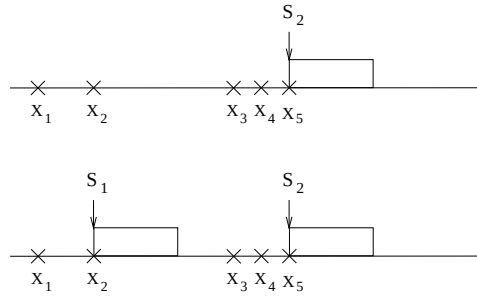


Figure 4.4: Latest Available Receiver (LAR).

Figure 4.4 illustrates an example of the LAR heuristic. A transmission is first scheduled at time X_5 . Transmissions scheduled at times X_4 or X_3 would conflict with the transmission at time X_5 , and therefore are not scheduled. Nodes 3 and 4 receive the transmission at time X_5 . A transmission at time X_2 would not conflict with the previously-scheduled transmission; therefore, a transmission may be scheduled at X_2 . A transmission at time X_1 would conflict with the transmission at X_2 and is not scheduled. Following is the algorithm for LAR.

LATEST AVAILABLE RECEIVER

Given: X_i, L, M

Step 0

Set $S'_1 = X_M$, $start = S'_M$, $i = M - 1$, $j = 2$

Step 1

If $X_i + L \leq start$

$S'_j = X_i$, $start = S'_j$

$j = j + 1$

$i = i - 1$

If $i \geq 1$ go to **Step 1**

Step 2

$S_k = S'_{j-k}$, $k = 1, \dots, j - 1$

end

4.4.3 Best Available Receiver (BAR)

Best Available Receiver (BAR) chooses each receiver i in turn, and for the chosen receiver, it schedules a transmission starting at time X_i . The EAR heuristic is then used to schedule transmissions for receivers available after $X_i + L$, and the LAR heuristic is used to schedule transmissions for receivers available before $X_i - L$. The average waiting time $\bar{w}_i$ is then calculated for the schedule. The heuristic then chooses over $i = 1, 2, \dots, M$ the schedule which has the minimum waiting time. This heuristic runs in $O(M^2)$ time. Figure 4.5 illustrates the iteration of BAR for $i = 2$, and its pseudocode follows.

BEST AVAILABLE RECEIVER

Given: X_i, L, M

Step 0

Set $i = 1$

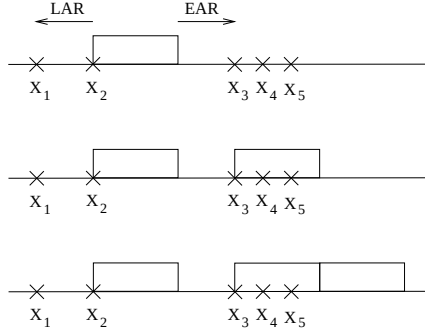
Step 1

$S_{start} = X_i$

Run EAR with $X'_k = X_{i+k-1}$, $k = 1, \dots, M - i + 1$

Run LAR with $X'_{M-k+1} = X_{i-k+1}$, $k = 1, \dots, i$

Merge the schedules produced by EAR and LAR and calculate the waiting time

Figure 4.5: Best Available Receiver (BAR) for $i=2$.

$Sched_i$ = schedule produced

w_i = waiting time

$i = i + 1$

If $i \leq M$ go to **Step 1**

Step 2

Choose $Sched_i$ for which w_i is minimum

end

4.4.4 Reschedule for Next Receiver (RNR)

Reschedule for Next Receiver (RNR) is similar to EAR, except that it allows rescheduling of the previously-scheduled transmission if doing so would decrease the waiting time for the next receiver. In this heuristic, transmissions are scheduled one at a time starting at time X_1 . For each subsequent receiver i , a decision is made to either move up the previously-scheduled transmission to time X_i so that it may be received by the receiver i , or to start a new transmission at the earliest possible time after the previous transmission has completed. The decision is made based on which option leads to the smaller waiting time for receivers 1 through i . A counter n is used by the heuristic to keep track of the number of receivers which are scheduled to receive the previously-scheduled transmission. If this transmission is scheduled at time S_j and is then moved up to accommodate the next receiver at time X_i , the increase in waiting time for the receivers in this group would be $n \cdot (X_i - S_j)$. On the other hand, if $S_j + L > X_i$ and a new transmission is scheduled for receiver i , the waiting time incurred by receiver i is $S_j + L - X_i$. These two waiting times are compared, and the schedule which results in the smaller waiting time is selected. This heuristic runs in time $O(M)$.

Figure 4.6 illustrates an example of the RNR heuristic. The first transmission is scheduled when the first receiver becomes available at time X_1 . For the second receiver (Figure 4.6(b)), we consider delaying the scheduled transmission until time X_2 , or scheduling a new transmission at time $X_1 + L$. The latter choice results in smaller average waiting time,

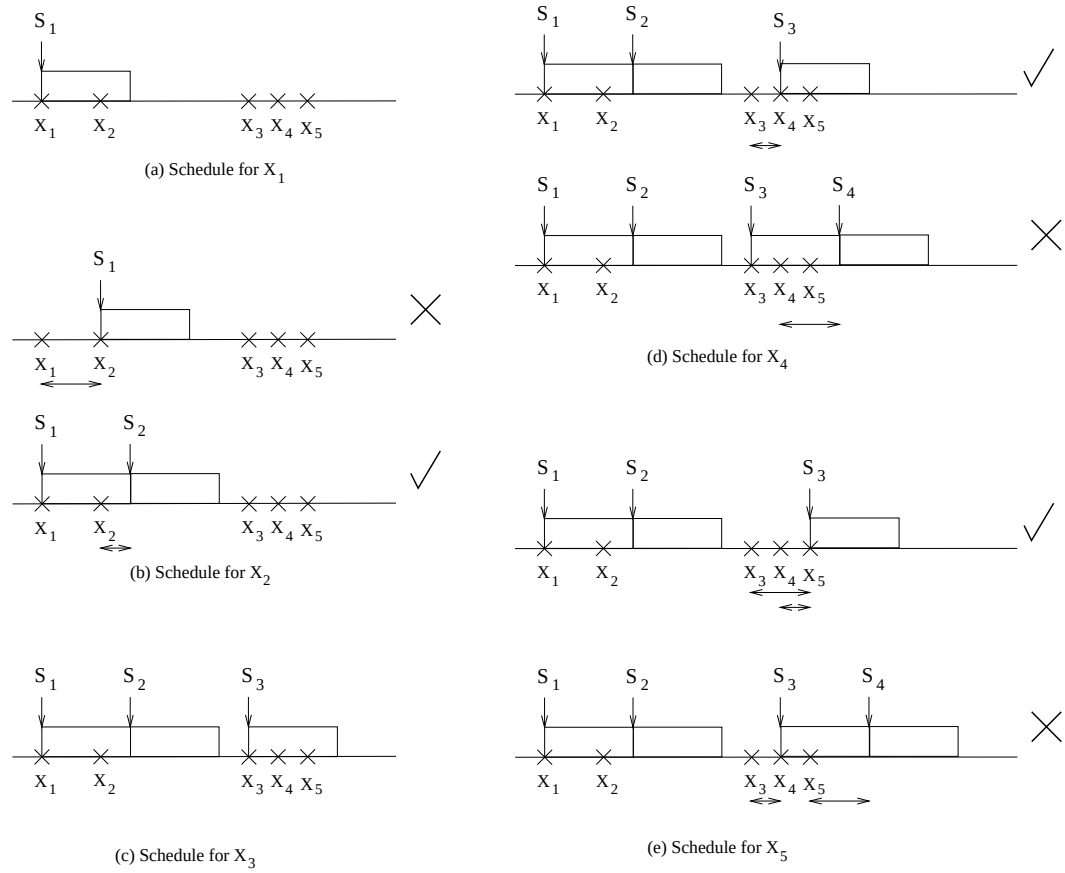


Figure 4.6: Reschedule for Next Receiver (RNR).

so a new transmission is scheduled. For the next receiver, delaying transmission S_2 would result in added waiting time for Node 2, but scheduling a new transmission at time X_3 would not result in any wait for Node 3. Thus, a transmission is scheduled at time X_3 . In Figure 4.6(d), delaying the transmission S_3 results in a small wait for Node 3 of $(X_4 - X_3)$, while scheduling a new transmission for Node 4 would result in a higher wait for Node 4 of $(X_3 + L - X_4)$. We then choose to delay transmission S_3 until time X_4 . At this point n is incremented to 2, since two nodes are now receiving transmission S_4 . Finally, in Figure 4.6(e), delaying the transmission until time X_5 increases the waiting time for Nodes 3 and 4 by $2 \cdot (X_5 - X_4)$. Scheduling a new transmission will create a wait of $(X_4 + L - X_5)$. Let us say that $2 \cdot (X_5 - X_4) < (X_4 + L - X_5)$. Then, we choose to reschedule the transmission at time X_5 . The pseudocode for the RNR heuristic is shown below.

RESCHEDULE FOR NEXT RECEIVER

Given: X_i, L, M

Step 0

Set $S_1 = X_1$, $endtx = S_1 + L$, $i = 2$, $j = 2$, $n = 1$

Step 1

If $X_i \geq endtx$

$S_j = X_i$, $endtx = S_j + L$, $n = 1$

$j = j + 1$

If $X_i < endtx$ and $n \cdot (X_i - (endtx - L)) \leq (endtx - X_i)$

$S_{j-1} = X_i$, $endtx = S_{j-1} + L$, $n = n + 1$

If $X_i < endtx$ and $n \cdot (X_i - (endtx - L)) > (endtx - X_i)$

$S_j = endtx$, $n = 1$

$endtx = S_j + L$

$j = j + 1$

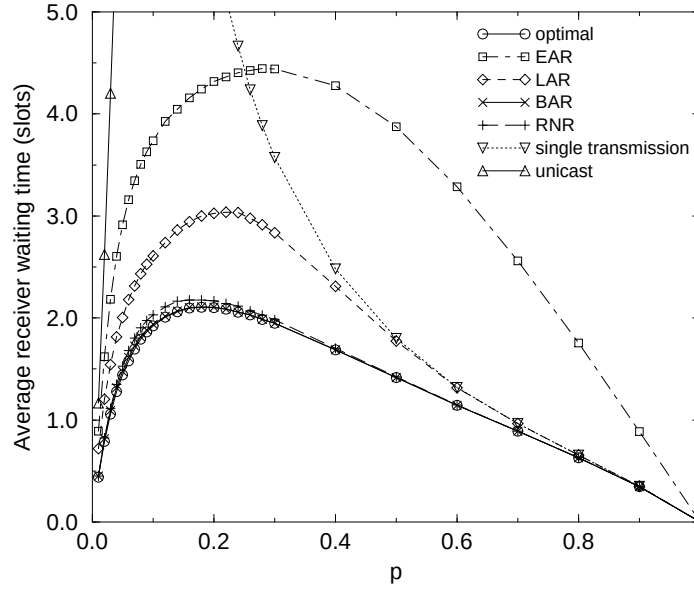
$i = i + 1$

If $i \leq M$ go to **Step 1**

end

4.5 Performance Results

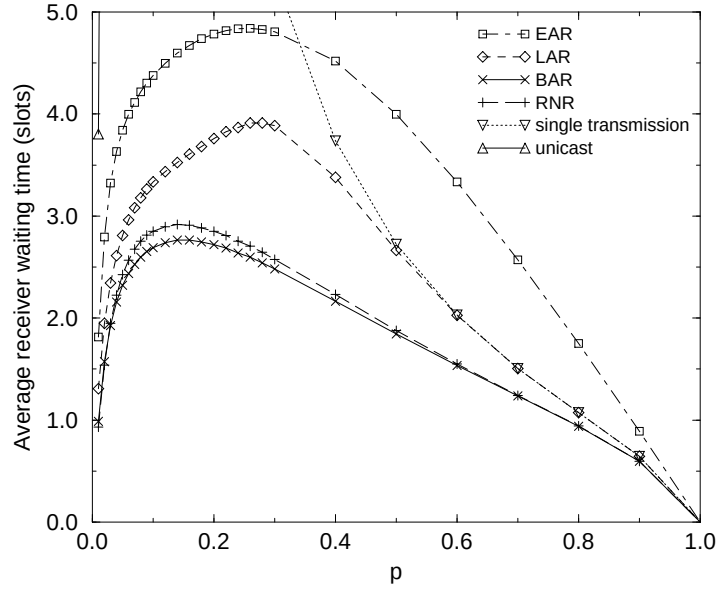
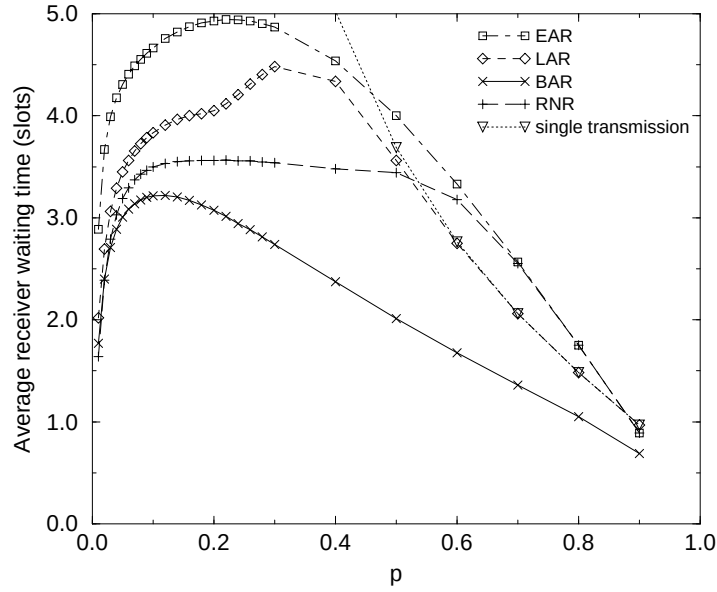
The various heuristics were tested on a number of randomly-generated test cases. Each test case consisted of generating $X_1, X_2, \dots, X_M$ randomly according to a geometric distribution with parameter p , and fixing L and M . In our examples, we set the message length $L = 10$ and let $M = 5, 10, 20$, and 50. Time is slotted, and the units for X_i and L are in slots.

Figure 4.7: Average waiting time vs. p for $L=10$ and $M=5$.

4.5.1 Waiting Time

In Fig. 4.7, we study the performance of the heuristics for $L = 10$ and $M = 5$. We compare the heuristics to optimal partitioning (obtained through exhaustive search), unicast transmissions, and traditional multicast transmissions (i.e., a single transmission to the multicast destinations). We varied the parameter for the geometric distribution, p . $p = 1$ is the case in which all receivers become available at the same time, while $p = 0$ is the case in which the receivers become available at intervals infinitely spaced apart. We observe that, for the traditional multicast and for multiple unicasts, the average waiting time can be very high. For traditional multicast, we note that the average waiting time goes to infinity as the receiver availability times become spaced further apart. For unicasts, the average waiting time increases to a fixed value as the receivers become available at the same time. This fixed value can be calculated as $\frac{L \cdot (M-1)}{2}$. Comparing the heuristics, we see that LAR offers significant improvement over both the single multicast transmission case and the unicast case. EAR compares well to the unicast case, but performs worse than the single multicast case when the receiver available times are closely spaced. This is because, while scheduling a transmission for a given receiver, EAR ignores future receiver ready times. Thus, if these receivers are available soon after the first receiver becomes available (as p approaches 1), a scheduled transmission pushes the transmission for these receivers further into the future. The more complex heuristics such as BAR and RNR offer performance near the optimum.

In Figures 4.8, 4.9, and 4.10, we examine the average waiting time for $L = 10$ and

Figure 4.8: Average waiting time vs. p for $L = 10$ and $M = 10$.Figure 4.9: Average waiting time vs. p for $L = 10$ and $M = 20$.

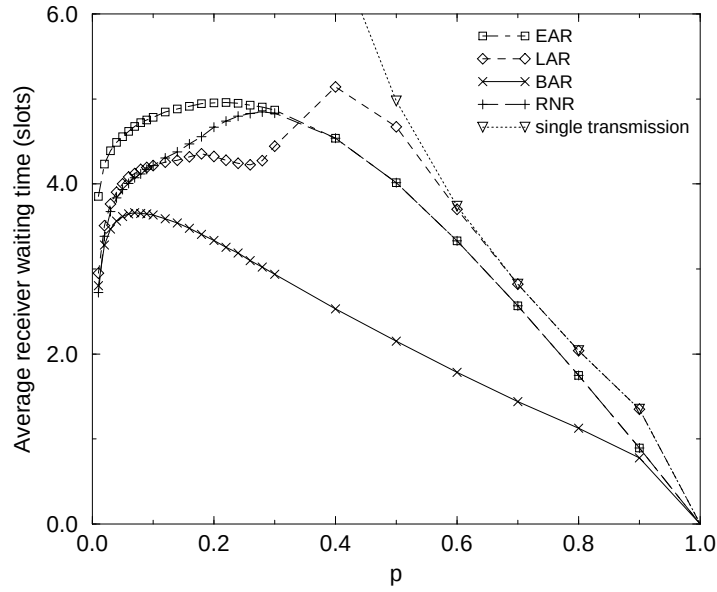


Figure 4.10: Average waiting time vs. p for $L = 10$ and $M = 50$.

$M = 10, 20$, and 50 . In these results, we do not examine the waiting times for optimal partitioning, because the required exhaustive search grows rapidly as M increases and becomes intractable for large M . In general, we see that, as M increases, the average waiting time increases for all heuristics. Except for very low values of p , the BAR heuristic consistently outperforms the other heuristics. RNR provides the next best performance, but the relative performance seems to degrade rapidly as M is increased. In the limit, it appears that RNR will approach the performance of EAR. The performance of LAR remains between those of BAR and EAR, except for large M and p close to one. In this situation, LAR approaches the performance of traditional multicast.

4.5.2 Transmitter and Channel Resources

Another measure of interest is the number of transmissions scheduled by the heuristics. In Fig. 4.11, we show the number of transmissions vs. p for each heuristic when $L = 10$ and $M = 5$. Unicasts require M separate transmissions while standard multicasts require only one transmission. (These plots are not shown in Fig. 4.11.) We observe that, if the receiver available times are further apart, the number of transmissions becomes M . When the receivers are ready at the same time, the number of transmissions is one. Compared to the number of transmissions used in the optimal case, EAR tends to schedule more transmissions while LAR tends to schedule fewer transmissions. RNR and BAR schedule

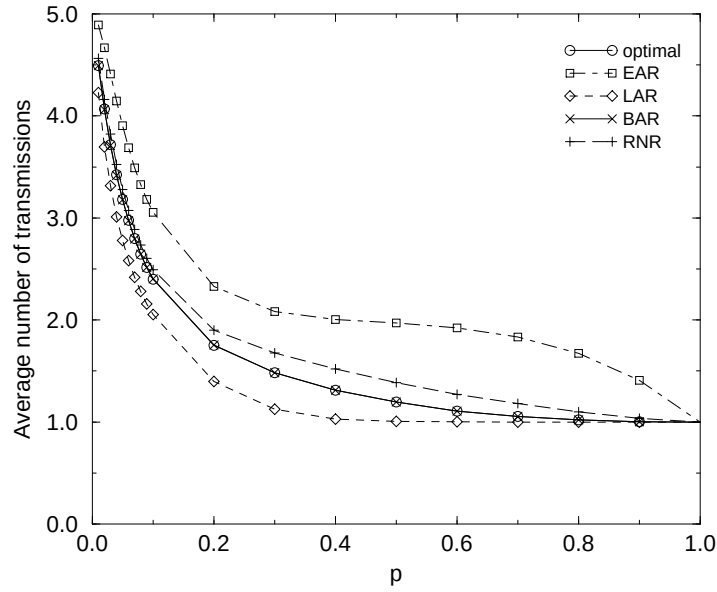


Figure 4.11: Number of transmissions vs. p for $L=10$ and $M=5$.

about the same number of transmissions as the optimal² case.

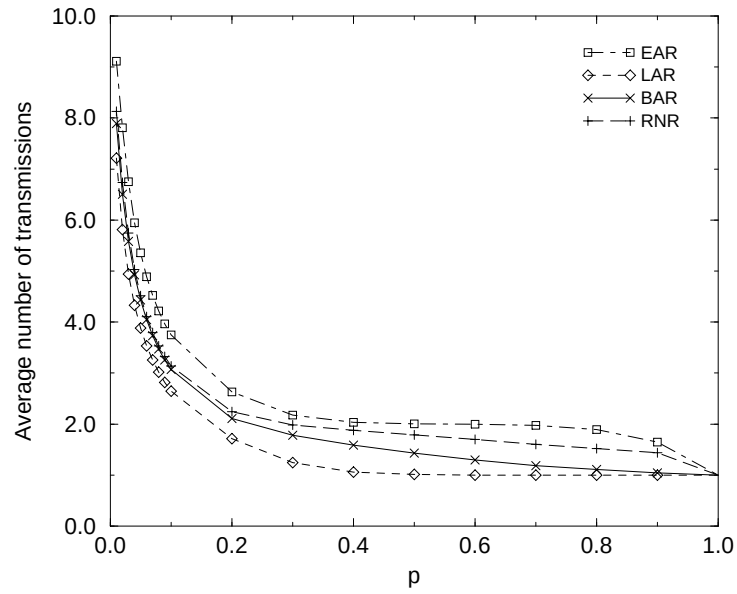
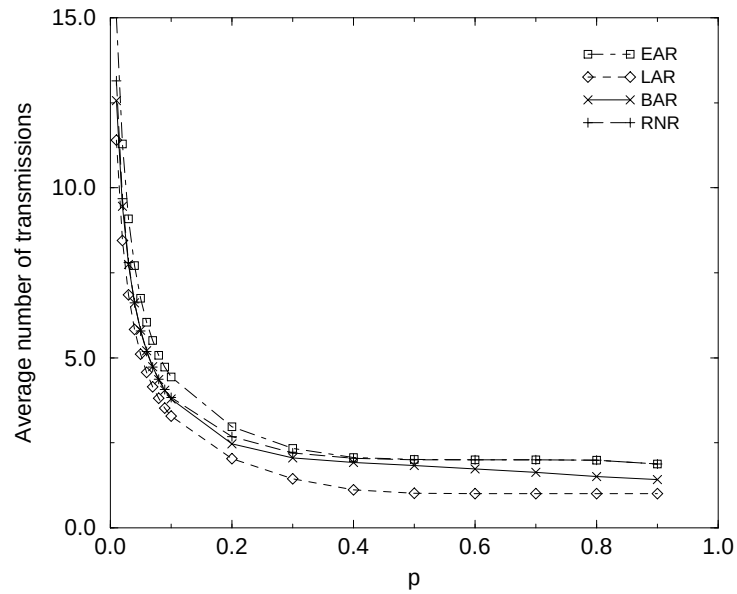
In Figures 4.12, 4.13, and 4.14, we examine the number of transmissions for $L = 10$ and $M = 10, 20$, and 50 . In these graphs we see that the relative performance of the heuristics stays about the same. Overall, we observe that the ratio of the number of transmission to M for a given p decreases as M increases. This means that, for high M , more nodes are receiving any given transmission.

4.6 Simulation Results

To compare the performance of the heuristics in a practical setting, we incorporated the heuristics into the network protocol described in [6]. We consider the following system.

- $N = 50$: Number of nodes in the network.
- $W = 50$: Number of data channels.
- Messages arrive to the system according to a Poisson process with rate λ messages per time slot.
- Message length is geometrically distributed with mean $= L$ slots.

²Note that “optimal” in this case is with respect to the utilization of receiver resources rather than utilization of transmitter resources.

Figure 4.12: Number of transmissions vs. p for $L = 10$ and $M = 10$.Figure 4.13: Number of transmissions vs. p for $L=10$ and $M=20$.

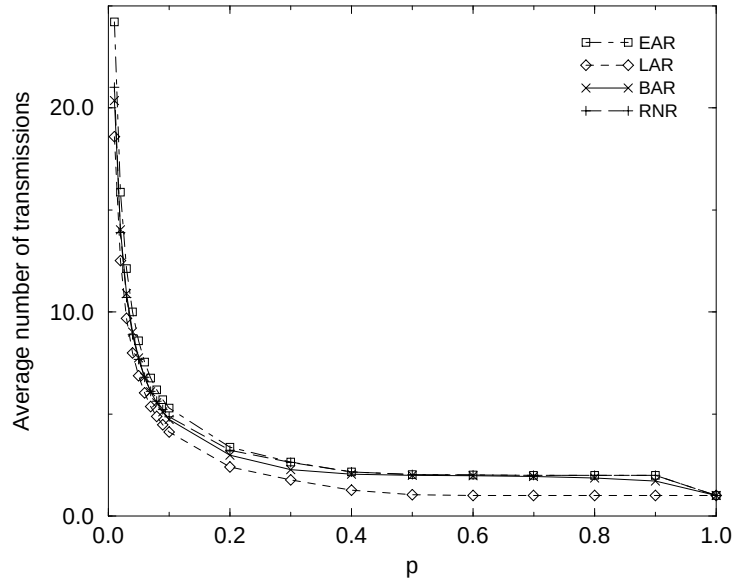


Figure 4.14: Number of transmissions vs. p for $L=10$ and $M=50$.

- M_{max} is the maximum multicast group size. The multicast group size M is uniformly distributed over $(1, 2, \dots, M_{max})$. $M_{max} = 50$ in the examples below.
- Tuning time and propagation delay are assumed to be negligible, since they are expected to be small relative to message lengths.

In addition to a tunable transmitter and a tunable receiver for sending and receiving data, each station is equipped with a fixed transmitter and a fixed receiver which are used to communicate on a control channel. When a message arrives at a station, the station sends a request packet on the control channel. The request packet contains the length of the message and the intended destinations. Each station in the network then runs a common algorithm to schedule the message. The scheduling algorithm incorporates the multicast partitioning heuristics and considers transmitter, receiver, and channel availability. The transmitter and channel availability times are accommodated by setting the inputs to the partitioning heuristic to $X_i = \max(T_j, R_i, C_k)$ for each destination i in the multicast group.

The measures of interest are receiver throughput, average message delay, and average receiver waiting time. The *normalized receiver throughput* is defined as the average fraction of busy receivers at steady state. The *average message delay* is defined as the average amount of time between a message's arrival to the system and the completion of the message's transmission to each of its destination nodes. The *receiver waiting time* is defined as the amount of time that a receiver must wait before it begins to receive a message. The waiting time is measured either from the point at which the receiver becomes available, or

the instant at which the message arrives to the system, whichever is later.

Figure 4.15 plots the delay versus throughput performance for an increasing arrival rate. We observe that partitioning the multicast groups provides significant gains over unicasting, and in some cases, gains over single-transmission multicasting. Under higher loads, LAR provides the best delay-throughput performance even though it does not provide the best partitioning for minimizing receiver waiting times. This behavior illustrates that, at some load, the availability of transmitter resources may begin to affect the system performance. We note that, for the given protocol, since each node only maintains the time at which a transmitter completes its last transmission, the key measure for the usage of transmitter resources is not necessarily the number of transmissions scheduled for the multicast, but the time at which the last transmission of the multicast is completed. The LAR heuristic guarantees that the last scheduled transmission in a multicast will end at the earliest possible time (L time units after the latest receiver becomes available); thus, the LAR heuristic outperforms the BAR heuristic in conserving transmitter resources.

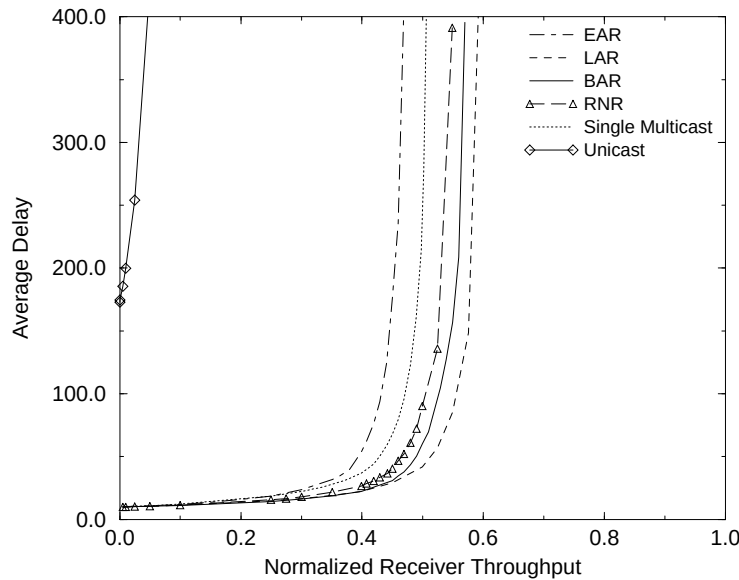
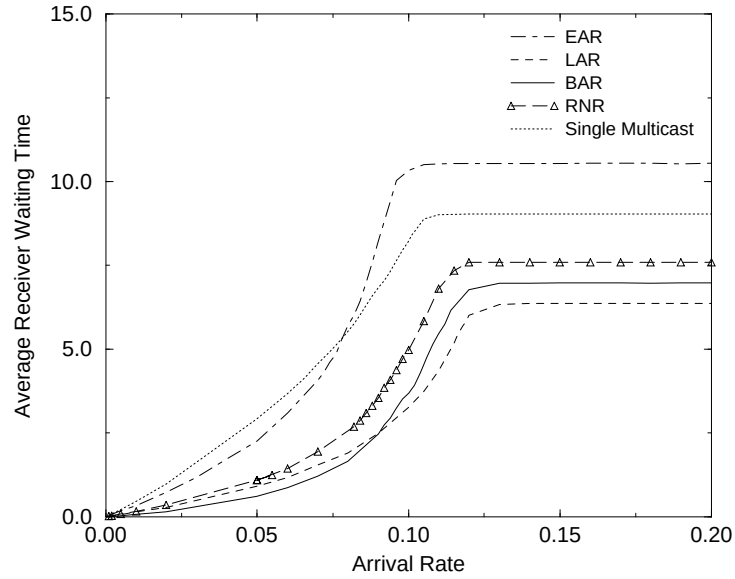
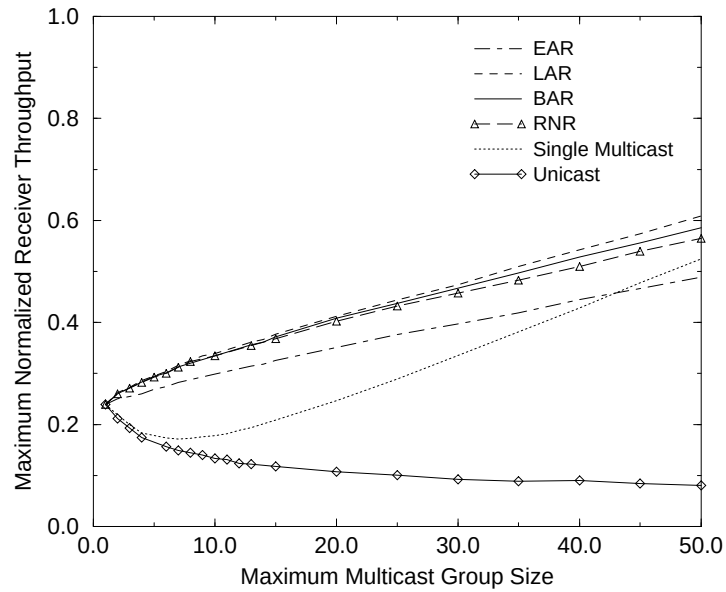


Figure 4.15: Delay vs. throughput ($M_{max} = 50$, $L = 10$).

Figure 4.16 shows the average receiver waiting time vs. arrival rate. For low arrival rates, BAR provides the lowest waiting times, which agrees with the results in Fig. 4.7. For higher arrival rates, LAR provides the lowest average receiver waiting time. We also observe that for high loads, the average receiver waiting time approaches a constant value. In the absence of transmitter and channel limitations, the receiver waiting time for the EAR, LAR, and BAR heuristics is not expected to be greater than the length of the message. If the waiting time for a receiver were to exceed the message length, then it would be possible to

Figure 4.16: Receiver waiting time vs. λ ($M_{max} = 50$, $L = 10$).Figure 4.17: Throughput vs. M_{max} for $\lambda = \infty$ and $p = 0.1$.

schedule another transmission for that receiver. At higher loads, the transmitter availability may also become a limiting factor, resulting in slightly higher receiver waiting times. For the case of a single-transmission multicast, the maximum waiting time is limited by the difference in time between the earliest and the latest available receiver in the multicast group. From Fig. 4.16, we observe that, when the load is increased beyond a certain point, the times at which the receivers become available remain within a limited distance of one another.

In Fig. 4.17, we examine the effect of the multicast group size on the maximum receiver throughput. Single-transmission multicasting tends to create more gaps in the receiver schedules, resulting in a lower throughput than for partitioned multicasts. However, when the multicast group size is increased, there is a higher probability that a given receiver will be busy at any time, reducing the volume of gaps in the schedule for single-transmission multicasting. We see that, as the multicast group size increases, single-transmission multicasting begins to approach the throughput of the partitioned-multicast schemes.

4.7 Conclusion and Future Work

It is important to consider the conservation of transmitter, receiver, and channel resources when scheduling multicast transmissions in a WDM LAN. Unnecessary multiple transmissions of a multicast message may result in a waste of bandwidth, while poor scheduling may result in excessive receiver waiting times. By partitioning a multicast transmission into multiple subgroups, an efficient balance between transmitter resources and receiver waiting times may be achieved. Simple heuristics such as LAR offer significant gains over protocols in which only unicast or single-transmission multicasts are considered.

As seen in the previous section, the conservation of receiver resources alone does not guarantee optimal performance. In future work, the multicast partitioning problem may be modified to incorporate transmitter and channel resources into the cost function. Also, our heuristics only provide the number of transmissions and the time at which each transmission should be scheduled. They do not provide the channels on which the transmissions should take place. In general, the channels should be chosen in a manner that minimizes the gaps in the channels' schedules. Another open problem is to generalize this work to accommodate multiple transmitters and receivers per node, especially multiple receivers since a multicast application will be more efficient if nodes have more receivers than transmitters.

Chapter 5

Switch Architectures for WDM Networks

5.1 Introduction

In a wavelength-routed WDM network, a connection may be established from a source node to a destination node entirely in the optical domain. To establish such a connection, also known as a *lightpath*, we require optical circuit switching at intermediate nodes. Most research on wavelength-routed WDM networks has been based on the wavelength-routing optical node architecture (known as a wavelength-routing switch (WRS) or a wavelength-selective crossconnect (WSXC) in the literature) as shown in Fig. 5.1 [35].

In such a node, at each input fiber port, the incoming signals are demultiplexed into individual wavelengths. Each wavelength is then sent to a different reconfigurable space switch. The signals are then routed by the space switch to different output fiber ports with the constraint that no two signals going to the same output fiber port may be on the same wavelength. Before being sent out on the output fiber port, each signal is multiplexed with other signals on different wavelengths which are headed to the same output fiber port.

A limitation of this type of node architecture is that it is not scalable with respect to the number of wavelengths. For example, in Fig. 5.1, the node can handle up to three wavelengths, but if additional wavelengths were to be added, an additional space switch element would be required for each new wavelength.

An alternative node architecture consists of adding to the traditional node architecture a single space switch which is *wavelength-insensitive* (Fig. 5.2). Now, all signals entering the new wavelength-insensitive component on a given input port are routed to the same output port without demultiplexing signals which are on different wavelengths. Since the wavelength-insensitive component does not distinguish individual wavelengths, it can gracefully accommodate any increase in the number of wavelengths without any additional space-switch components. The tradeoff is that, in this wavelength-insensitive component,

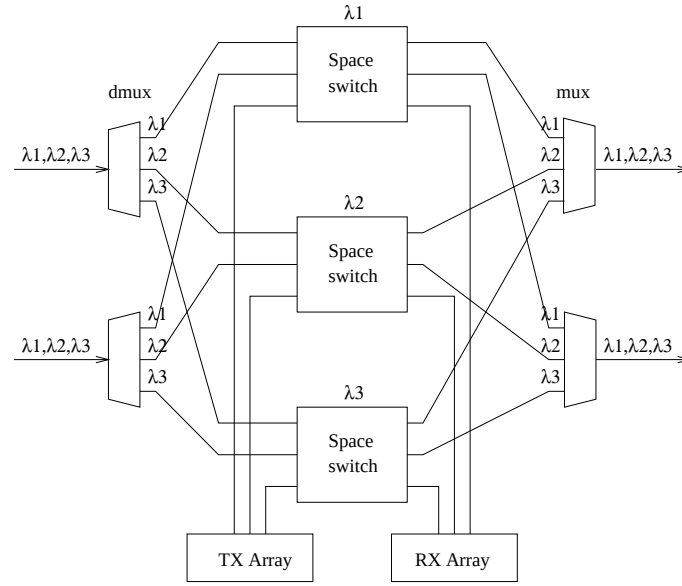


Figure 5.1: Traditional wavelength-routing node architecture.

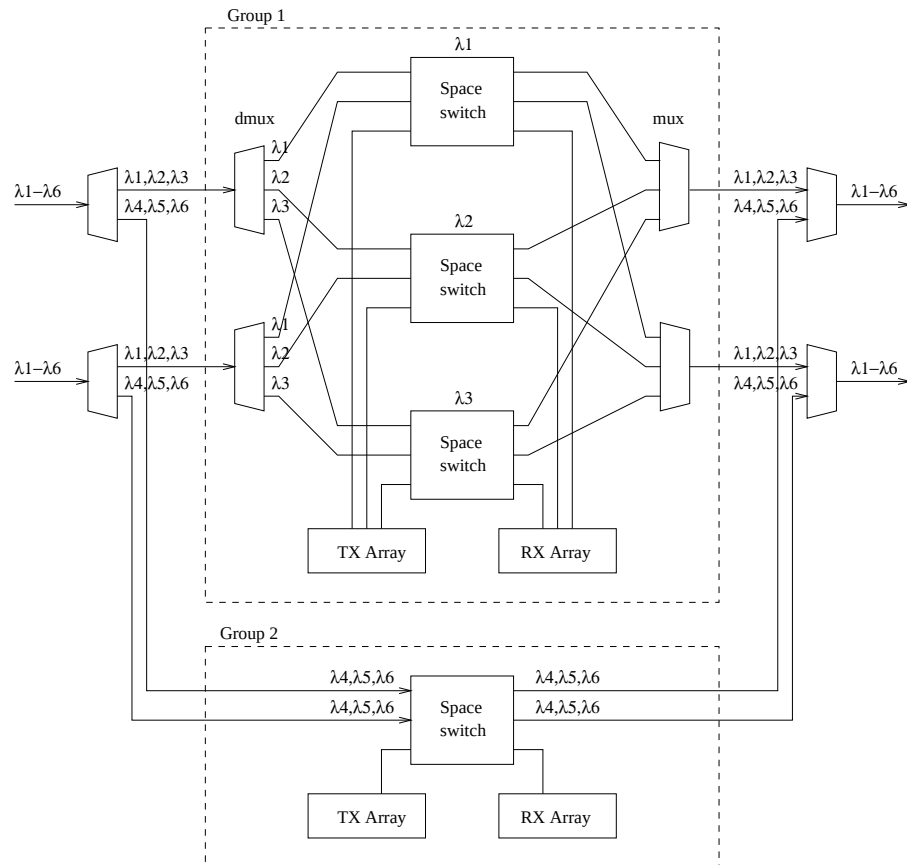


Figure 5.2: New scalable node architecture.

wavelengths entering the node on a given input port cannot be individually routed to different output ports. This inflexibility may lead to higher call blocking probabilities in the network. However, this limitation may be partially alleviated by allowing the parallel transmission of a call over several wavelengths, thus reducing the overall duration of the call. Examination of this scalability versus flexibility tradeoff is an important objective of this study.

5.1.1 Hybrid Node Architecture and its Benefits

We propose a hybrid node architecture (Fig. 5.2) which combines both wavelength-sensitive and wavelength-insensitive components. This architecture offers a number of possible advantages which are listed below.

- A call may be transmitted over a number of channels in parallel, resulting in shorter call durations.
- The traditional wavelength-routing part of the node allows for a high degree of switching flexibility, while the new wavelength-insensitive component allows for future growth capability, namely, increases in the number of wavelengths without excessive costs.
- If the spacing between the different wavelengths or channels changes in the future, then the wavelength-insensitive component would not need to be upgraded.
- The hybrid node architecture may be able to better support various classes of traffic, e.g., the wavelength-routed portion of the node may support connection-oriented traffic in the same way as traditional wavelength-routed WDM networks, while the wavelength-insensitive portion of the node may be able to effectively handle packet-switched traffic using an approach such as photonic slot routing¹ [13].

We analyze the call-blocking performance of the proposed node architecture, study policies for deciding whether to route traffic over the wavelength-sensitive or wavelength-insensitive part of the node, and develop a simple cost model to illustrate the cost-performance benefits of the proposed node architecture.

In Section 5.2, we describe the protocol and architecture for a network which utilizes the proposed node elements. Section 5.3 presents the blocking probability analysis and cost models for the network, and Section 5.4 presents some numerical examples. Section 5.5 concludes the chapter.

¹In photonic slot routing, all wavelengths in a given time slot are routed as a single unit to the same output fiber of a switch, thus eliminating the need for the demultiplexing of wavelengths at intermediate nodes.

5.2 Network Protocol and Architecture

The network to be investigated is made up of a number of nodes interconnected by optical fiber, with each node consisting of optical elements as shown in Fig. 5.2. Each fiber is capable of carrying multiple wavelengths, and the wavelengths are partitioned into two groups, Group 1 and Group 2. Each of the Group 1 wavelengths are routed individually through the wavelength-sensitive component of the node, with each wavelength utilizing a separate space switch. All of the Group 2 wavelengths are routed through the wavelength-insensitive component of the node, consisting of a single space switch, and these wavelengths must therefore all be routed together from the same input fiber port to the same output fiber port of the node. Each wavelength requires a transmitter and a receiver at each node.

The network is assumed to be a backbone network with each node being a gateway to a regional network. Thus, the traffic between any pair of nodes is aggregated traffic and may consist of packets from multiple source hosts. The aggregated traffic demand is handled as a series of call arrivals and departures, and it is modeled as follows. Connection requests arrive to the network according to a Poisson process and are serviced by setting up an appropriate all-optical connection (lightpath) between the source node and the destination node. The connection is maintained for an exponentially distributed period of time before being released. Traffic is assumed to be homogeneous, i.e., there is a single class of traffic which may utilize either Group 1 wavelengths or Group 2 wavelengths to set up a connection. An alternative situation is one in which there exist two different classes of traffic, where the Group 1 wavelengths handle one class of traffic while the Group 2 wavelengths are employed for the other class of traffic; this alternative is not examined in the present study, and it is a topic of future research.

The connection setup procedure involves both route selection and wavelength assignment. When a call arrives, a route for a lightpath between the source and the destination node is determined. In this work, we will consider fixed shortest-path routing. Alternative routing approaches include least-loaded routing [10] and fixed-alternate path routing [37]. After selection of a route, a wavelength must be chosen. The lightpath may be established on a single Group 1 wavelength or on all of the Group 2 wavelengths. When setting up a connection on Group 2 wavelengths, it is possible for the call to use multiple wavelengths. When multiple wavelengths are used, the call holding time is reduced by a factor corresponding to the number of wavelengths used. This reduction in call holding time occurs because the call's packets are being transmitted in parallel across different wavelengths. For example, if a call (which normally has a call holding time of $1/\mu$ when routed over a single wavelength) utilizes four Group 2 wavelengths, then its call holding time will be reduced to $1/4\mu$. We will assume that, if a connection is established over Group 2 wavelengths, then the call will utilize all of the wavelengths in Group 2. The reasoning behind this assumption is that all Group 2 wavelengths are routed from the same source node to the same

destination node; thus, these wavelengths cannot be used by any other source-destination pair. We also assume that a call cannot move from a Group 1 wavelength to the Group 2 wavelengths, or vice versa, anywhere along the route of the call (i.e., there is no wavelength conversion at intermediate nodes). We consider the following wavelength-selection policies:

Group 1 First (*G1F*): Attempt to establish an incoming connection request first on a Group 1 wavelength. If the connection cannot be established on a Group 1 wavelength, attempt to establish the connection on the Group 2 wavelengths. When selecting a Group 1 wavelength, a *first-fit* approach is utilized, i.e., the wavelengths are ordered, and the first available wavelength in the ordering is selected. A connection which cannot be established on either group of wavelengths is blocked.

Group 2 First (*G2F*): Attempt to establish an incoming connection request on Group 2 wavelengths first. If a connection cannot be established on the Group 2 wavelengths, try to establish it on a Group 1 wavelength. A first-fit approach is used to select a Group 1 wavelength. If a call cannot be routed on either group of wavelengths, then it is blocked.

5.3 Blocking and Cost Analysis

5.3.1 Blocking Model

Consider a network with N nodes (each with identical node architectures) and $W = W_1 + W_2$ wavelengths. The wavelength-sensitive components (Group 1) support W_1 wavelengths which can be routed independently, while the wavelength-insensitive part of the network (Group 2) supports W_2 wavelengths which are routed as a group. Each of the individual wavelengths in Group 1 can support a separate channel, while the combined W_2 wavelengths in Group 2 will support a single channel with W_2 times the bandwidth of an individual wavelength². Each channel may support a single connection at any point in time, and the total number of channels is $C = W_1 + u(W_2)$, where $u(W_2) = 0$ if $W_2 = 0$, and $u(W_2) = 1$ if $W_2 > 0$. The channels C_k , with $k = 1, 2, \dots, C$, are ordered such that calls are first attempted on Channel C_1 , and if this channel is blocked, the call is then attempted on Channel C_2 , etc. In a *G1F* approach, Channels C_1 to C_{W_1} correspond to the Group 1 wavelengths, while Channel C_{W_1+1} corresponds to the Group 2 wavelengths. In a *G2F* approach, Channel C_1 corresponds to the Group 2 wavelengths, while Channels C_2 to C_{W_1+1} correspond to the Group 1 wavelengths (see Fig. 5.3).

The network consists of a number of nodes which are interconnected by optical fiber links. Routes between source-destination pairs are defined by the links that are traversed in the route, and the arrival of calls to each route R is Poisson with rate a_R calls per second. The call holding time is assumed to be exponentially distributed with a mean of $1/\mu_k$ time units. If a call is established on a Group 1 wavelength, then μ_k is assumed to

²In this work, the term *channel* will be used to denote a group of one or more wavelengths which are routed from the same source node to the same destination node and which are used for a single connection.

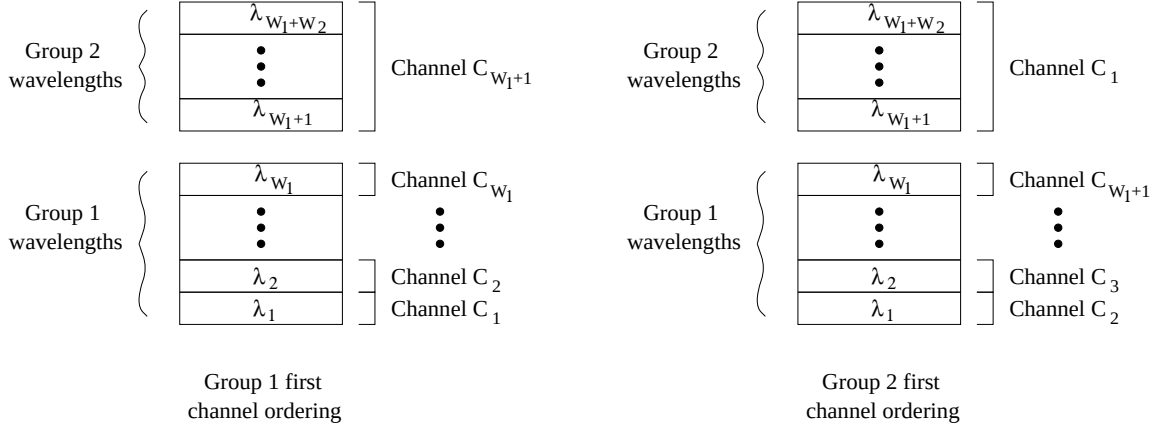


Figure 5.3: Channel orderings for the wavelength- and group-selection policies.

be 1 (normalized unit), while if a call is established on the Group 2 wavelengths, then μ_k is assumed to be W_2 .

The blocking probability is computed sequentially, starting with Channel C_1 . The blocking probability on Channel C_1 is then used in the calculation of blocking probability on Channel C_2 , the blocking probability on Channel C_2 is used to calculate the blocking probability on Channel C_3 , etc. For each channel, we iteratively calculate the load on each link, the blocking probability on each link, and the call blocking probabilities on each route.

Let $\lambda_{j,k}$ be the load on Link j on Channel k . Initially, the load on each link on Channel 1 is approximated using the following equation:

$$\lambda_{j,1} = \sum_{R:j \in R} a_R. \quad (5.1)$$

For example, in Fig. 5.4, the load on Link j on Channel C_1 is $\lambda_{j,1} = a_{R_1} + a_{R_2} + a_{R_3}$. Since some calls may be blocked, the actual load on a given channel on a given link is lower than the load indicated by the call arrival rate specified in the above equation. We use a reduced-load approximation scheme [4] to better approximate the load for a given call arrival rate.

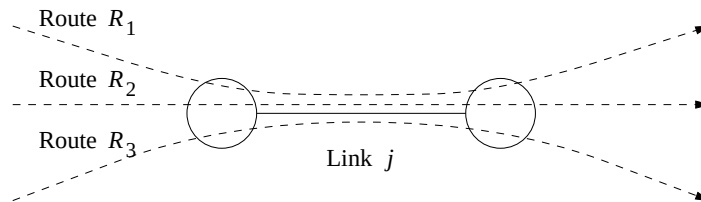


Figure 5.4: A link which is utilized by three different source-destination pairs.

The blocking probability on link j is given by:

$$P_{link}(j, k) = \frac{\lambda_{j,k}/\mu_k}{1 + \lambda_{j,k}/\mu_k} \quad (5.2)$$

which is based on the Erlang loss formula for a single channel.

The blocking probability on route R is the probability that at least one of the links is blocked, or alternatively, one minus the probability that none of the links are blocked:

$$P_{route}(R, k) = 1 - \prod_{j \in R} (1 - P_{link}(j, k)) \quad (5.3)$$

The call arrival rate to a given route R on a given channel k is approximated by multiplying the blocking probability on channel $k - 1$ by the call arrival rate on channel $k - 1$, since calls which are blocked on a given channel will be attempted on the next available channel in the sequence:

$$a_{R,k} = P_{route}(R, k - 1) \cdot a_{R,k-1} \quad (5.4)$$

Note that $a_{R,1} = a_R$ since all calls will first attempt to establish calls on Channel C_1 .

The load on Link j on Channel k can then be calculated as:

$$\lambda_{j,k} = \sum_{R: j \in R} a_{R,k}. \quad (5.5)$$

Again, a reduced-load approximation scheme can be used to yield a more accurate value for the load.

The overall call blocking probability for a given call is the probability that the call is blocked on all channels. Averaging the call blocking probability over all routes yields:

$$P_b = \sum_R \frac{a_R}{\sum_R a_R} \prod_{k=1}^C P_{route}(R, k). \quad (5.6)$$

5.3.2 Cost Model

One of the benefits of the proposed node architecture is reduced upgrade costs. To illustrate these benefits, a simple cost model is developed. In the node, the most expensive elements are the space switches and the transmitters and receivers. The cost of a space switch will be measured by the number of 2×2 crosspoint elements contained in the space switch, i.e., the cost of a 2×2 crosspoint element is normalized to unity. We will assume that the switches are implemented using a Banyan switching fabric which requires $\frac{N}{2} \log_2 N$ crosspoint elements, where N is the number of input and output ports on the space switch. For a strictly non-blocking space switch, the number of required 2×2 crosspoint elements

may be as high as N^2 . The normalized cost of a transceiver pair is given by α , which is the ratio of the cost of a transceiver element to the cost of a 2×2 crosspoint element. The total cost of a node (see Fig. 5.2) is then given by:

$$Cost = (W_1 + 1) \frac{N}{2} \log_2 N + \alpha(W_1 + W_2) \quad (5.7)$$

5.4 Numerical Examples

Consider the simple 6-node network shown in Fig. 5.5 for illustration purposes. The net-

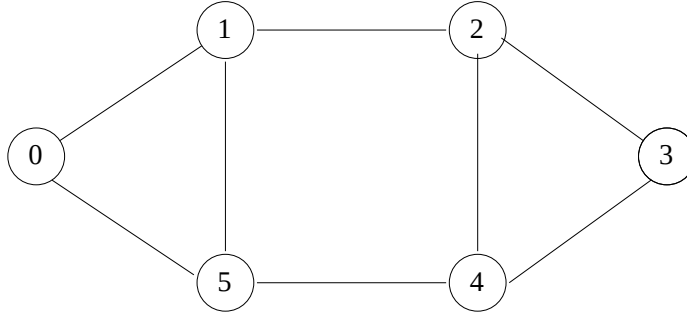


Figure 5.5: An example network topology.

work supports a total of 8 wavelengths which are partitioned into Group 1 and Group 2 wavelengths. Traffic is assumed to be uniform with the call arrival rate being the same for each source-destination pair.

In Fig. 5.6, we plot blocking probability versus call arrival rate for different group partitionings of the wavelengths. A $G2F$ policy is implemented. We observe that the network with only Group 1 wavelengths (a network consisting of traditional wavelength-routing switches) yields lower blocking probability than the network with only Group 2 wavelengths (a network with wavelength-insensitive switches). The performance difference is due to the added flexibility of routing individual wavelengths in the network with only Group 1 wavelengths. It also indicates that the compression of service times cannot fully compensate for the increase in blocking caused by fewer available alternative paths for the Group 2 wavelengths. The hybrid network with both Group 1 and Group 2 wavelengths yields a blocking probability which is between those of the other two networks.

Simulation results are also shown in Fig. 5.6 to determine the accuracy of the analysis. We observe that the analysis is accurate for the case in which only Group 2 wavelengths are used, but underestimates the blocking probability for the other two cases, particularly under low load situations. The reason for this discrepancy is that the link load values, $\lambda_{j,k}$, are only approximate values, and, when calculating the blocking probabilities for each channel, any inaccuracies in the blocking probability calculations for one channel will be compounded in the calculations of blocking probabilities for subsequent channels. Thus, the

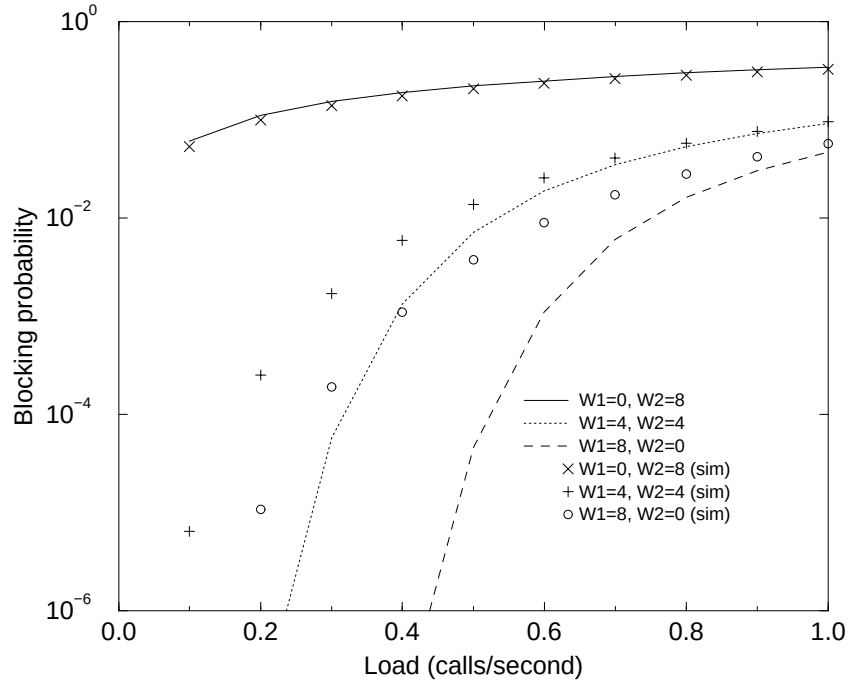


Figure 5.6: Blocking probability vs. load for various wavelength groupings ($W = 8$).

analysis becomes less accurate for systems in which there is a greater number of channels.

5.4.1 Effect of Channel Ordering in Wavelength Selection

Figure 5.7 compares blocking probabilities for the two different channel-ordering schemes. The results show that, by attempting to use Group 2 wavelengths before using Group 1 wavelengths, the blocking probability can be reduced. The Group 2 wavelengths essentially act as a single higher-speed channel, and by establishing calls on the Group 2 wavelengths first, a larger number of calls will be completed in a shorter amount of time.

In Fig. 5.8, we plot blocking probability versus W_1 for various values of load and different channel orderings. The number of wavelengths in Group 2 is equal to $W - W_1$. Note that the case in which $W_1 = 7$ and $W_2 = 1$ is identical to the case in which $W_1 = 8$ and $W_2 = 0$. When there is only one wavelength in Group 2, that wavelength behaves similarly to a Group 1 wavelength, since it is routed independently of all other wavelengths. We also observe that the $G2F$ policy outperforms the $G1F$ policy, and that the advantage of establishing calls on Group 2 wavelengths first is greater for lower loads. Obviously, if there are only Group 1 wavelengths ($W_1 = 8$) or only Group 2 wavelengths ($W_1 = 0$), then the ordering of the channels doesn't matter.

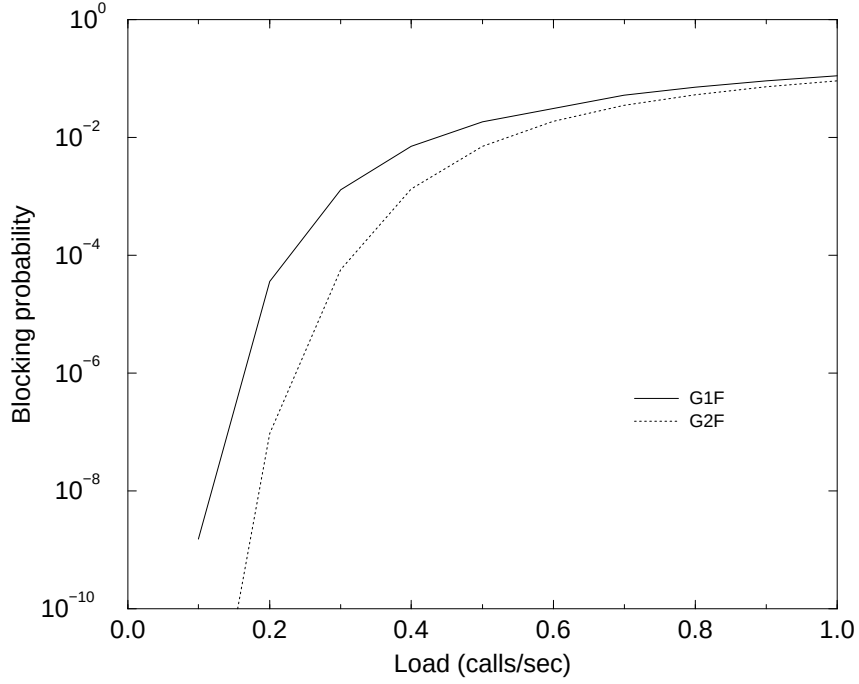


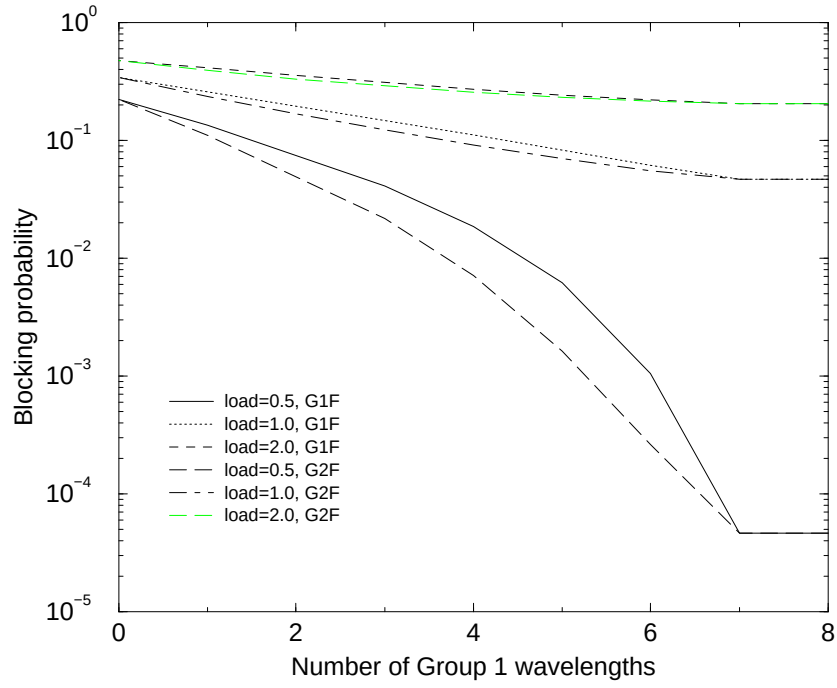
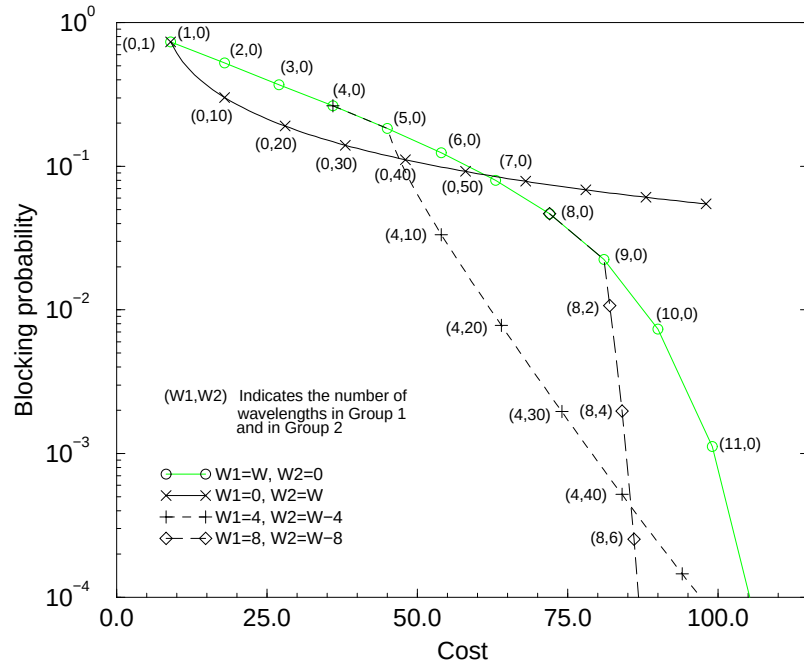
Figure 5.7: Comparison of different channel orderings with $W_1 = 4$ and $W_2 = 4$.

5.4.2 Cost Benefit

In Fig. 5.9, we plot blocking probability versus cost for various node configurations with $\alpha = 1$ (i.e., a transceiver has the same cost as a 2×2 crossconnect element) and a load of 1.0 calls per second for each source-destination pair. The total number of wavelengths, W , is varied to obtain different cost-performance points. The number of wavelengths in Groups 1 and 2 are shown for various points in the graph.

When W_2 is increased from 0 to 1, an additional space switch as well as an additional transceiver needs to be added to each node, resulting in a jump in the cost. However, as W_2 is increased beyond 1, only transceivers need to be added to each node, resulting in much lower values for cost increments. This effect causes a sharp change in the slope of the blocking probability-cost curves at the points where $W_2 = 1$. Note that, although adding a single Group 2 wavelength is less costly than adding a single Group 1 wavelength, each additional Group 1 wavelength results in a greater performance improvement.

For a desired level of blocking performance, we observe that a network containing both wavelength-sensitive and wavelength-insensitive components is potentially less costly than a network consisting only of wavelength-sensitive components. For example, a system in which four Group 1 wavelengths and twenty Group 2 wavelengths are used ($W_1 = 4, W_2 = 20$) performs about as well as a system in which there are only ten Group 1 wavelengths

Figure 5.8: Comparison of different channel orderings for different loads with $W = 8$.Figure 5.9: Blocking probability vs. cost for an arrival rate of 1.0 call/second and $\alpha = 1$.

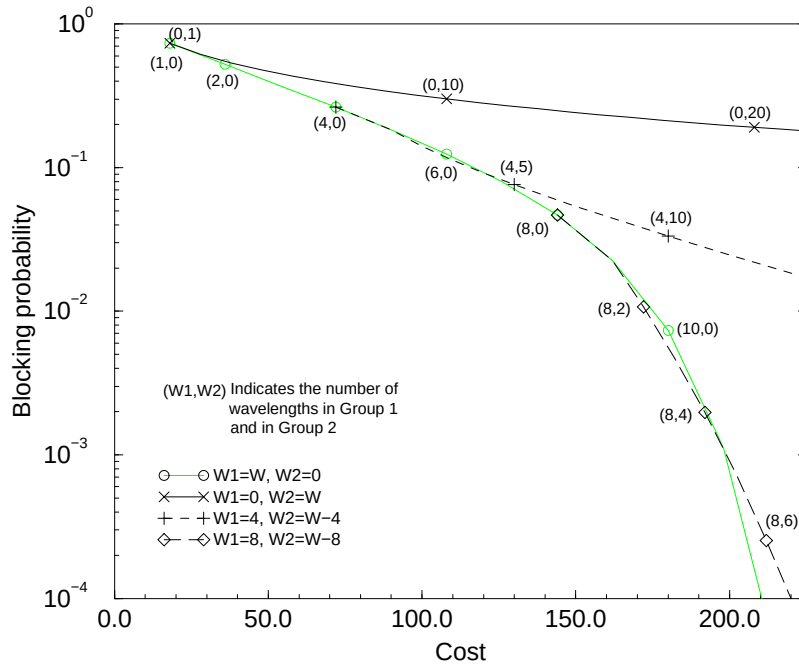


Figure 5.10: Blocking probability vs. cost for an arrival rate of 1.0 call/second and $\alpha = 10$.

($W_1 = 10, W_2 = 0$). Although the first system utilizes a greater number of wavelengths, it utilizes fewer switching elements, and thus has a lower overall cost. Therefore, the ($W_1 = 4, W_2 = 20$) system is much more preferable than the ($W_1 = 10, W_2 = 0$) system, assuming that constraints on the available number of wavelengths are insignificant.

Initially, when adding wavelengths to a network, it may be less costly to simply add a number of new wavelengths to Group 2. However, at a certain point, adding wavelengths to Group 2 will not provide significant performance enhancements. There may also be physical issues which may limit the maximum number of wavelengths that can be added to the system. At this point, it would be more cost effective to add wavelengths to Group 1 using additional switching components.

We also note that introducing an additional wavelength to Group 2 results in higher performance benefits when there is a larger number of Group 1 wavelengths. For example, adding ten Group 2 wavelengths to a system with no Group 1 wavelengths results in a small improvement in blocking performance, while adding ten Group 2 wavelengths to a system with four Group 1 wavelengths results in a much higher performance improvement. This result indicates that it may be beneficial to place as many wavelengths as possible into Group 1 when initially designing the system.

When transceiver costs are much higher than crossconnect costs, the advantage of having Group 2 wavelengths is not as significant. In Fig. 5.10, we show blocking probability

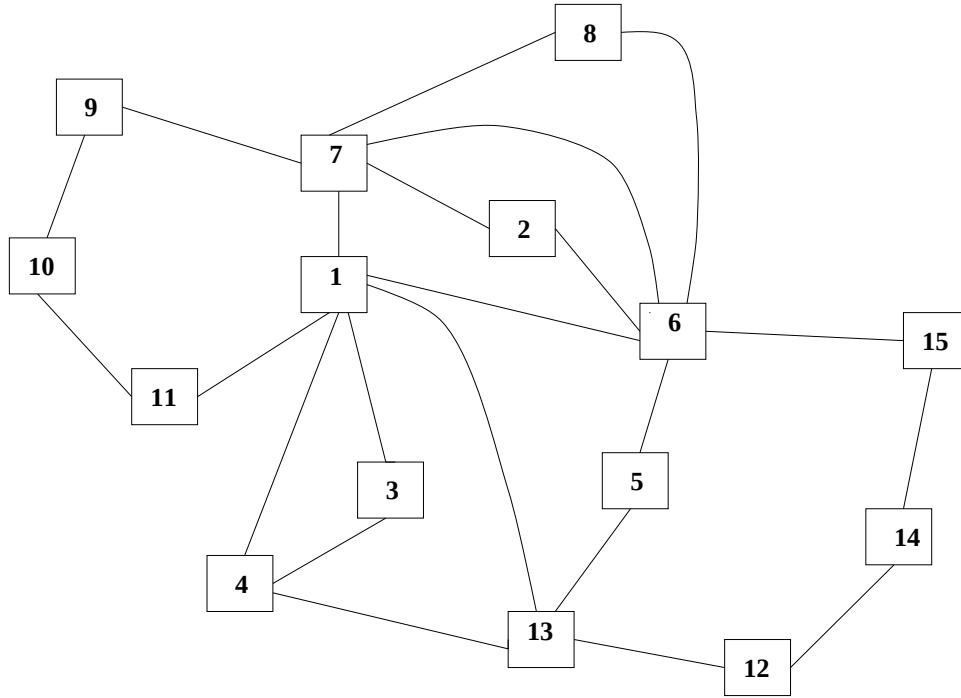
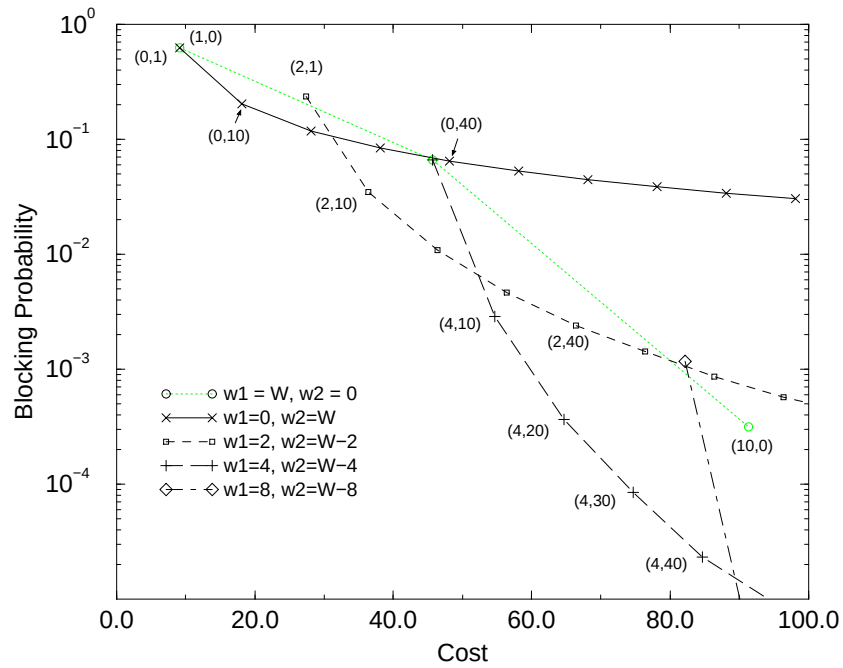


Figure 5.11: Topology of a typical 15-node telecommunications network.

Figure 5.12: Blocking probability vs. cost for a 15-node network with arrival rate = 0.1 calls per second and $\alpha = 1$ (simulation).

versus cost for $\alpha = 10$ (i.e., a transceiver is 10 times as costly as a 2×2 crossconnect element). We note that the system with only Group 2 wavelengths offers no advantage, while systems with fewer Group 2 wavelengths offer only marginal benefits over the system in which there are only Group 1 wavelengths.

In order to study the cost-performance benefits of the node architecture in larger networks, we consider the 15-node network shown in Fig. 5.11. Since the analysis for such a large network is computationally intensive, we show only simulations results. Figure 5.12 shows simulation results for blocking probability versus cost for an arrival rate of 0.1 calls per second and $\alpha = 1$. We observe that the results are similar to those for the 6-node network.

5.5 Conclusion

When designing a wavelength-routed WDM network, it is important not only to consider performance, but also to consider cost and scalability. In order to be scalable, a WDM network should not only be able to accommodate additional wavelengths to meet increasing traffic demands, but it should also be able to accommodate changes in the spacing between wavelengths. The traditional wavelength-routing node architecture offers good performance, but lacks in terms of cost and scalability. The traditional wavelength-routing node utilizes a separate space switch for each wavelength, an approach which provides a high degree of flexibility and minimizes blocking probability. However, designing network nodes in this way may limit the future expandability of the network and may result in high costs due to the large number of switching elements and the wavelength-specific nature of each switching element. An alternative approach is to allow multiple wavelengths to use the same space-switch component. Since wavelengths share the same space-switch component, they cannot be routed independently, resulting in increased blocking probability in spite of the compressed call durations during Group 2 transmissions. However, introducing new wavelengths to the system is relatively inexpensive, requiring only an additional transceiver pair at each node and no additional switching elements. We have shown, using simple blocking and cost models, that, by including a wavelength-insensitive component in the node, new wavelengths may easily be introduced to the network at a reasonable cost, and the network may have a much favorable performance-cost tradeoff when compared to a network with a traditional wavelength-routing node architecture. Another potential benefit of the new node architecture and a topic for future research is the possibility of a node supporting multiple classes of traffic, with one class of traffic traversing the wavelength-routed part of the node, and another class of traffic traversing the wavelength-insensitive part of the node.

Chapter 6

Conclusion

The amount of traffic being transmitted over existing networks is rising at an unprecedented rate, and telecommunications and data communications companies are racing to provide the means for meeting these demands. Wavelength-division multiplexing is an approach which recently has been receiving much attention. Existing WDM systems are being used only to provide higher bandwidth; however, WDM has the potential to provide many additional benefits. The primary challenge is to design systems which take full advantage of the capabilities of WDM components. In this dissertation, we presented just a few examples of protocols and architectures which effectively and efficiently utilize various optical components. In this chapter, we will summarize the contributions of this work, and we will provide some directions for future research.

6.1 Summary of Contributions

A number of WDM local area networks which are based on the passive-star coupler have been implemented as experimental testbeds. In Chapter 2, we presented an analysis of one testbed, the Rainbow network, which was developed at IBM. It was found that, for given network parameters, there is a value of the timeout duration which optimizes the network performance. If the timeout duration is too low, stations timeout too quickly, leading to fewer connections being established, while if the timeout duration is too high, there may be a greater number of deadlocks. It was also shown that the high connection setup time also makes the Rainbow system impractical for packet-switched traffic.

Traditionally, WDM networks based on the passive-star coupler have been characterized by the node architecture, and whether the protocol is single-hop or multihop. In Chapter 3, we developed a framework for further characterizing WDM networks based on the number of transceivers per node, the number of wavelengths, and the amount of tuning required. A new class of protocols, called multiconfiguration multihop protocols (MMPs), was developed which makes use of both tuning and multiple hops to establish connectivity

in the network. It was shown that such an approach can offer performance benefits over traditional single-hop and multihop protocols running over the same architecture. Further work may be required to determine the optimal logical topologies for given network parameters, as well as to extend the protocols to accommodate dynamic variations in traffic patterns.

In Chapter 4, heuristics for the efficient partitioning of multicast groups in a WDM LAN was presented. The heuristics attempt to conserve receiver resources by minimizing the amount of time a receiver has to wait for a multicast message. The proposed partitioning approach was shown to outperform both unicasting and traditional single-transmission multicasting. This work may be extended to also consider the amount of transmitter resources required for multicast transmissions, and to include the case in which each node has multiple transmitters and receivers.

In Chapter 5, a new node architecture for providing scalability in wavelength-routed networks was presented. By introducing wavelength-insensitive components into the network, it was shown that the number of wavelengths may be increased without significant additional costs in the switching components.

6.2 Future Work

Although significant progress has been made over the past several years, much additional work in WDM research is still needed before WDM succeeds in fulfilling its promise. Below, we outline some possible future directions for WDM research.

6.2.1 Multi-Gigabit LANs

The LAN architectures of today have evolved rapidly in terms of the bandwidth offered to the end-user, with current and emerging LANs such as FDDI and Gigabit Ethernet utilizing high-speed optical links. We expect that WDM-based optical technologies will take the rapid evolution in LANs even further. Although WDM networks offer huge bandwidth improvements over current LANs, no WDM LAN has been commercially deployed yet. In order to be cost-effective and practical, next-generation LANs must meet a number of requirements. First, a LAN must be scalable with respect to both bandwidth and the number of nodes, i.e., it should be relatively easy and inexpensive to upgrade a LAN to handle a greater amount of traffic and a greater number of end-users. Second, future LANs must also be able to provide quality of service (QoS) in terms of bandwidth, delay, and jitter (variations in the delay), especially for multimedia applications. Finally, it is important that next-generation LANs be compatible with legacy LAN equipment, so as to minimize upgrade costs.

Work must be done to incorporate traditional protocols, such as Ethernet, into WDM networks in order to ensure compatibility with existing infrastructure. Such compatibility

is essential for situations in which a large investment has already been made in legacy LAN equipment.

WDM LAN architectures and protocols which provide a greater variety of services to the end user must also be developed. In addition to multicasting services and connection-less packet-switched services, a WDM LAN may also need to provide connection-oriented services as well as varying levels of services for different applications.

6.2.2 WDM for the Next-Generation Internet

As the Internet continues to evolve, it will be expected to support a growing number of applications, such as Internet telephony, video conferencing, and video distribution. To support these applications, the network must be capable of not only providing the required bandwidth, but also providing guarantees with respect to quality of service, security, and fault tolerance. A WDM network can provide the necessary bandwidth to support a wide range of multimedia Internet services; however, further work is required towards the development of new switch and router architectures, along with appropriate protocols, to enable service guarantees. Implementing the Internet Protocol (IP) over a WDM network also requires the consideration of a number of additional factors. Approaches for setting up and taking down WDM lightpaths in an IP over WDM environment need to be investigated, and the interaction between lightpath reconfiguration and IP routing needs to be studied.

With today's networks experiencing enormous growth, it is important that we develop and take advantage of technologies to keep pace with these increasing demands. WDM is an extremely promising technology for supporting these needs, and by developing architectures and protocols which fully utilize the benefits of WDM, we can ensure that this tremendous growth will continue well into the future.

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